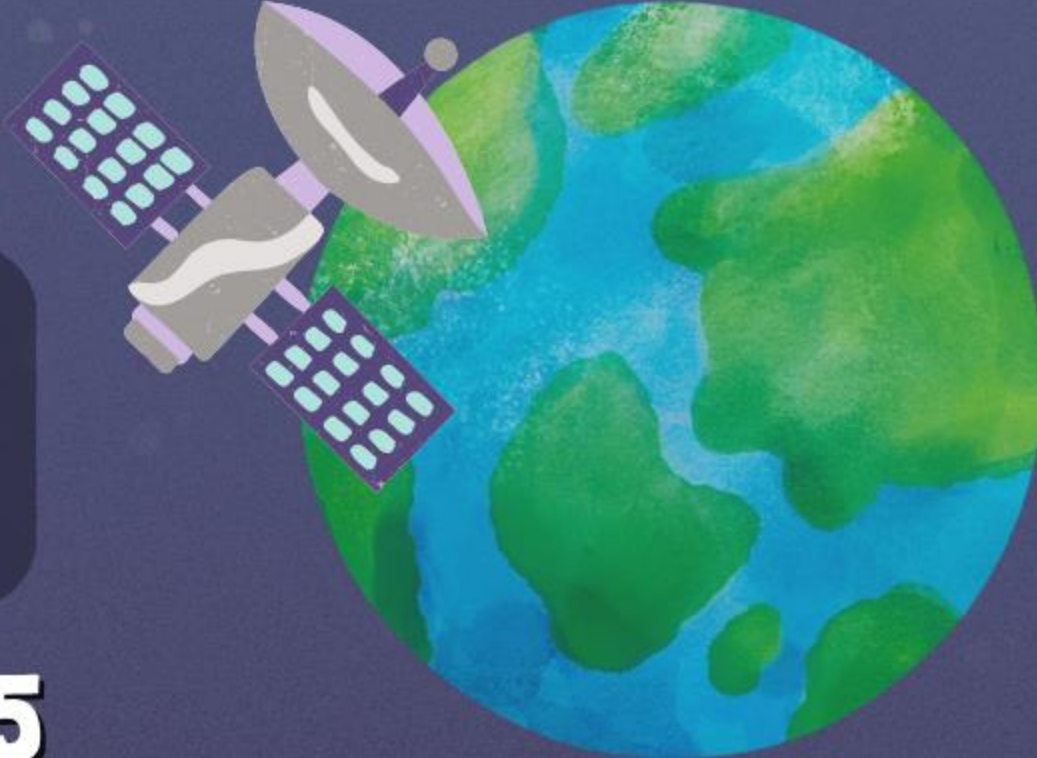




# COMPUTER NETWORKS

23CS/MC/CN55



# UNIT 1

BASIC COMPUTER CONCEPTS  
NETWORK MODELS <sup>2</sup>

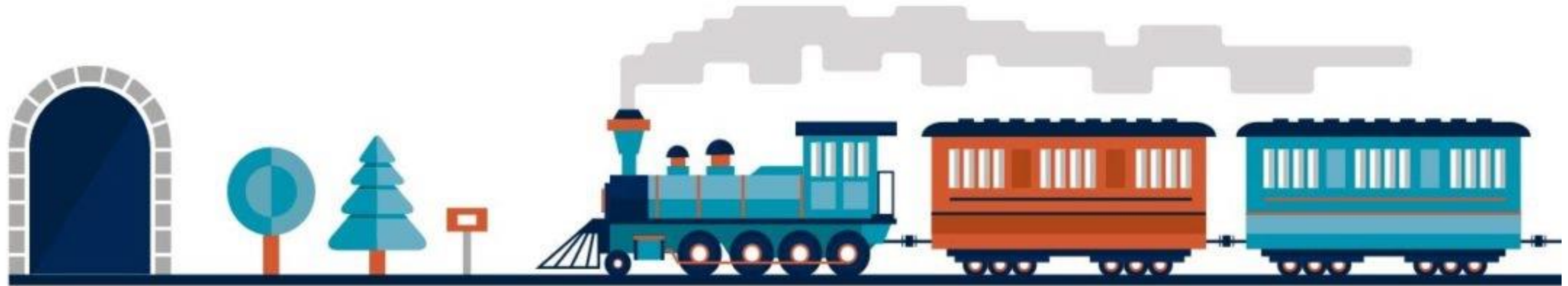


# What is a Network?

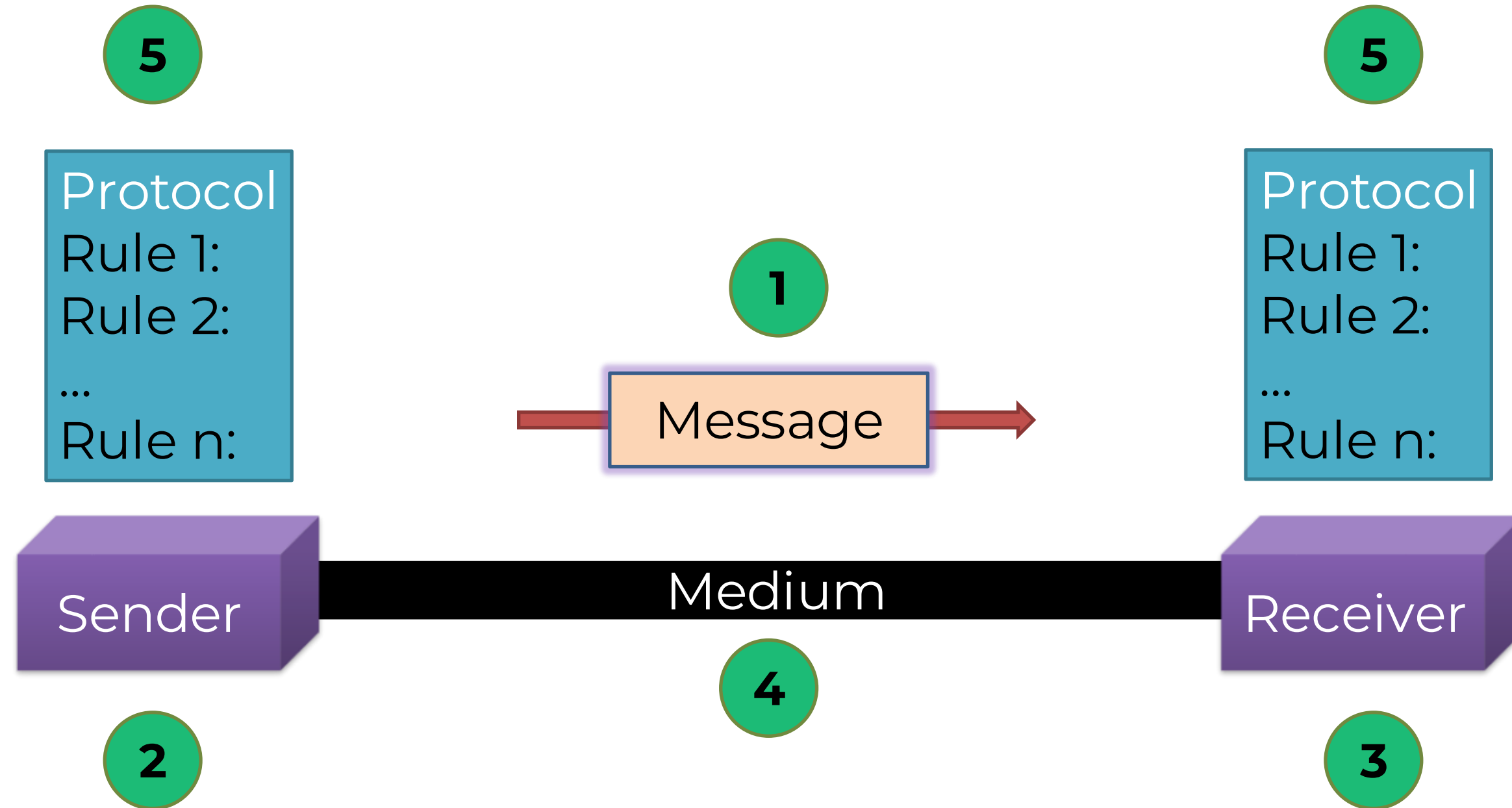


# What is a Network?

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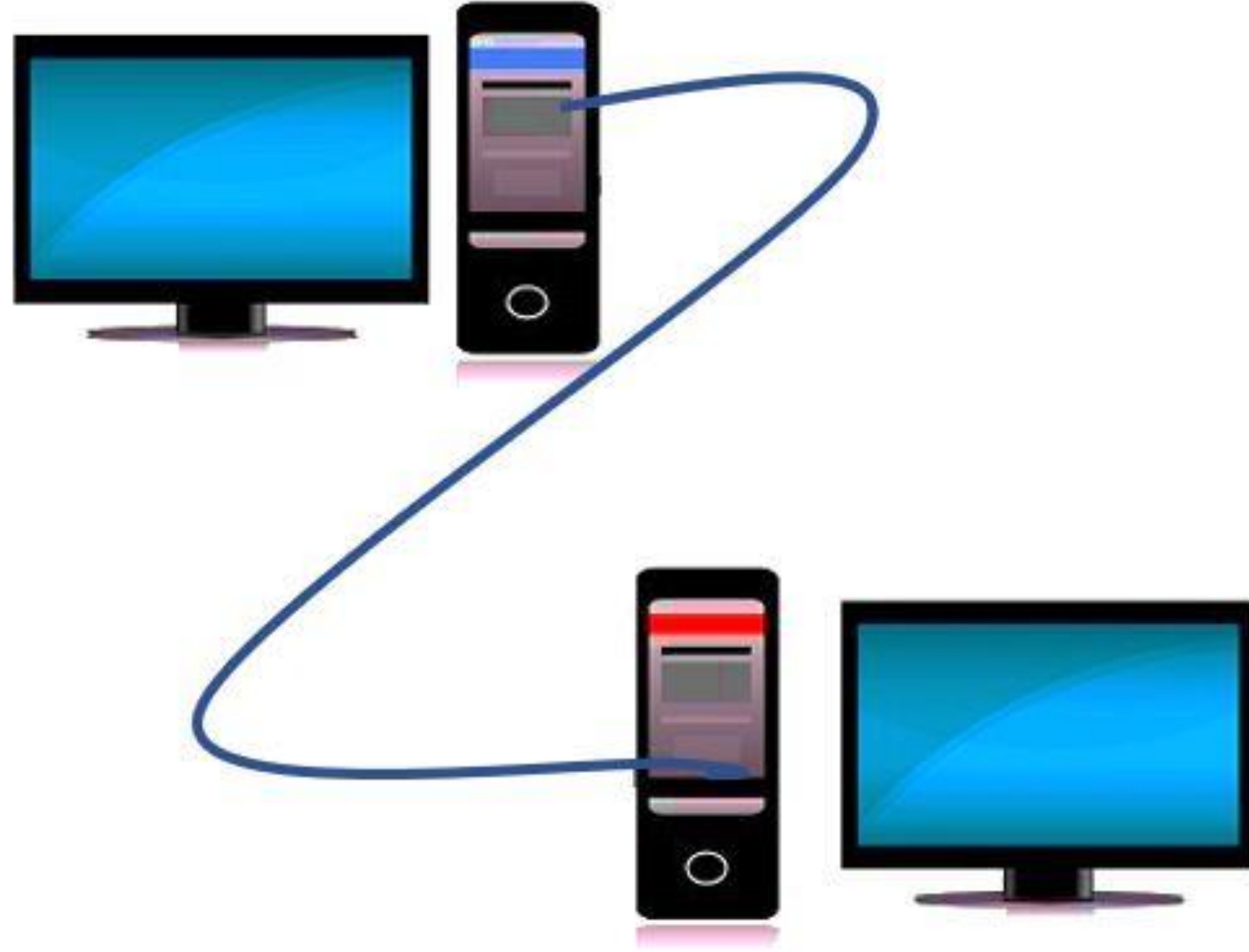


# Components



# Guess the Components

---



# Data Communications

---



- ▶ The exchange of data between two devices via some form of transmission medium.
- ▶ The communicating devices must be part of a communication system made up of a combination of hardware and software.

# Networks

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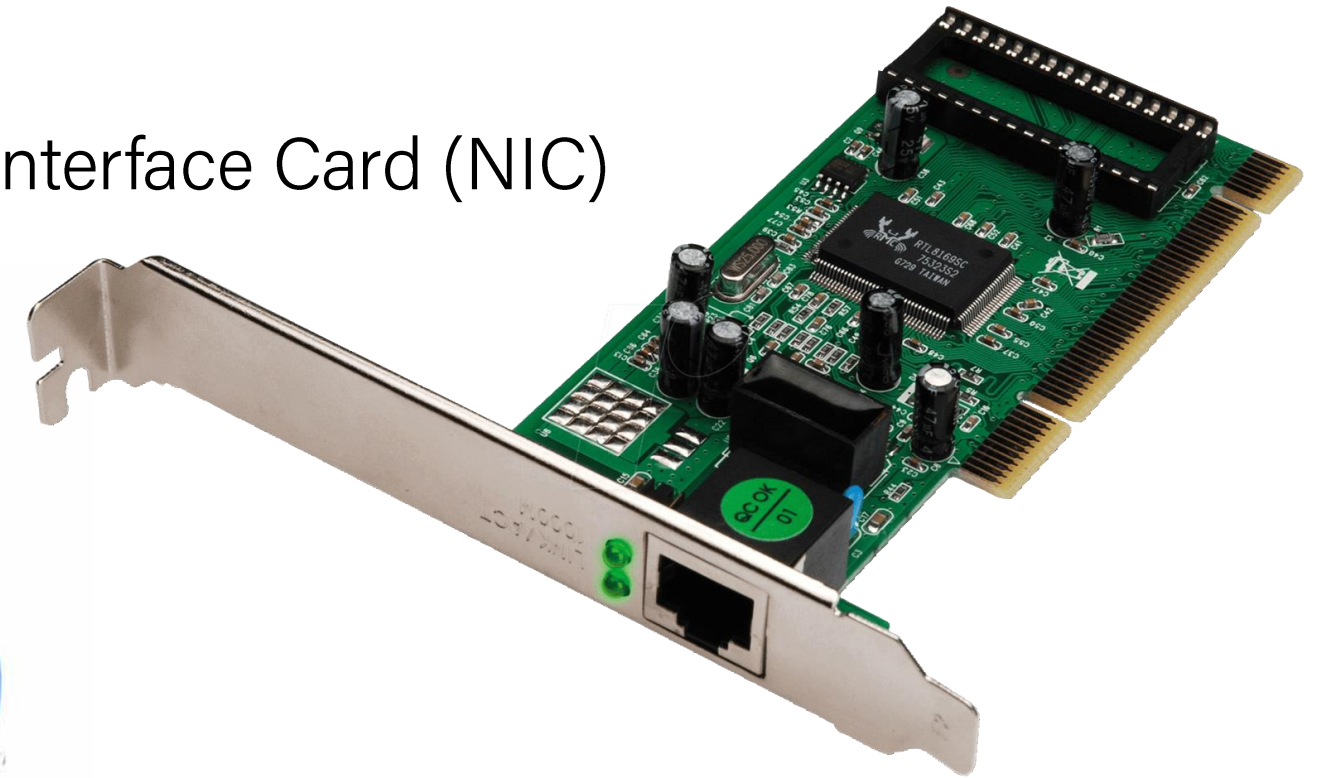
- ▶ A network is the interconnection of a set of devices capable of communication.
- ▶ A device can be a host
  - Examples: large computer, desktop, laptop, workstation, cellular phone, or security system.
- ▶ A device can also be a connecting device
  - Examples: router that connects the network to other networks, a switch that connects devices together, a modem (modulator-demodulator) that changes the form of data, and so on.
- ▶ These devices in a network are connected using wired or wireless transmission media

# Basic Computer Network



## ETHERNET – Wired Local Area Network (LAN)

Network Interface Card (NIC)



CAT 5 Cable



RJ45 Connector





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**amazon**



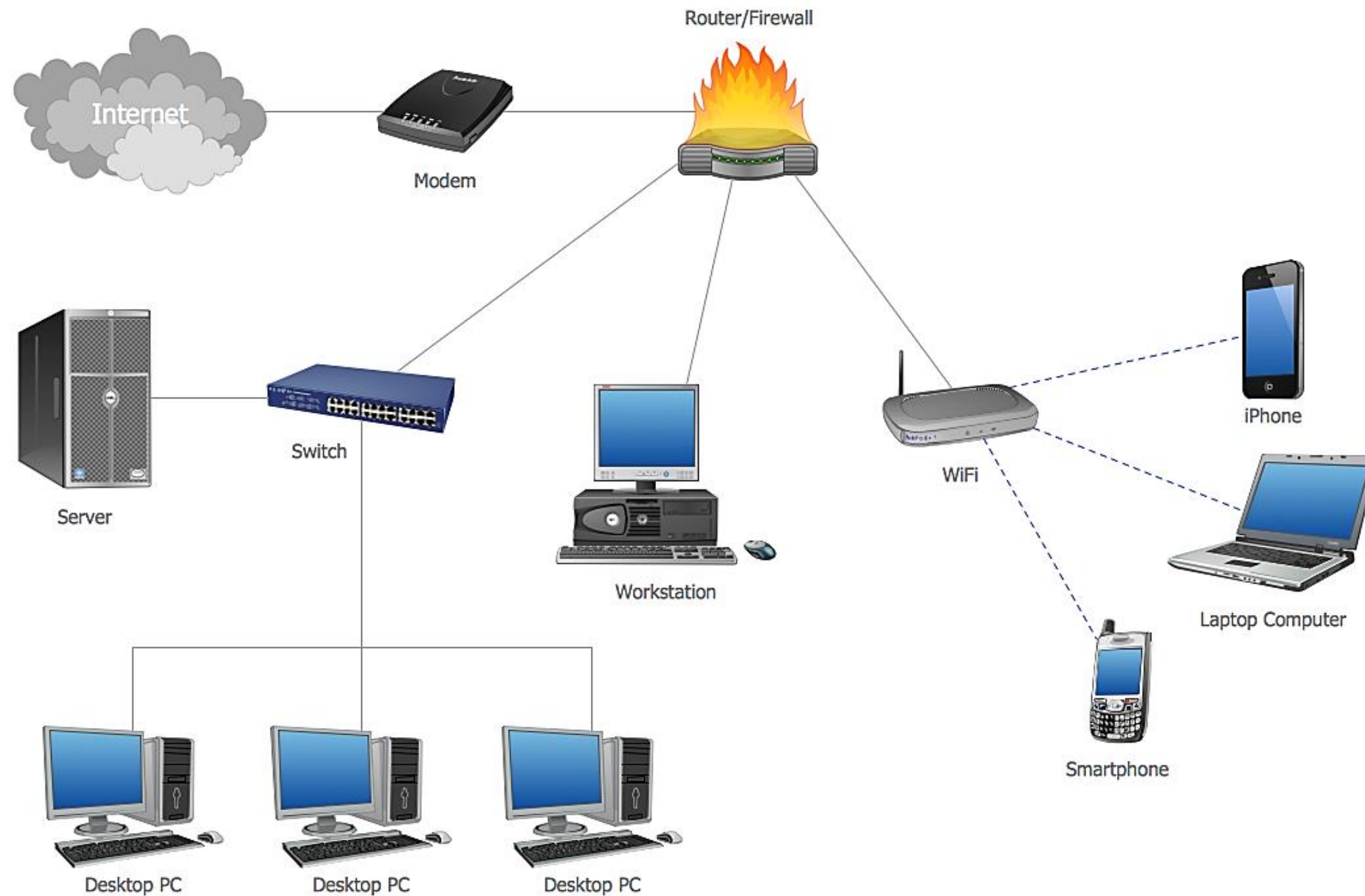
# Effective Data Communication

---



- ▶ 4 fundamental characteristics:
- ▶ Delivery
  - The system must deliver data to the correct destination.
- ▶ Accuracy
  - The system must deliver the data accurately.
- ▶ Timeliness
  - The system must deliver data in a timely manner.
- ▶ Jitter
  - Refers to the variation in the packet arrival time.

# Computer Network



# Data Representation

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- ▶ Text
- ▶ Numbers
- ▶ Images
- ▶ Audio
- ▶ Video

# Text

---



- ▶ Text is represented as a bit pattern, sequence of bits (0s or 1s).
- ▶ Different sets of bit patterns to represent text symbols.
- ▶ Each set is called a code.
- ▶ The process of representing symbols is called coding.
  - Unicode: 32 bits representation of any symbol/character used in most scripts of the world.
  - American Standard Code for Information Interchange (ASCII): has the first 127 characters in Unicode; also referred to as Basic Latin

# Text



ASCII/8859-1 Text

|   |           |
|---|-----------|
| A | 0100 0001 |
| S | 0101 0011 |
| C | 0100 0011 |
| I | 0100 1001 |
| I | 0100 1001 |
| / | 0010 1111 |
| 8 | 0011 1000 |
| 8 | 0011 1000 |
| 5 | 0011 0101 |
| 9 | 0011 1001 |
| - | 0010 1101 |
| 1 | 0011 0001 |
|   | 0010 0000 |
| t | 0111 0100 |
| e | 0110 0101 |
| x | 0111 1000 |
| t | 0111 0100 |

ASCII has 128  
characters

Unicode Text

|    |                     |
|----|---------------------|
| A  | 0000 0000 0100 0001 |
| S  | 0000 0000 0101 0011 |
| C  | 0000 0000 0100 0011 |
| I  | 0000 0000 0100 1001 |
| I  | 0000 0000 0100 1001 |
|    | 0000 0000 0010 0000 |
| 天  | 0101 1001 0010 1001 |
| 地  | 0101 0111 0011 0000 |
|    | 0000 0000 0010 0000 |
| س  | 0000 0110 0011 0011 |
| ل  | 0000 0110 0100 0100 |
| ا  | 0000 0110 0010 0111 |
| م  | 0000 0110 0100 0101 |
|    | 0000 0000 0010 0000 |
| a  | 0000 0011 1011 0001 |
| \$ | 0010 0010 0111 0000 |
| γ  | 0000 0011 1011 0011 |

Unicode has  
1,54,998  
characters

# Numbers



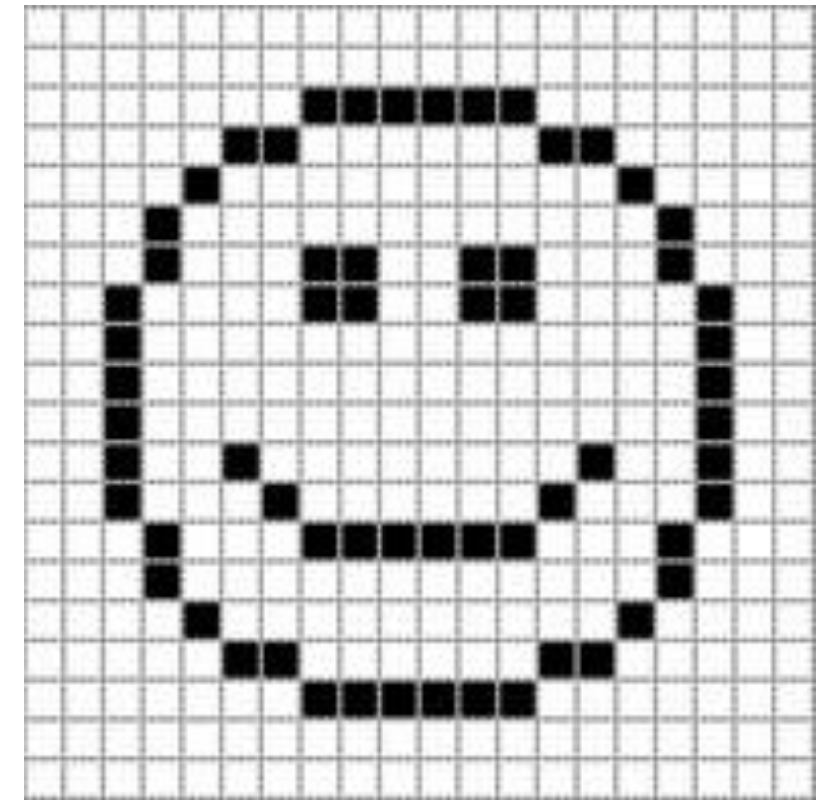
- ▶ Numbers are represented by bit patterns.
- ▶ The number is directly converted to a binary number to simplify mathematical operations.

| Binary Value | Decimal Representation |     |     |   | Decimal Value |
|--------------|------------------------|-----|-----|---|---------------|
|              | 8                      | 4   | 2   | 1 |               |
| 0 0 0 0      | 0 +                    | 0 + | 0 + | 0 | 0             |
| 0 0 0 1      | 0 +                    | 0 + | 0 + | 1 | 1             |
| 0 0 1 0      | 0 +                    | 0 + | 2 + | 0 | 2             |
| 0 0 1 1      | 0 +                    | 0 + | 2 + | 1 | 3             |
| 0 1 0 0      | 0 +                    | 4 + | 0 + | 0 | 4             |
| 0 1 0 1      | 0 +                    | 4 + | 0 + | 1 | 5             |
| 0 1 1 0      | 0 +                    | 4 + | 2 + | 0 | 6             |
| 0 1 1 1      | 0 +                    | 4 + | 2 + | 1 | 7             |
| 1 0 0 0      | 8 +                    | 0 + | 0 + | 0 | 8             |
| 1 0 0 1      | 8 +                    | 0 + | 0 + | 1 | 9             |
| 1 0 1 0      | 8 +                    | 0 + | 2 + | 0 | 10            |

# Images



- ▶ An image is composed of a matrix of pixels (picture elements), where each pixel is a small dot.
- ▶ The size of the pixel depends on the resolution.
- ▶ Image is represented using bit patterns
- ▶ 1-bit pattern – black: 0, white: 1

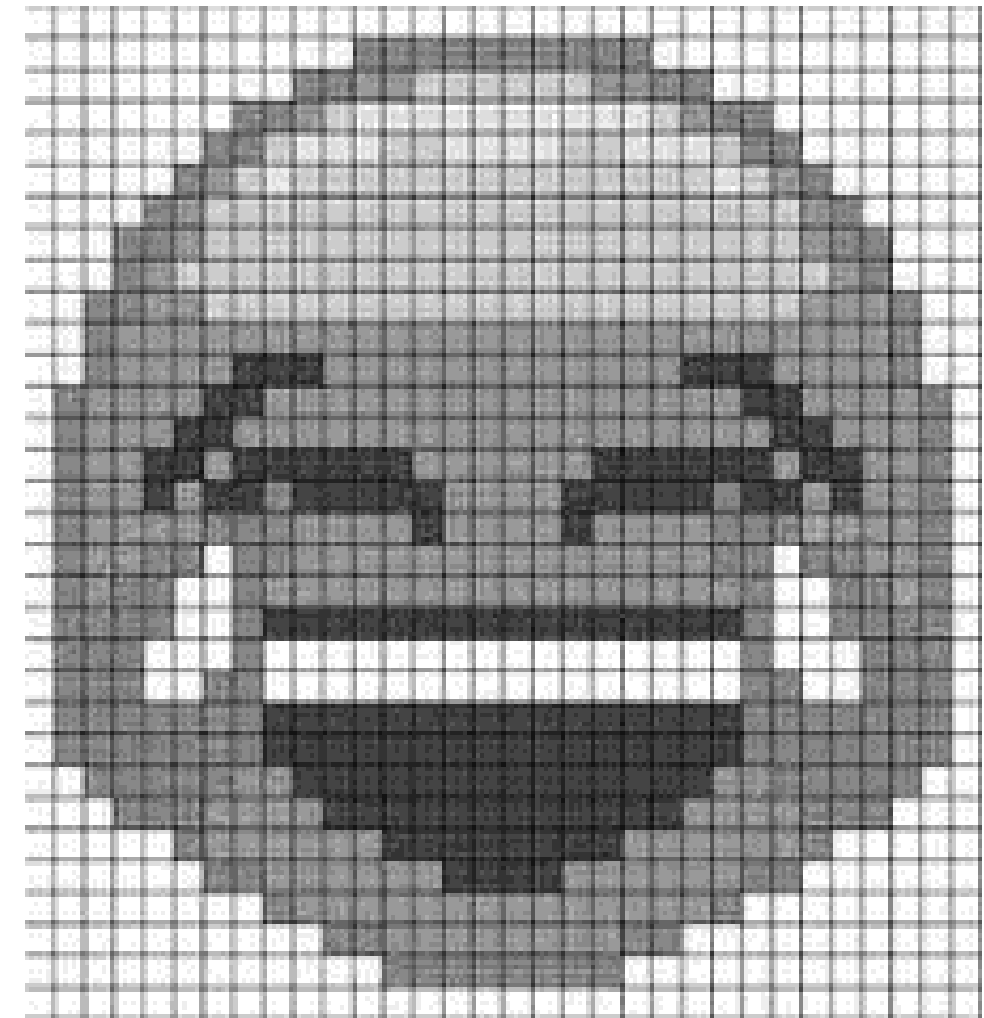


# Images

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- ▶ For grayscale, 2-bit pattern
- ▶ black: 00, dark gray: 01, light gray: 10, white pixel: 11.

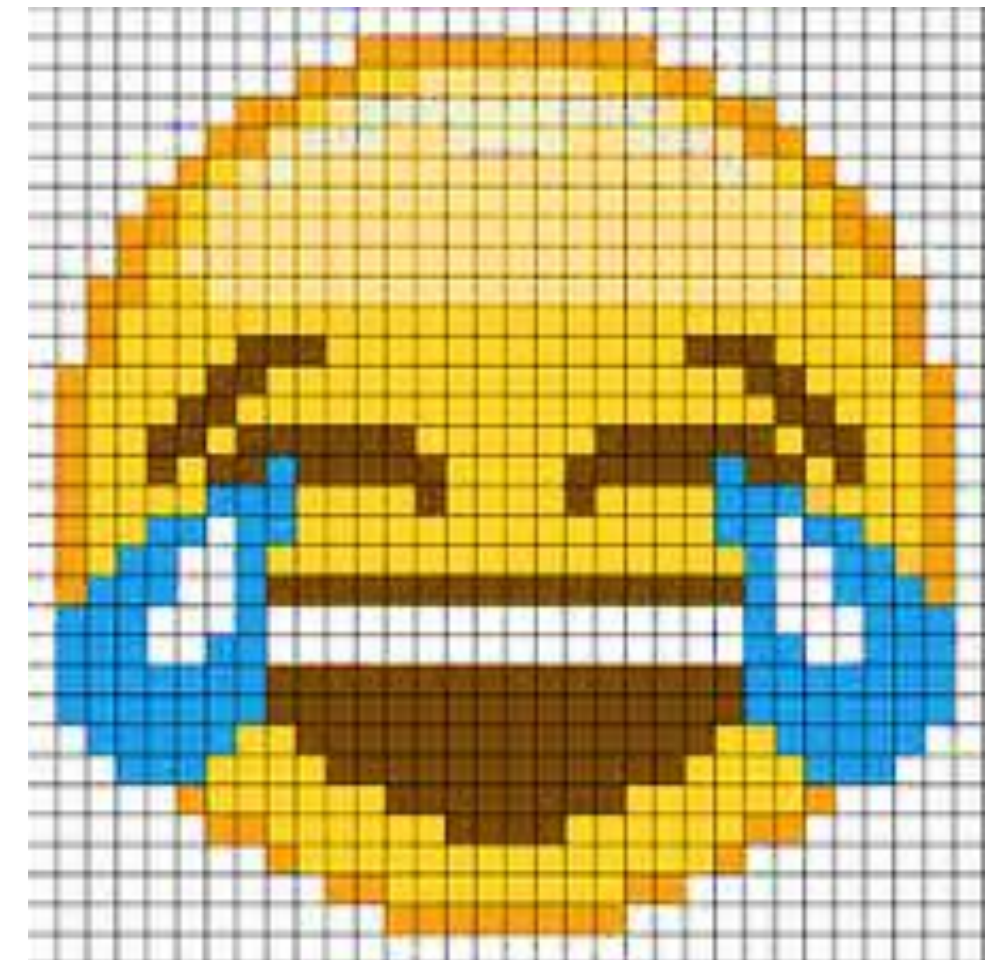


# Images

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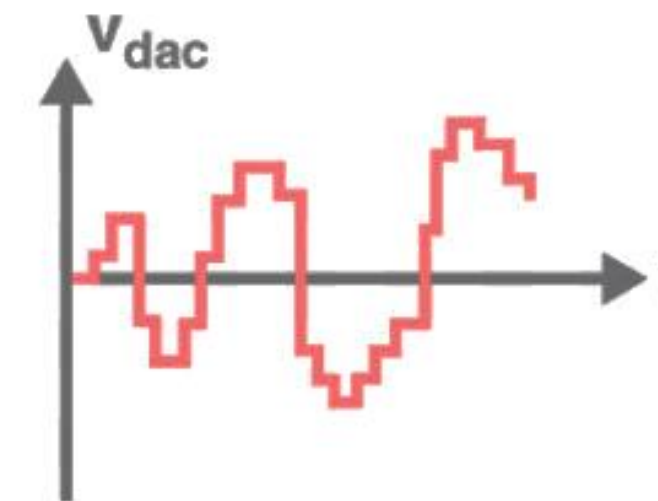
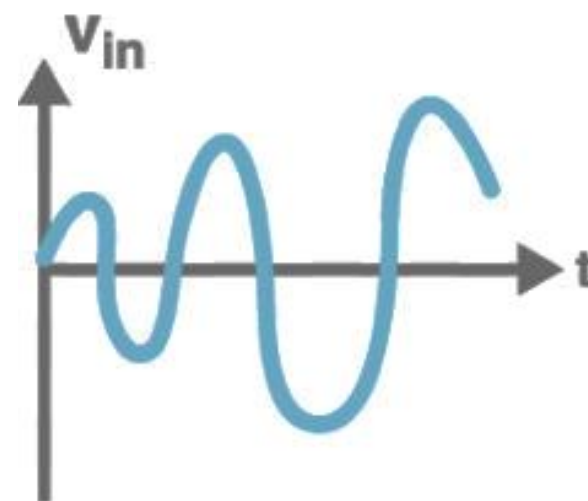
- ▶ Several methods to represent color images
- ▶ RGB (Red, Green, Blue)
  - Each color is made of a combination of three primary colors
  - A bit pattern is assigned based on intensity of each color
- ▶ YCM (Yellow, Cyan, Magenta)



# Audio



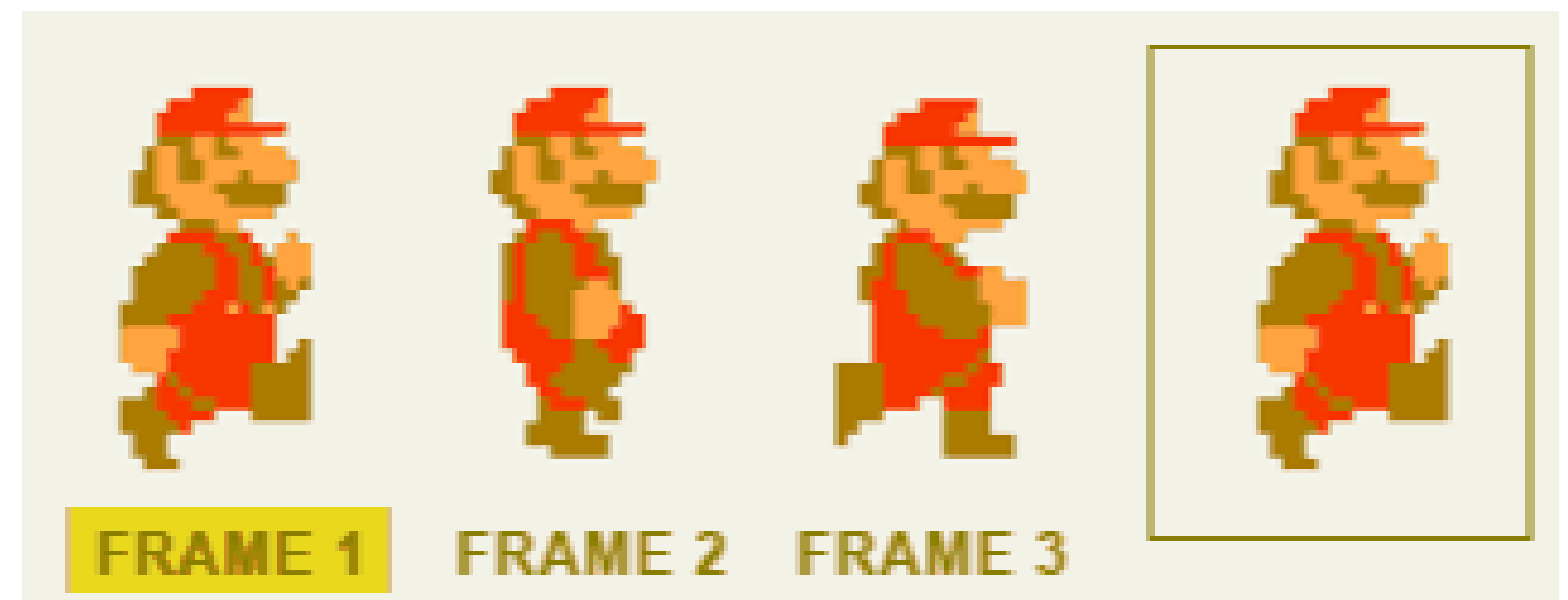
- Refers to the recording or broadcasting of sound or music.
  - It is continuous, not discrete.



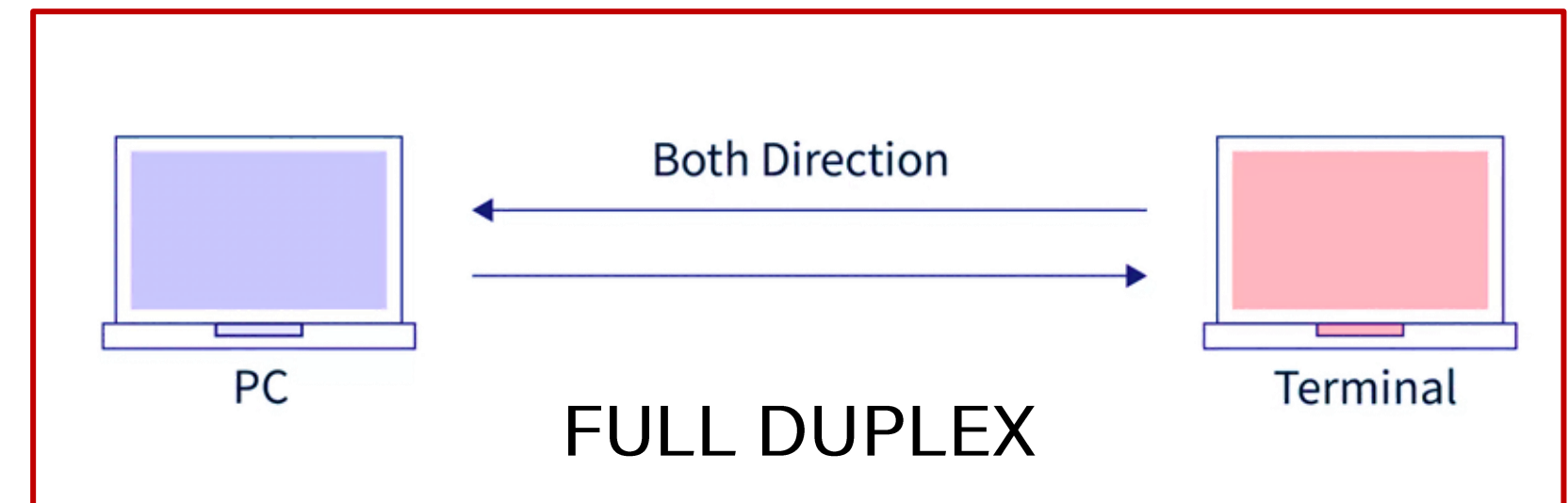
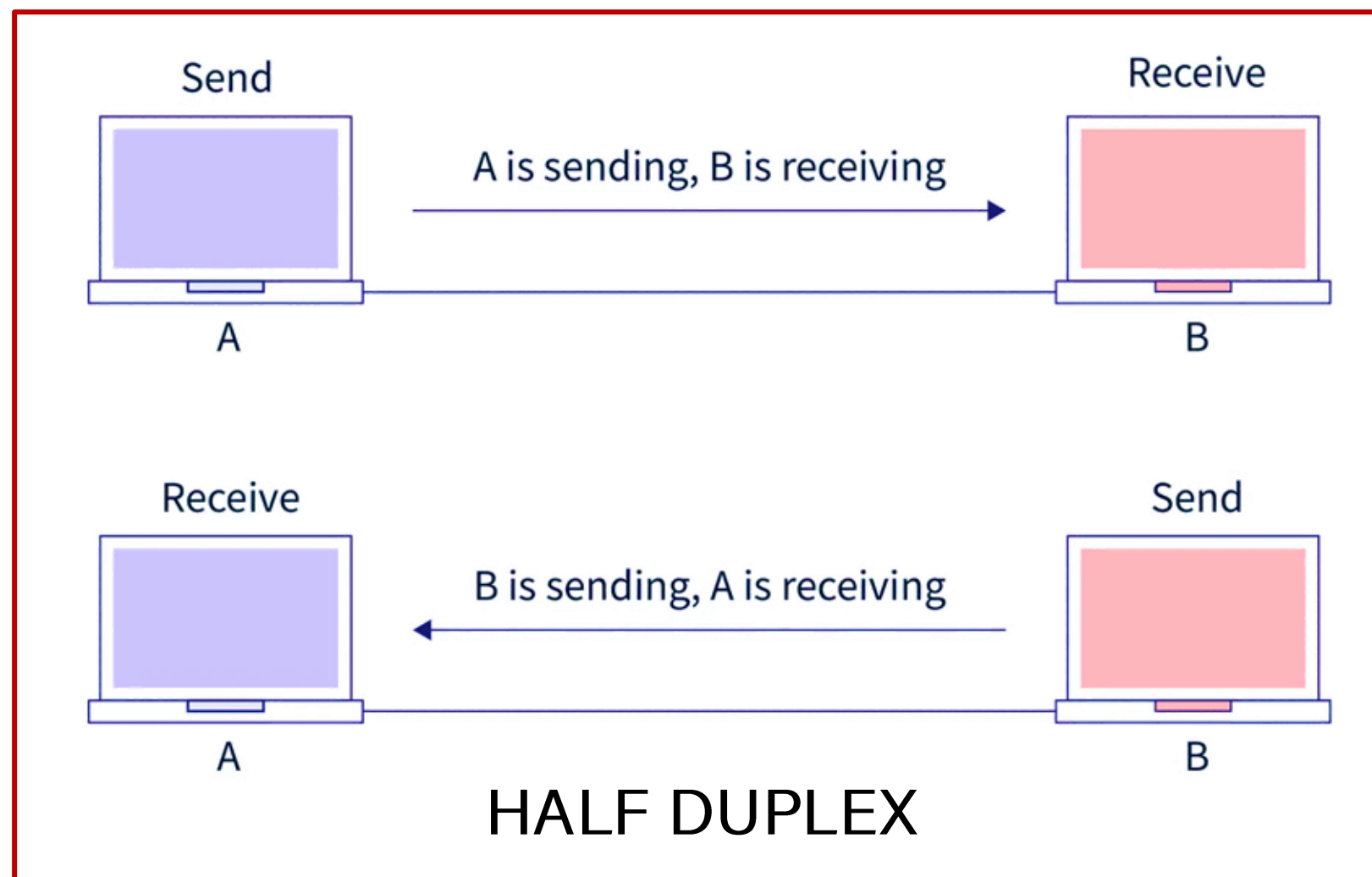
# Video



- ▶ Refers to the recording or broadcasting of a picture or movie.
- ▶ Video can either be produced as
  - a continuous entity (e.g., by a TV camera), or
  - a combination of images, each a discrete entity, arranged to convey the idea of motion



# Data Flow



# Identify the Data Flow!



HALF DUPLEX



FULL DUPLEX



SIMPLEX



# Imagine buying a car!



## PERFORMANCE

- ▶ Speed, Mileage

## RELIABILITY

- ▶ Long run, No big repairs

## SECURITY

- ▶ Passenger safety, Secure seats, Airbags



# Network Criteria

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- ▶ A network must be able to meet a certain number of criteria.



# Performance

---

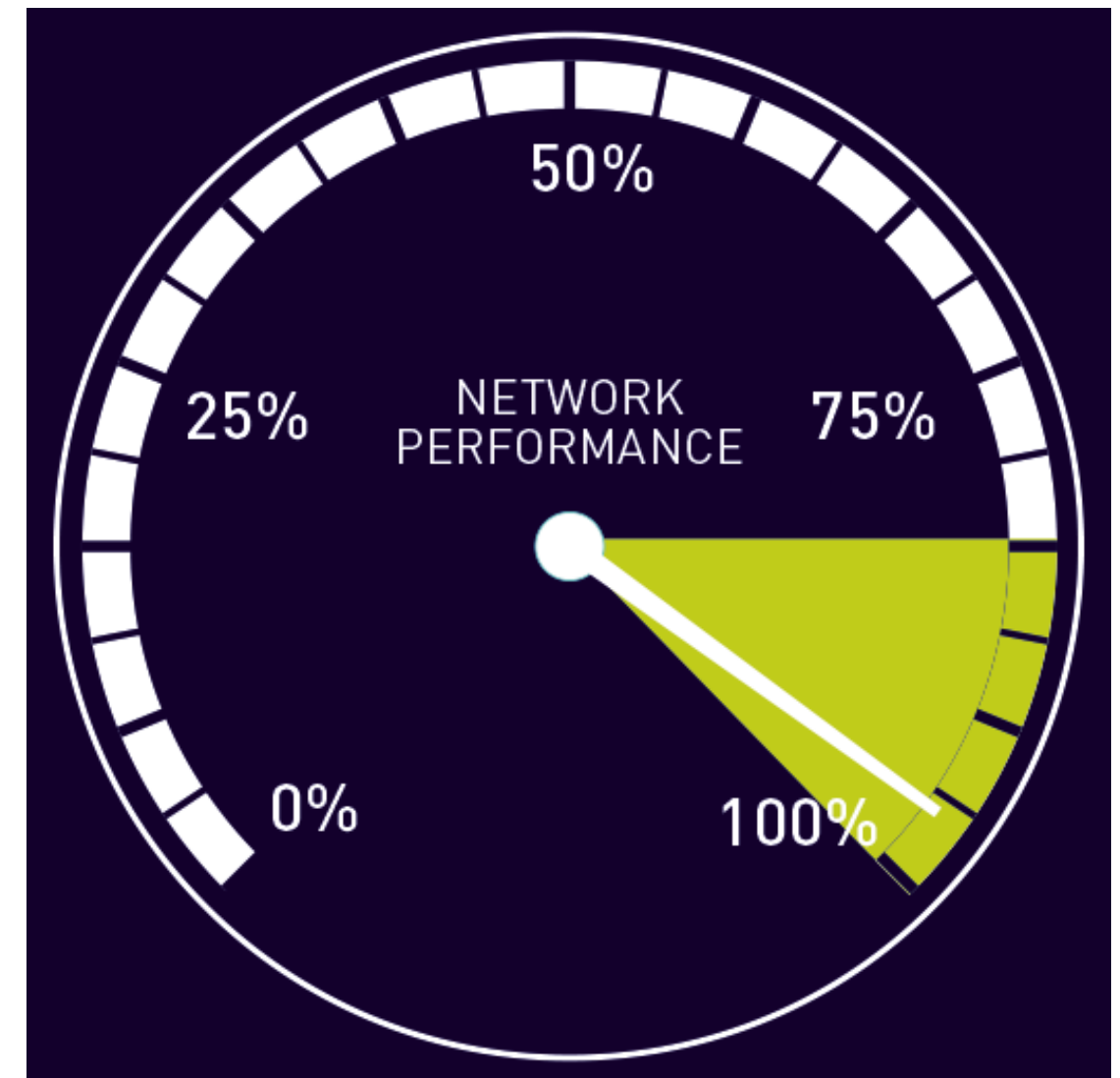


- ▶ Performance can be measured in many ways, including:
- ▶ Response time
  - Elapsed time between an inquiry and a response.
- ▶ Transit time
  - Amount of time required for a message to travel from one device to another.

# Performance



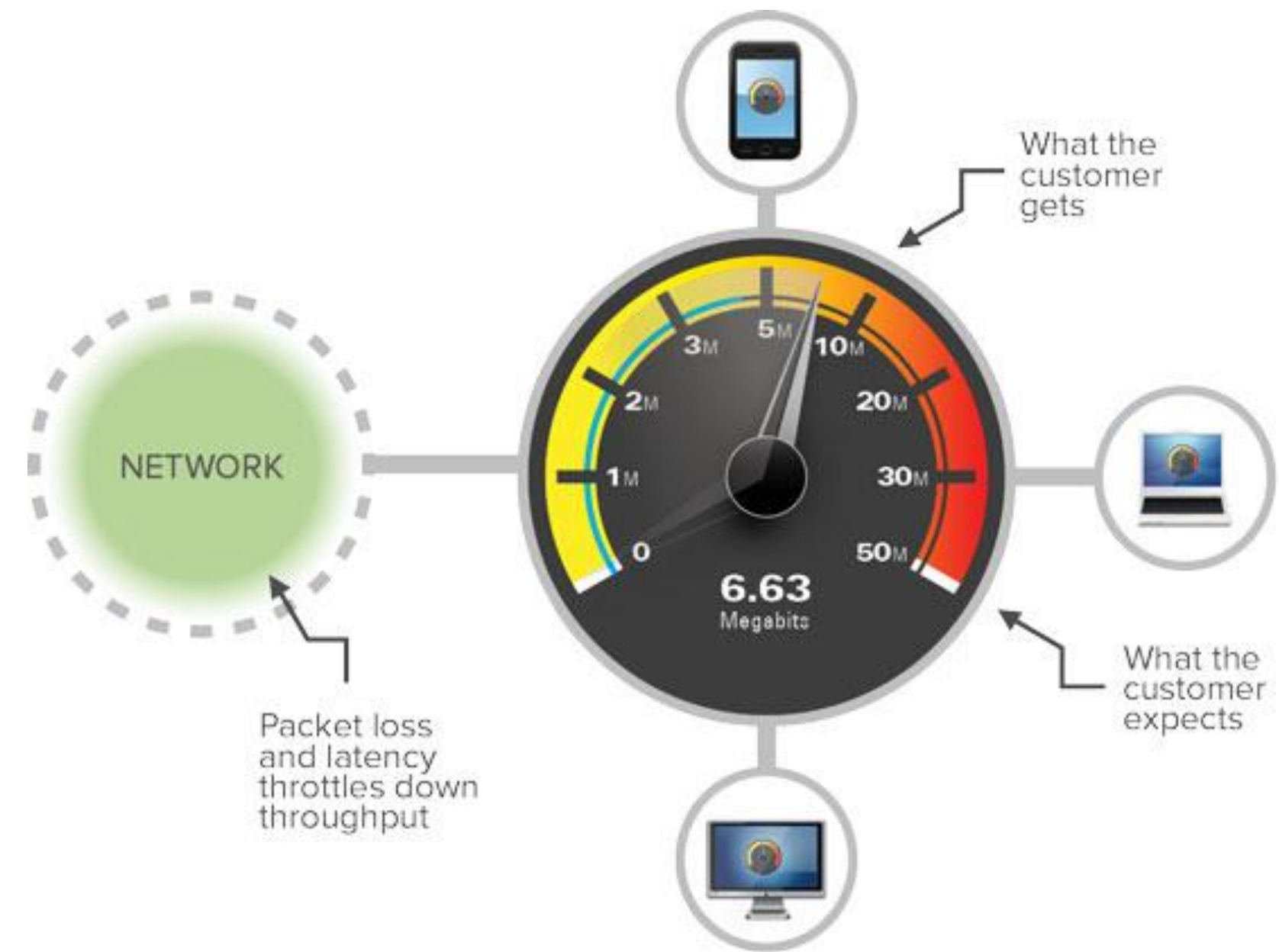
- ▶ The performance of a network depends on a number of factors:
  - number of users
  - type of transmission medium
  - capabilities of the connected hardware
  - efficiency of the software



# Performance



- Evaluated by 2 networking metrics: throughput (capacity) and delay (latency).
- Desired - more throughput and less delay (but contradictory?!)



# Reliability

---



- ▶ Accuracy of delivery
- ▶ Measured by
  - The frequency of failure
  - The time it takes a link to recover from a failure
  - The network's robustness in a catastrophe

# Security

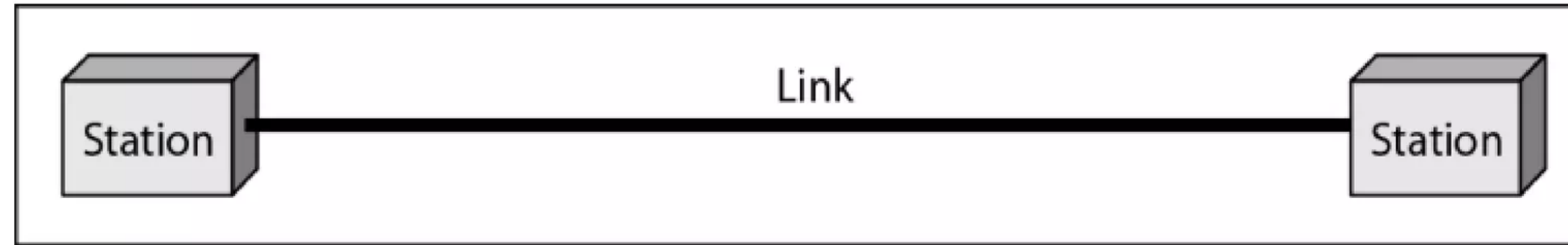
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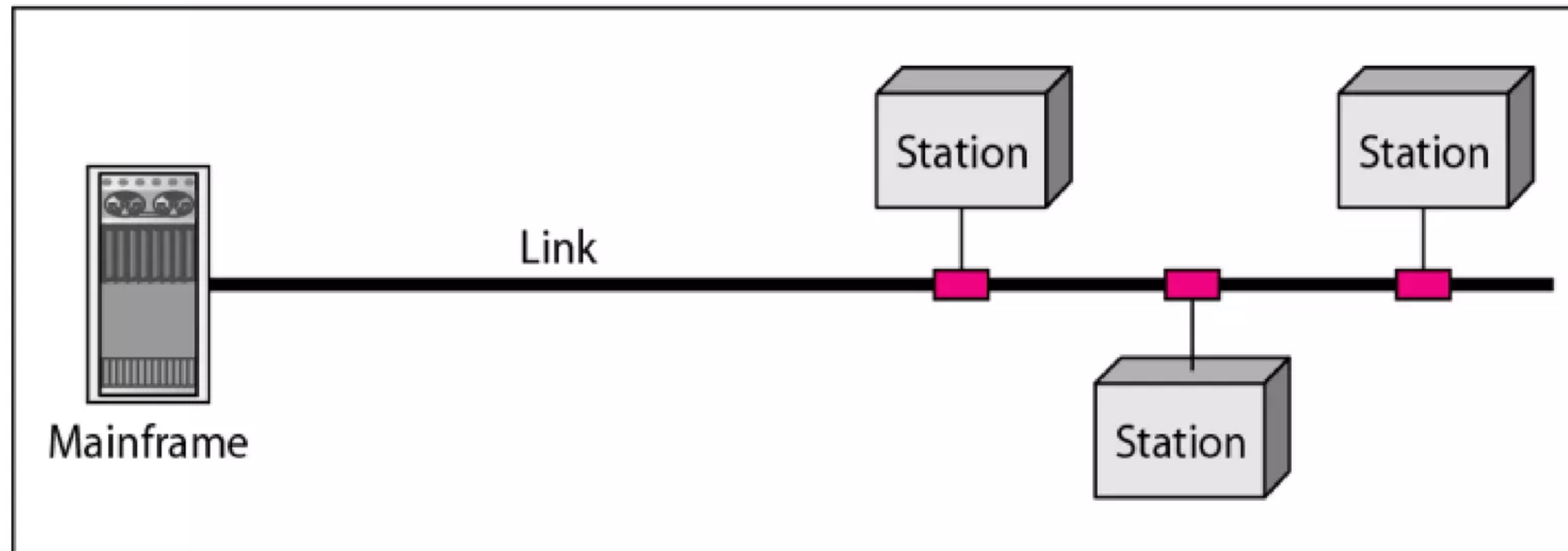
- ▶ Protecting data from unauthorized access
- ▶ Protecting data from damage and development
- ▶ Implementing policies and procedures for recovery from breaches and data losses



# Physical Structures - Type of Connection



a. Point-to-point



b. Multipoint

- Spatially shared
- Time shared

# Physical Topology

---



- ▶ Refers to the way in which a network is laid out physically.
- ▶ Two or more devices connect to a link; two or more links form a topology.
- ▶ Topology is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

# Physical Topology

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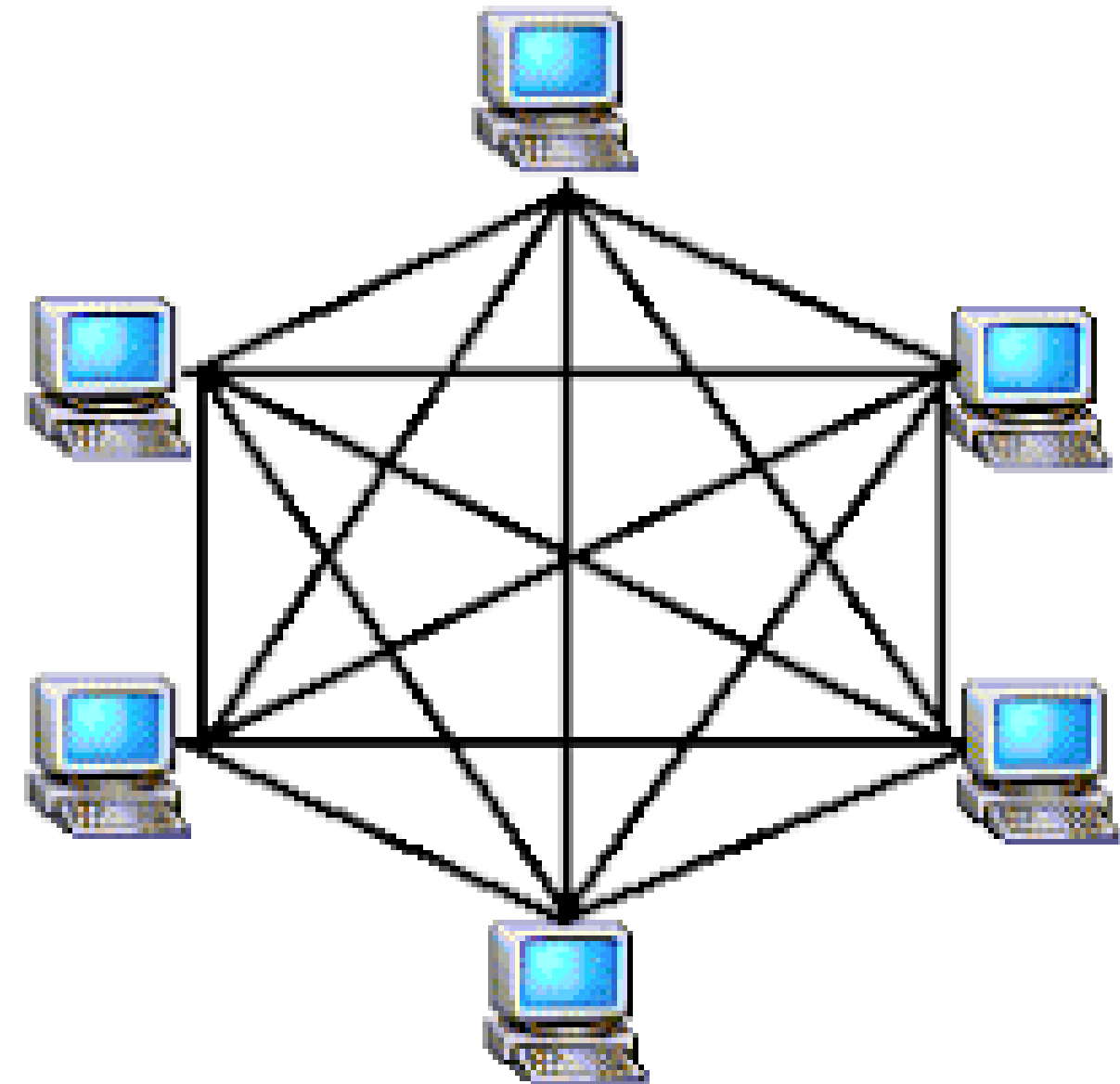


- ▶ There are four basic topologies possible:
- ▶ Mesh
- ▶ Star
- ▶ Bus
- ▶ Ring

# Mesh Topology



- ▶ Every device has a dedicated point-to-point link to every other device
- ▶ Each node must be connected to every other node. To connect 'n' nodes,  $n(n-1)$  links are required
- ▶ For duplex mode,  $n(n-1)/2$  links are required



# Mesh Topology - Advantages

---



- ▶ Eliminates traffic problems
- ▶ If one link becomes unusable, it does not disable the entire system - robustness
- ▶ When every message travels along a dedicated line, only the intended recipient sees it
- ▶ Physical boundaries prevent other users from gaining access to messages
- ▶ Fault identification and fault isolation is easy

# Mesh Topology - Disadvantages

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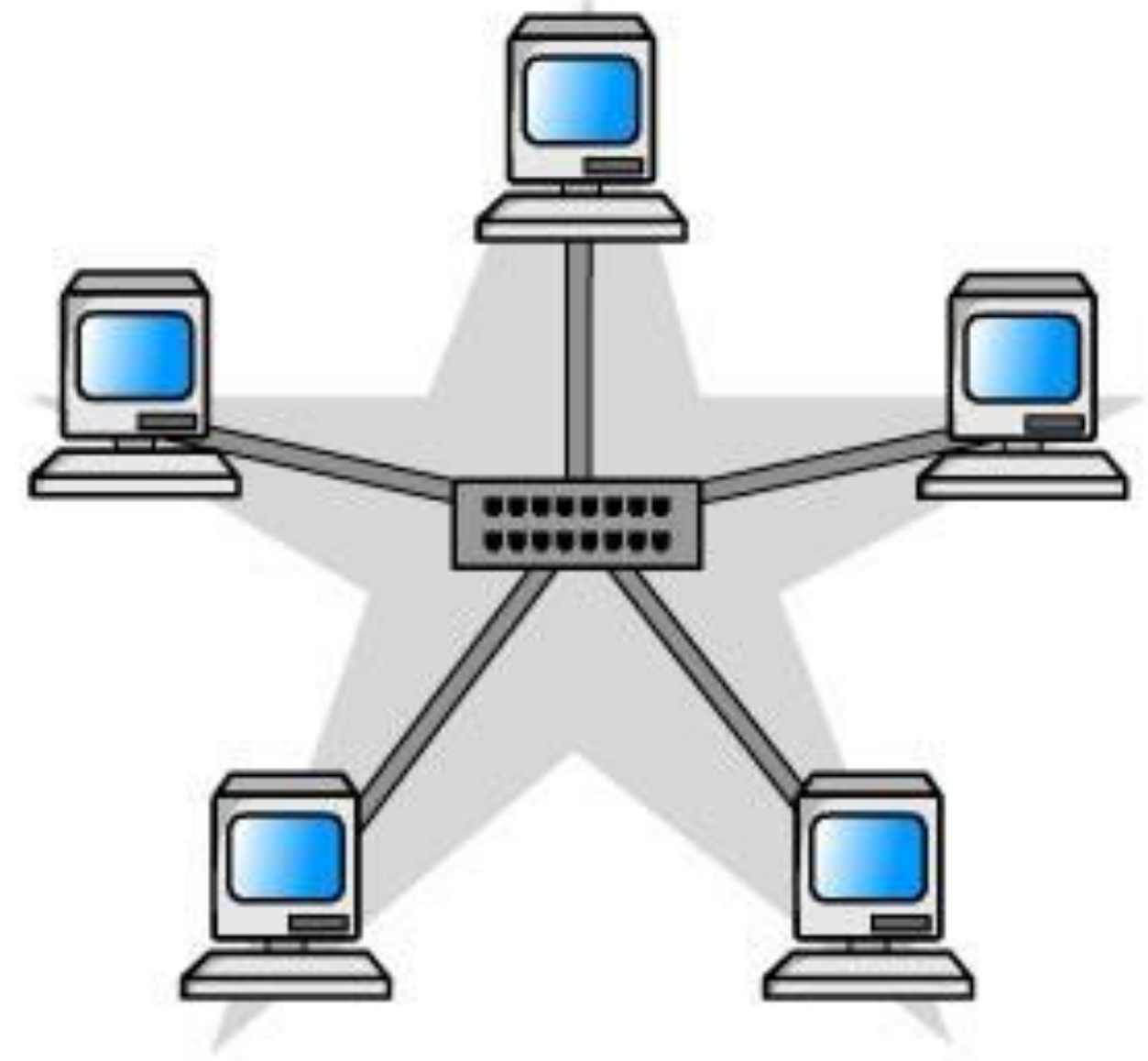


- ▶ Amount of cabling and the number of I/O ports required
- ▶ Installation and reconnection are difficult
- ▶ Sheer bulk of the wiring can be greater than the available space
- ▶ Hardware required to connect each link can be expensive
- ▶ So, mesh is used limitedly - as a backbone connecting the main computers of a hybrid network that can include several other topologies.

# Star Topology



- ▶ Each device has a dedicated point-to-point link only to a central controller, usually called a hub
- ▶ The devices are not directly linked to one another
- ▶ The controller acts as an exchange
- ▶ High-speed LANs often use a star topology with a central hub



# Star Topology - Advantages

---



- ▶ Less expensive than a mesh topology
- ▶ Easy to install and reconfigure - less cabling is needed
- ▶ If one link fails, only that link is affected and all other links remain active.
  - leads to easy fault identification and fault isolation
- ▶ The hub can be used to monitor link problems and bypass defective links

# Star Topology - Disadvantages

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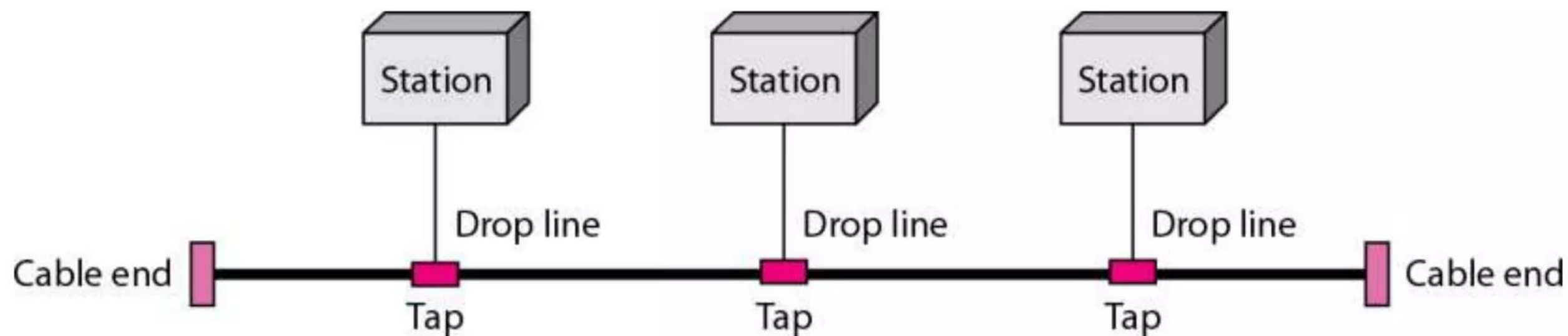


- ▶ The dependency of the whole topology on one single point, the hub
- ▶ More cabling is required in a star than in some other topologies (such as ring or bus)
  - Each node must be linked to a central hub

# Bus Topology



- ▶ A multipoint topology with one long cable that acts as a backbone to link all the devices in a network
- ▶ Nodes are connected to the bus cable by
  - Drop line - connection between the device and the main cable
  - Tap - connector that create a contact with the metallic core



# Bus Topology - Advantages

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- ▶ Ease of installation
- ▶ A bus uses less cabling than mesh or star topologies

# Bus Topology - Disadvantages

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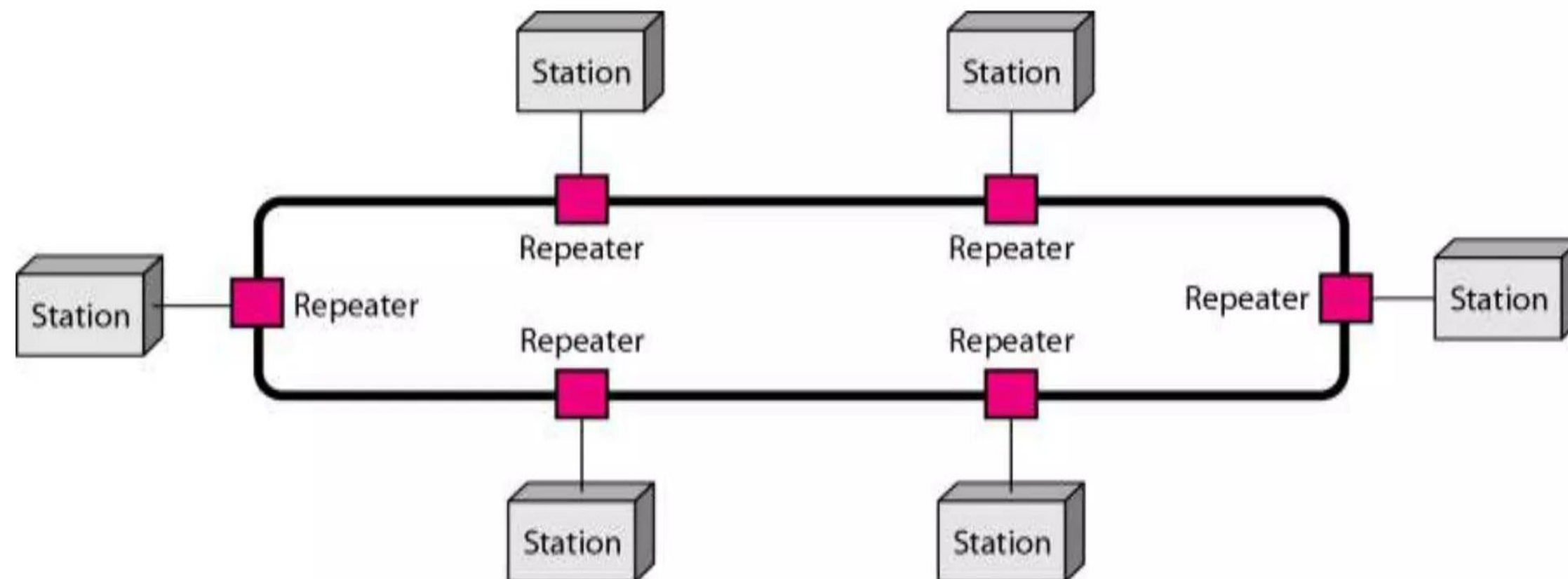


- ▶ Difficult reconnection and fault isolation
- ▶ Can be difficult to add new devices
- ▶ Signal reflection at the taps can cause degradation in quality
  - Can be controlled by limiting the number and spacing of devices connected to a given length of cable
- ▶ A fault or break in the bus cable stops all transmission

# Ring Topology



- ▶ Each device has a dedicated point-to-point connection with only the two devices on either side of it.
- ▶ Each device in the ring incorporates a repeater
  - Repeater regenerates the bits and passes them along



# Ring Topology - Advantages

---



- ▶ Relatively easy to install and reconfigure.
  - To add or delete a device requires changing only two connections.
- ▶ Simple fault isolation
  - If one device does not receive a signal within a specified period, it can issue an alarm.

# Ring Topology - Disadvantages

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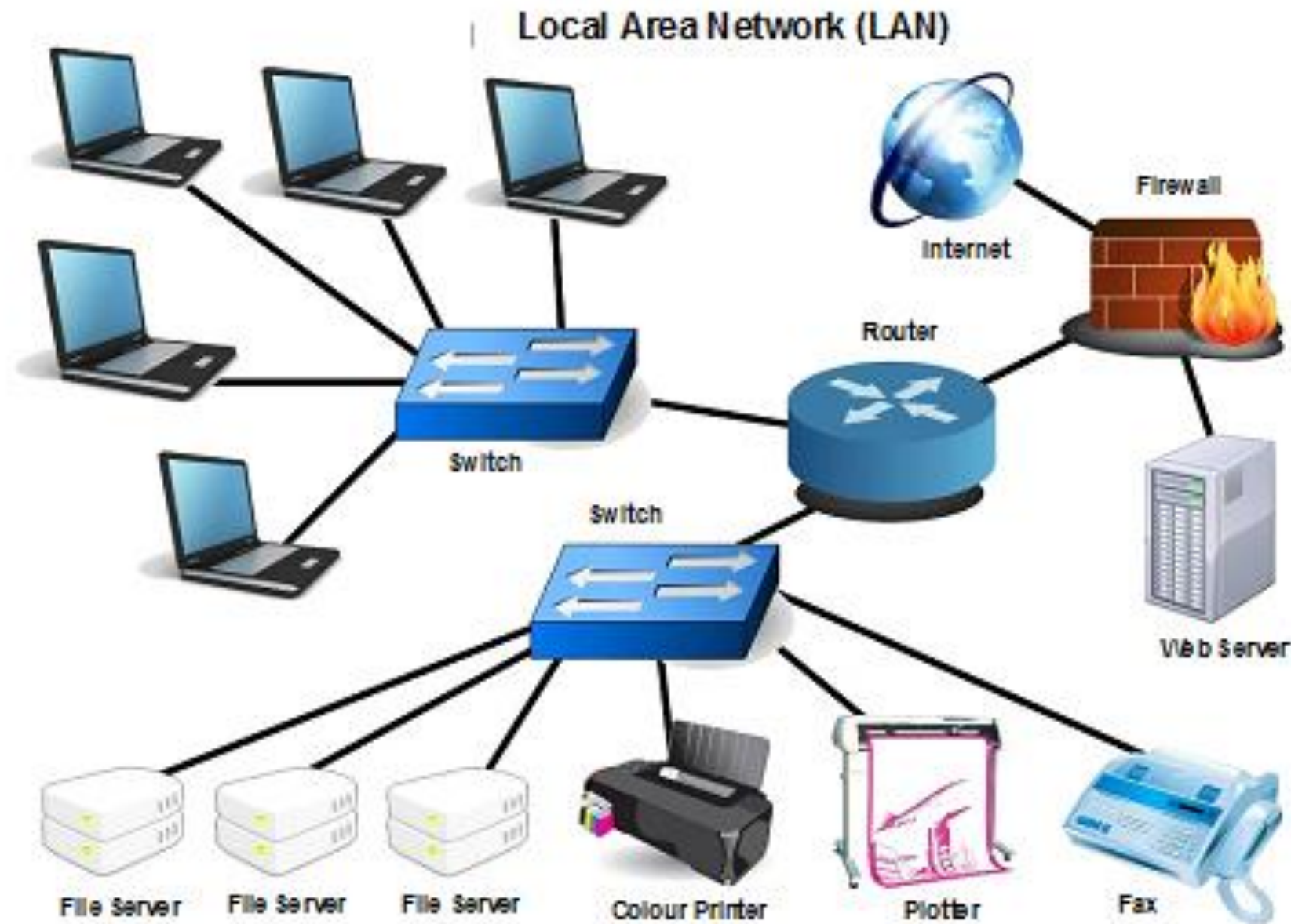


- ▶ Unidirectional traffic
- ▶ In a simple ring, a break in the ring (such as a disabled station) can disable the entire network.
  - This can be solved by using a dual ring or a switch capable of closing off the break.

# Local Area Network (LAN)



- Usually privately owned and connects some hosts in a single office, building, or campus.



# Local Area Network (LAN)

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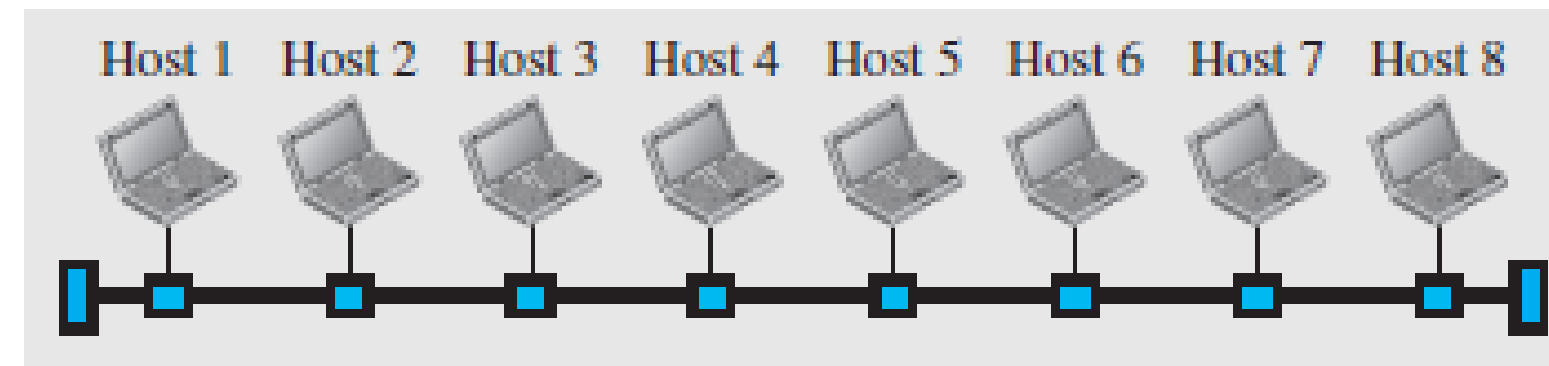


- ▶ Each host in a LAN has
  - an identifier
  - an address, that uniquely defines the host
- ▶ A packet sent by a host to another host carries both the source host's and the destination host's addresses.

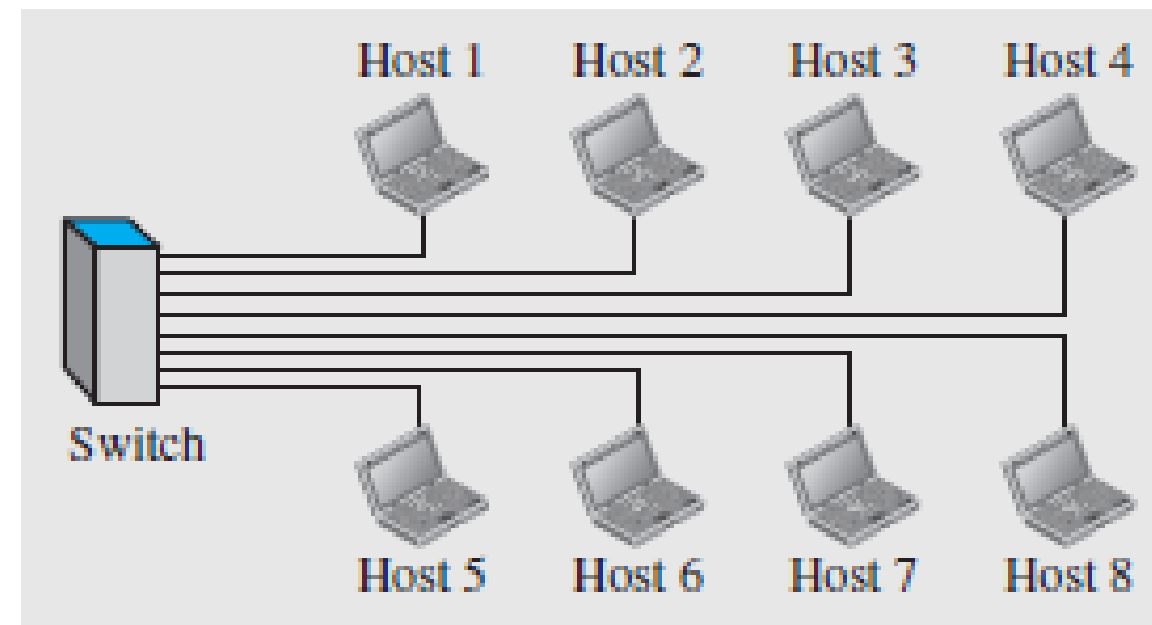
# Local Area Network (LAN)



## ► Past



## ► Today



# Wide Area Network (WAN)

---



- ▶ WAN has a wider geographical span, spanning a town, a state, a country, or even the world.
- ▶ A LAN interconnects hosts; a WAN interconnects connecting devices such as switches, routers, or modems.



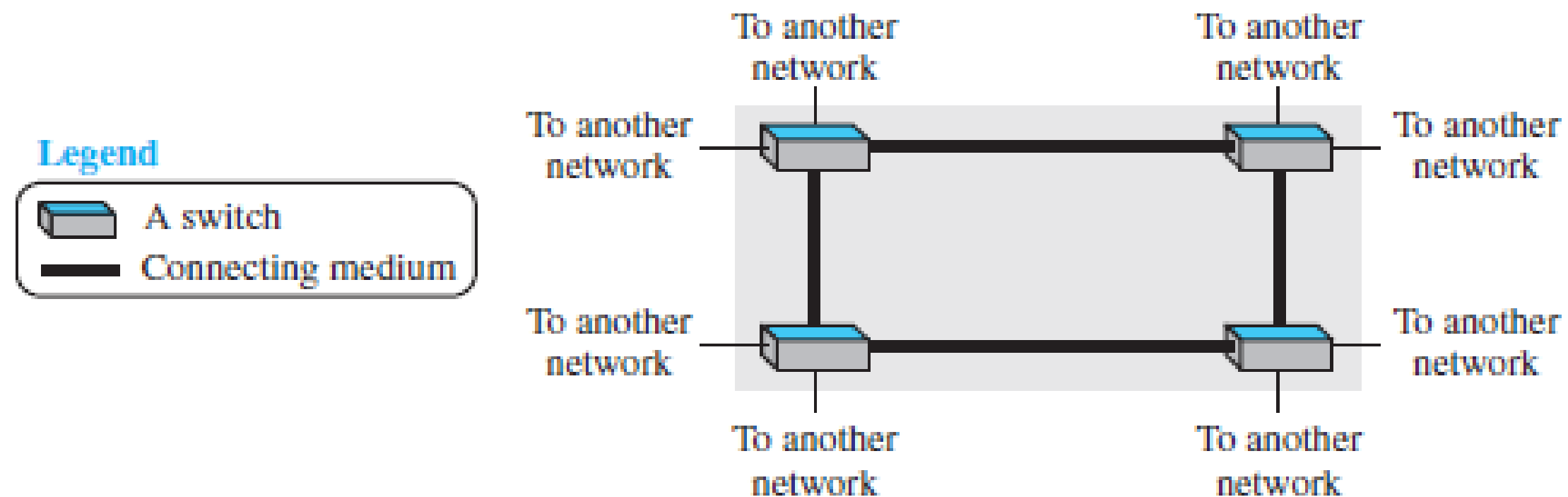
# Wide Area Network (WAN)



## ► Point-to-point WANs



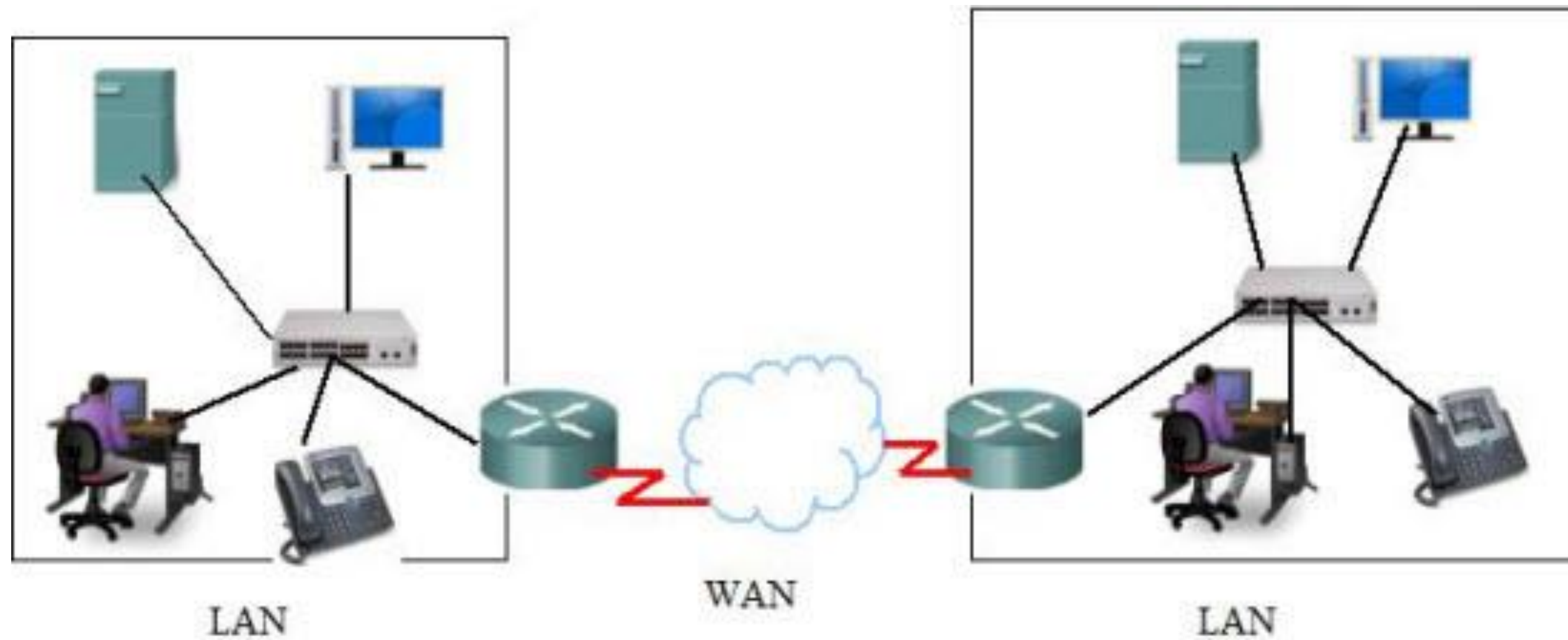
## • Switched WANs



# Internetwork



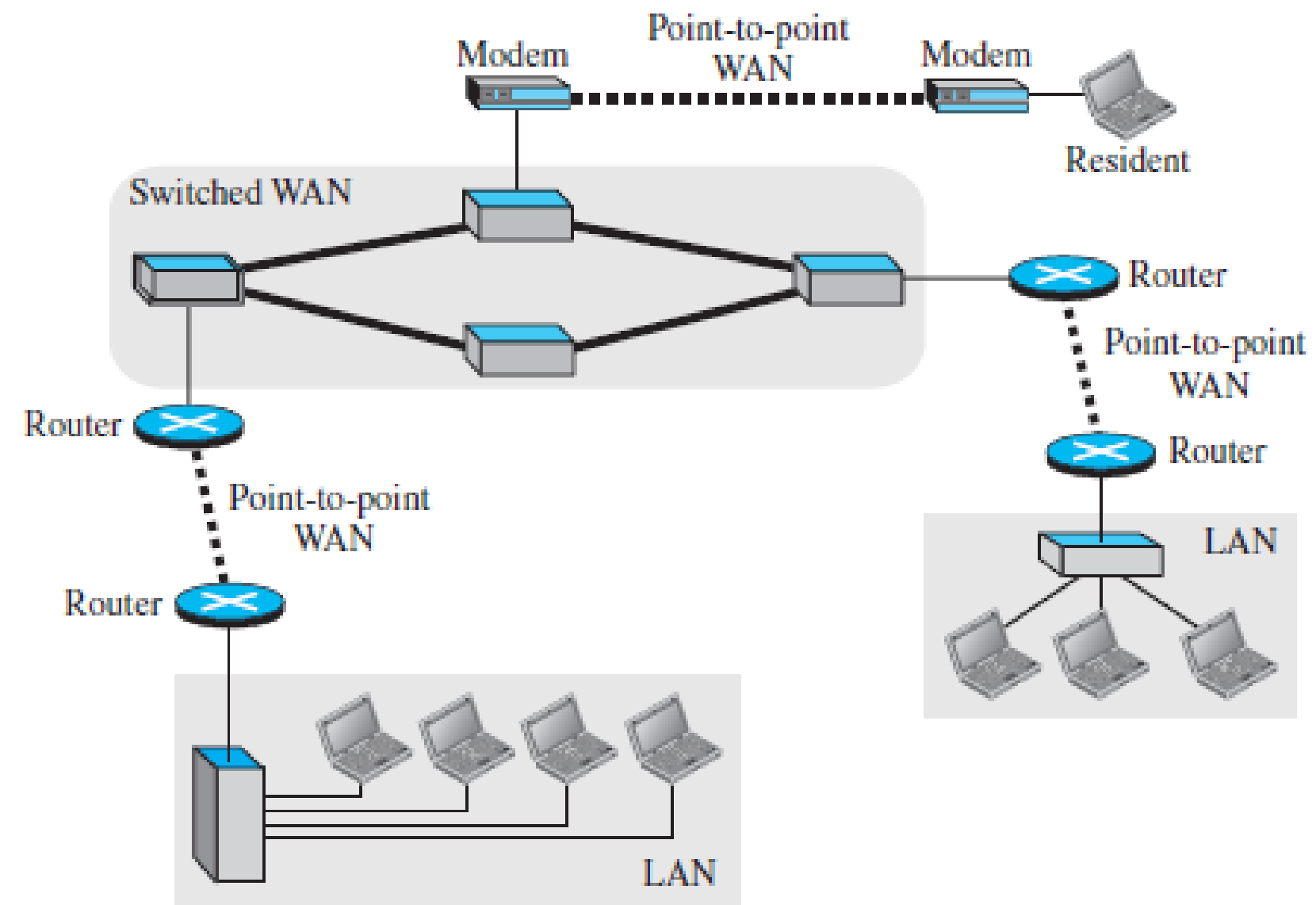
- ▶ When two or more networks are connected, they make an internetwork, or internet



# Switching



- ▶ An internet is a switched network in which a switch connects at least two links together.
- ▶ A switch needs to forward data from a network to another network when required



# Switching

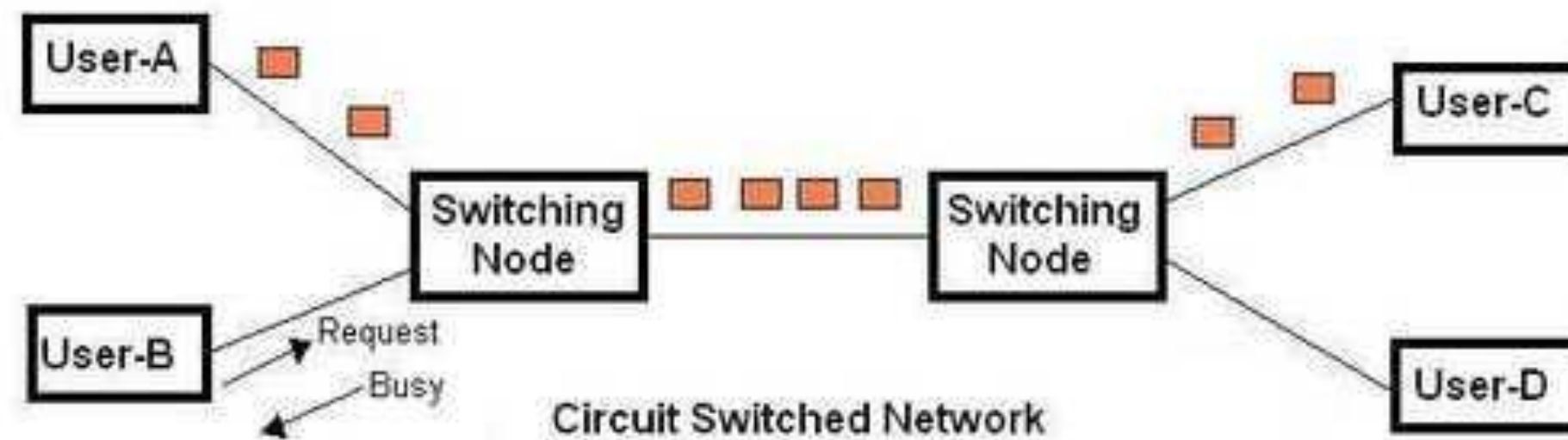
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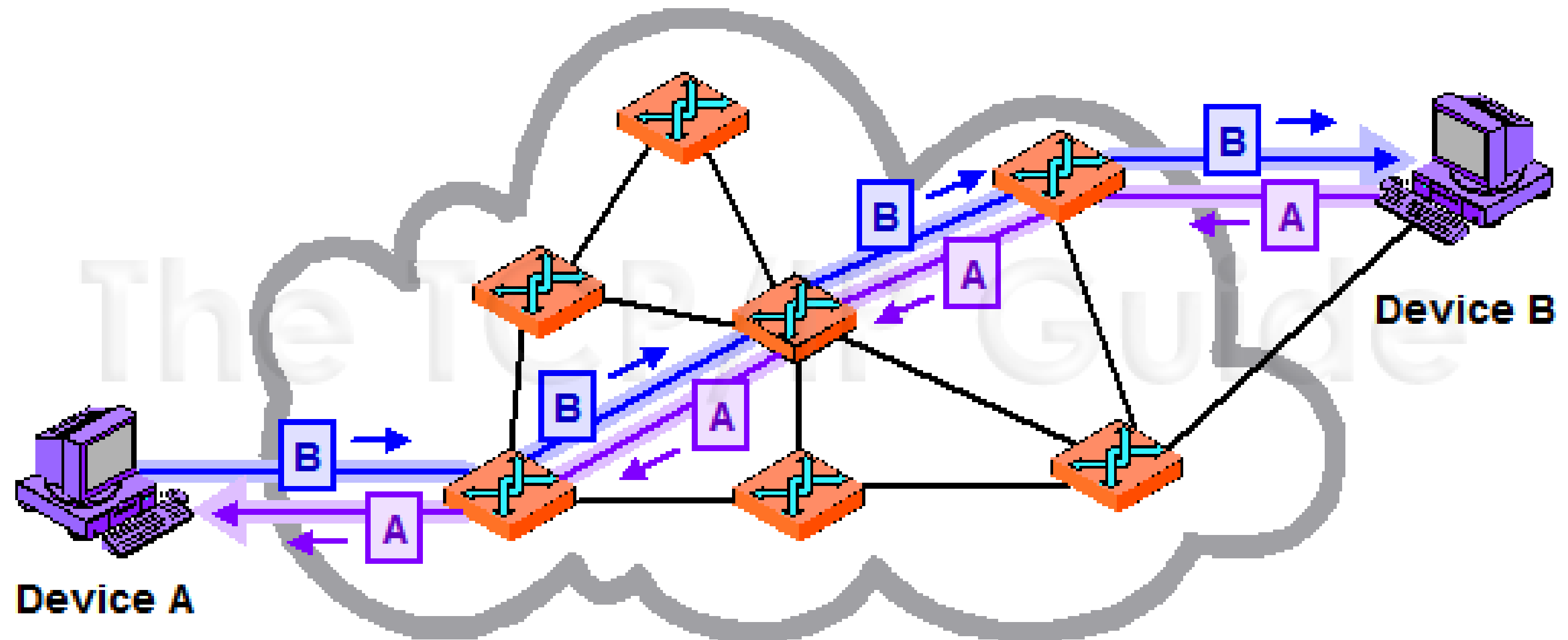
# Circuit Switched Network



- ▶ A dedicated connection, called a circuit, is always available between the two end systems; the switch can only make it active or inactive.



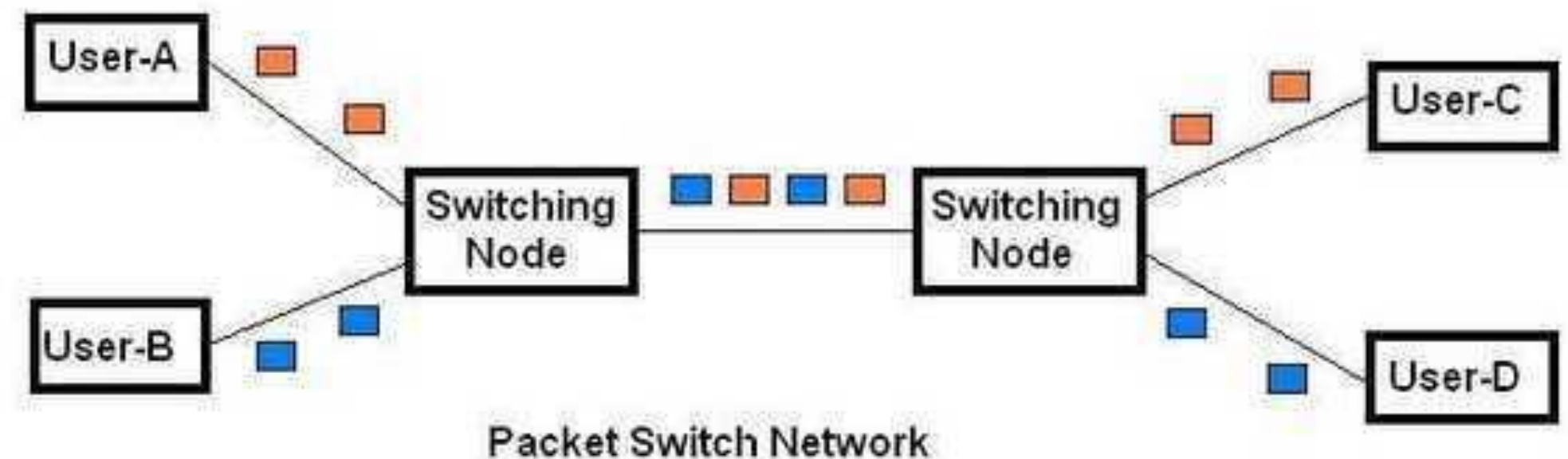
# Circuit Switched Network



# Packet Switched Network

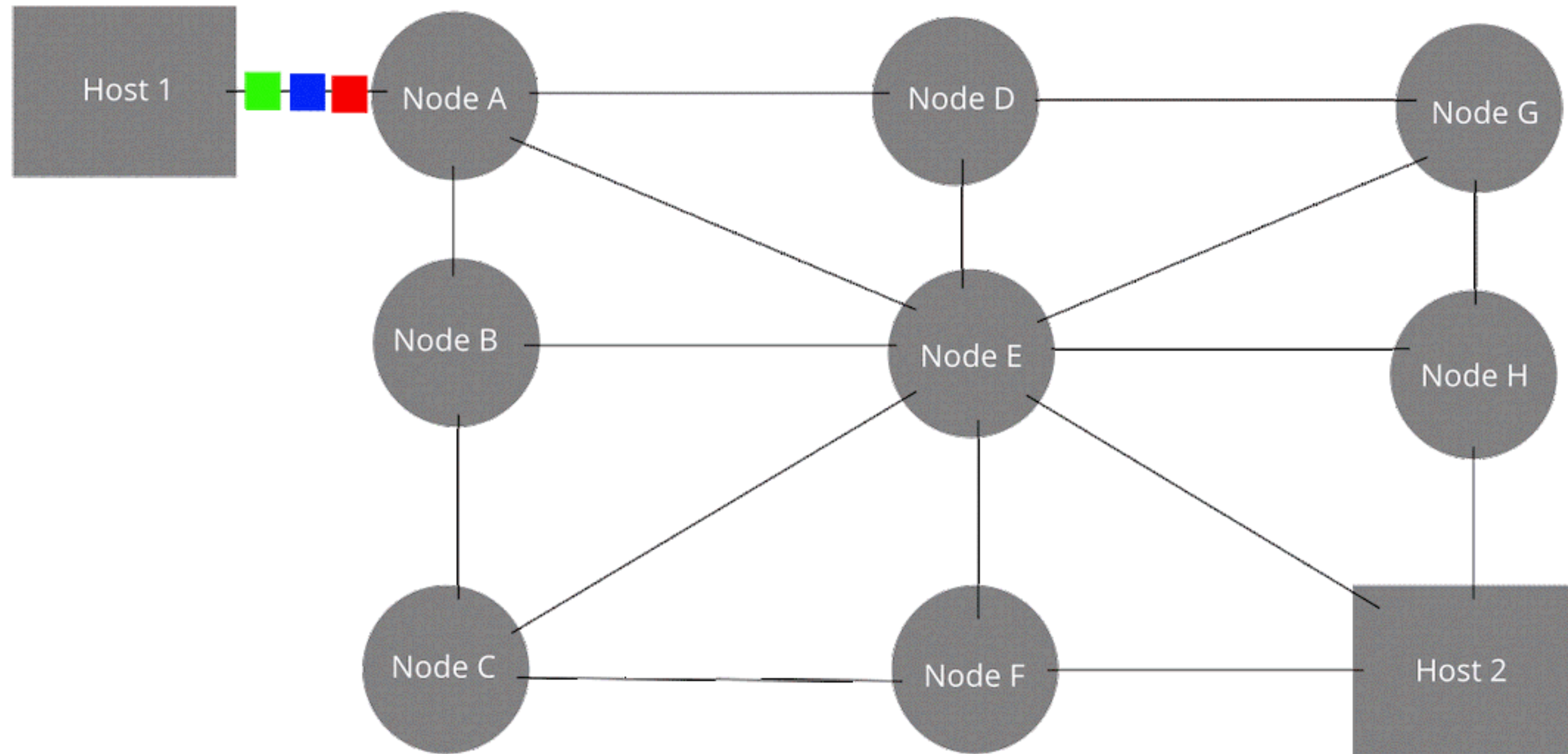


- ▶ The communication between the two ends is done in blocks of data called packets.
- ▶ Exchange of individual data packets between the two computers and not continuous communication.
- ▶ A router in a packet-switched network has a queue that can store and forward the packet.

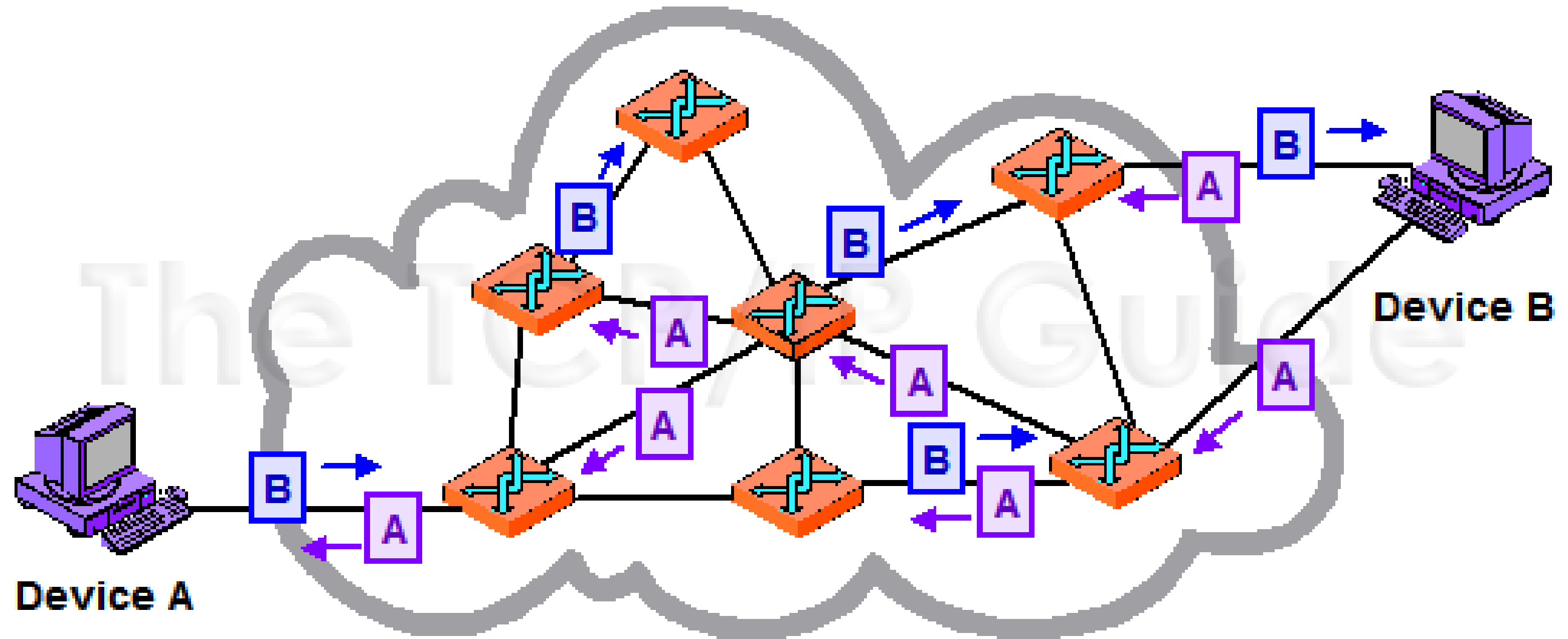




The original message is **Green**, **Blue**, **Red**.



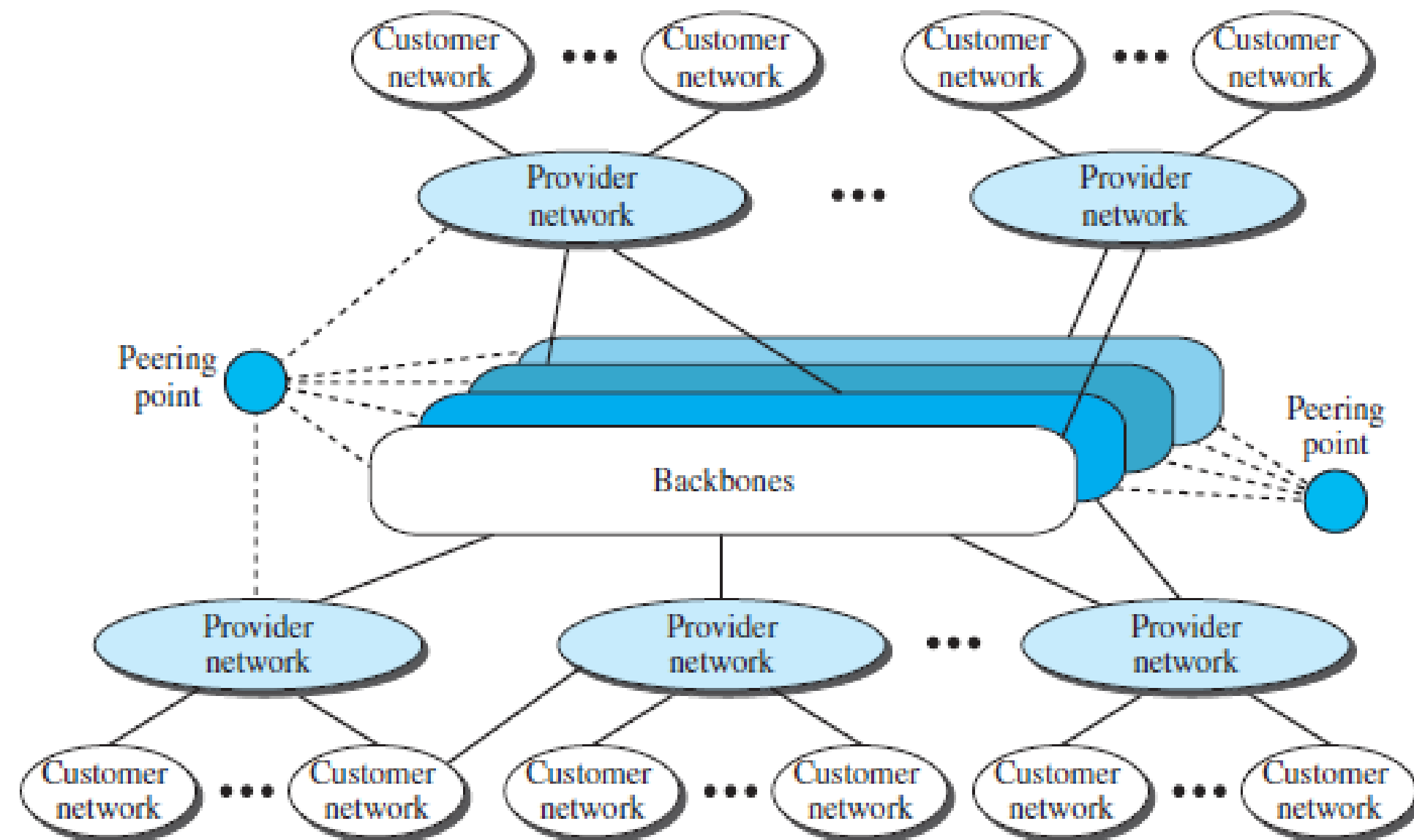
# Packet Switched Network



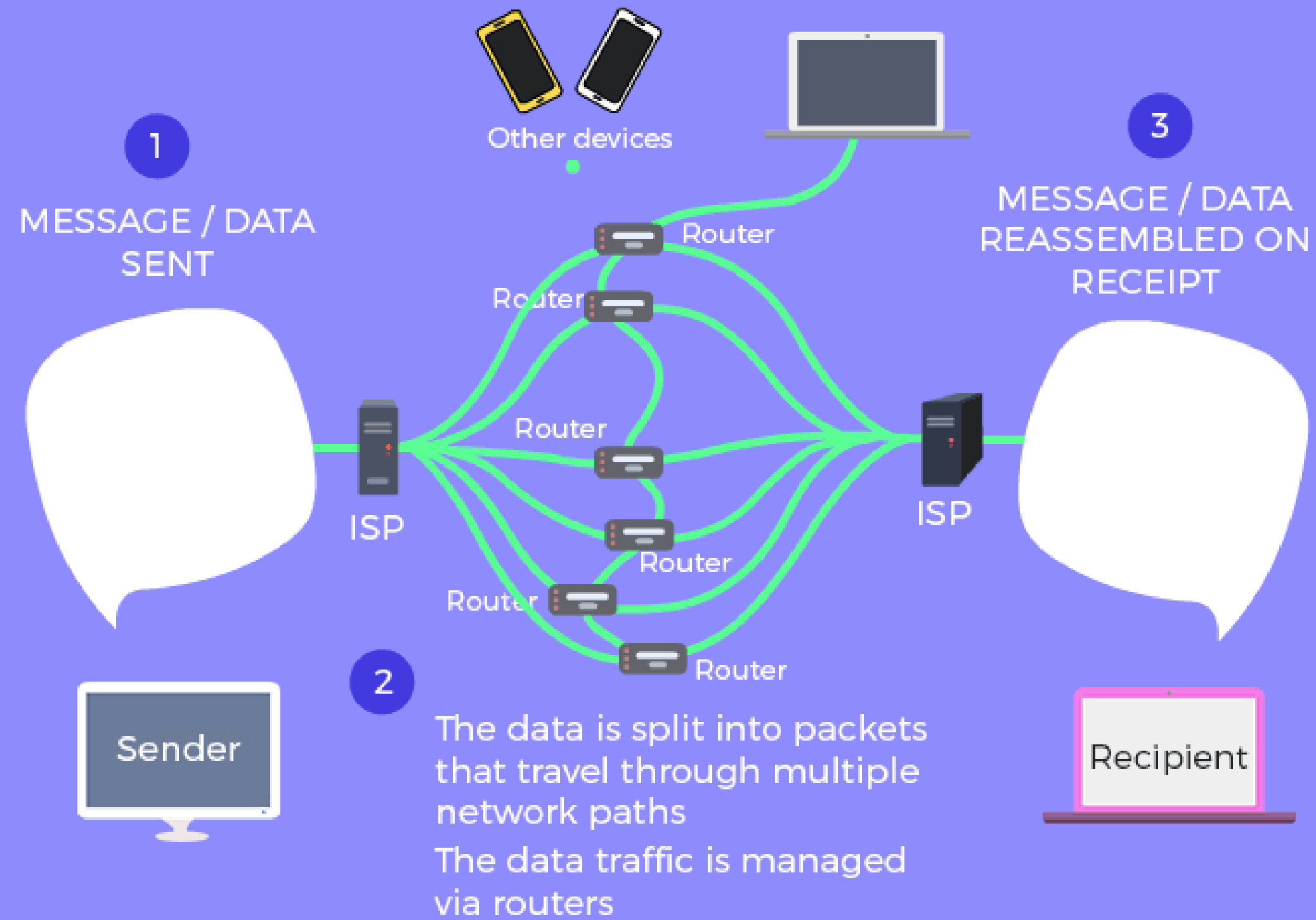
# The Internet



- The most notable internet is called the Internet (uppercase I ), and is composed of thousands of interconnected networks.



# HOW THE INTERNET WORKS



# Accessing the Internet

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- ▶ Using Telephone Lines
  - Dial-up Service
  - DSL (Digital Subscriber Line) Service
- ▶ Using Cable Networks
- ▶ Using Wireless Networks
- ▶ Direct Connection to the Internet

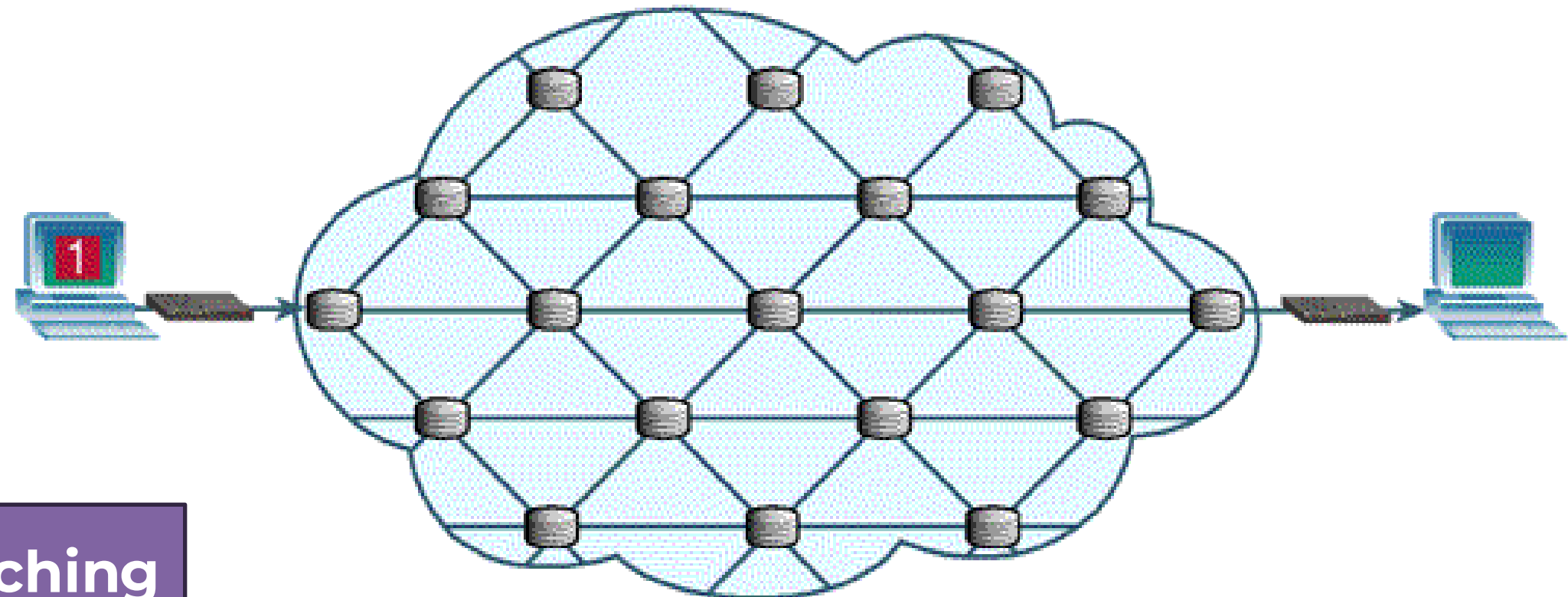


## Circuit Switching



Credit: Seattle Municipal Archives

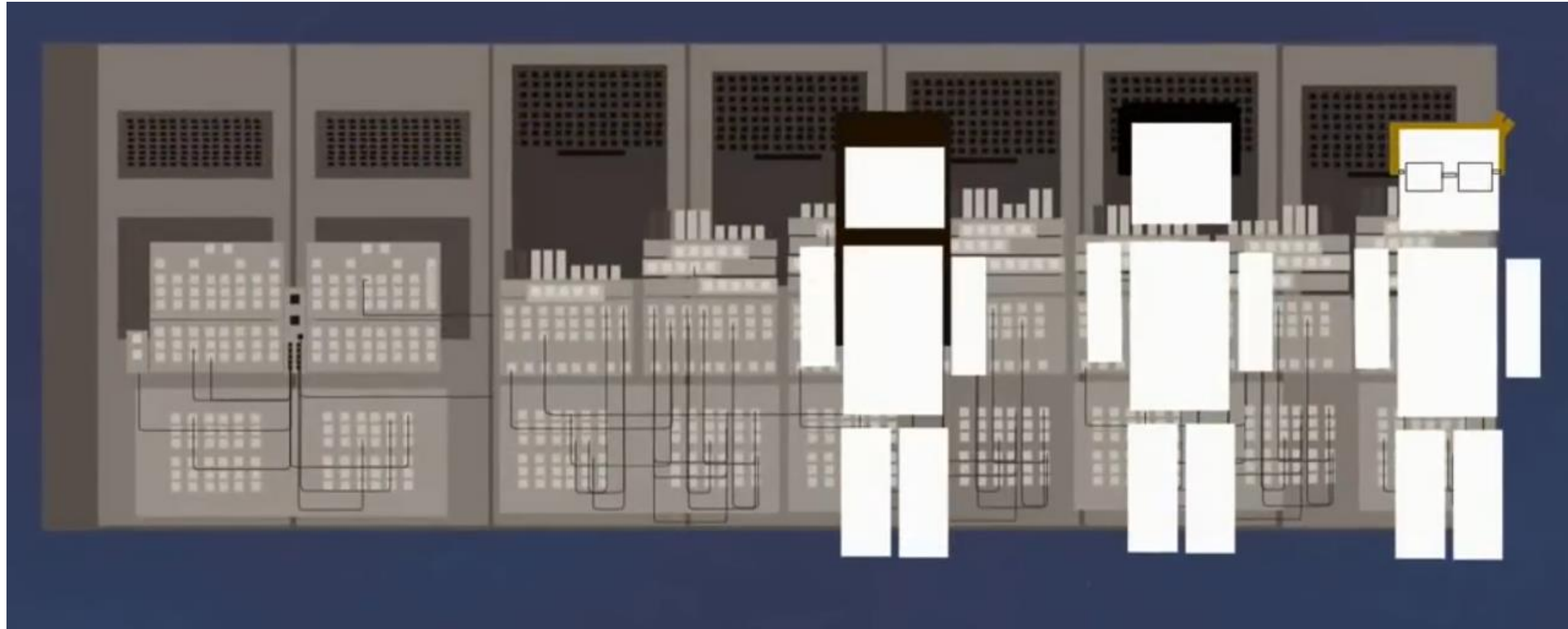
## Packet routing through WAN/Internet



## Packet Switching

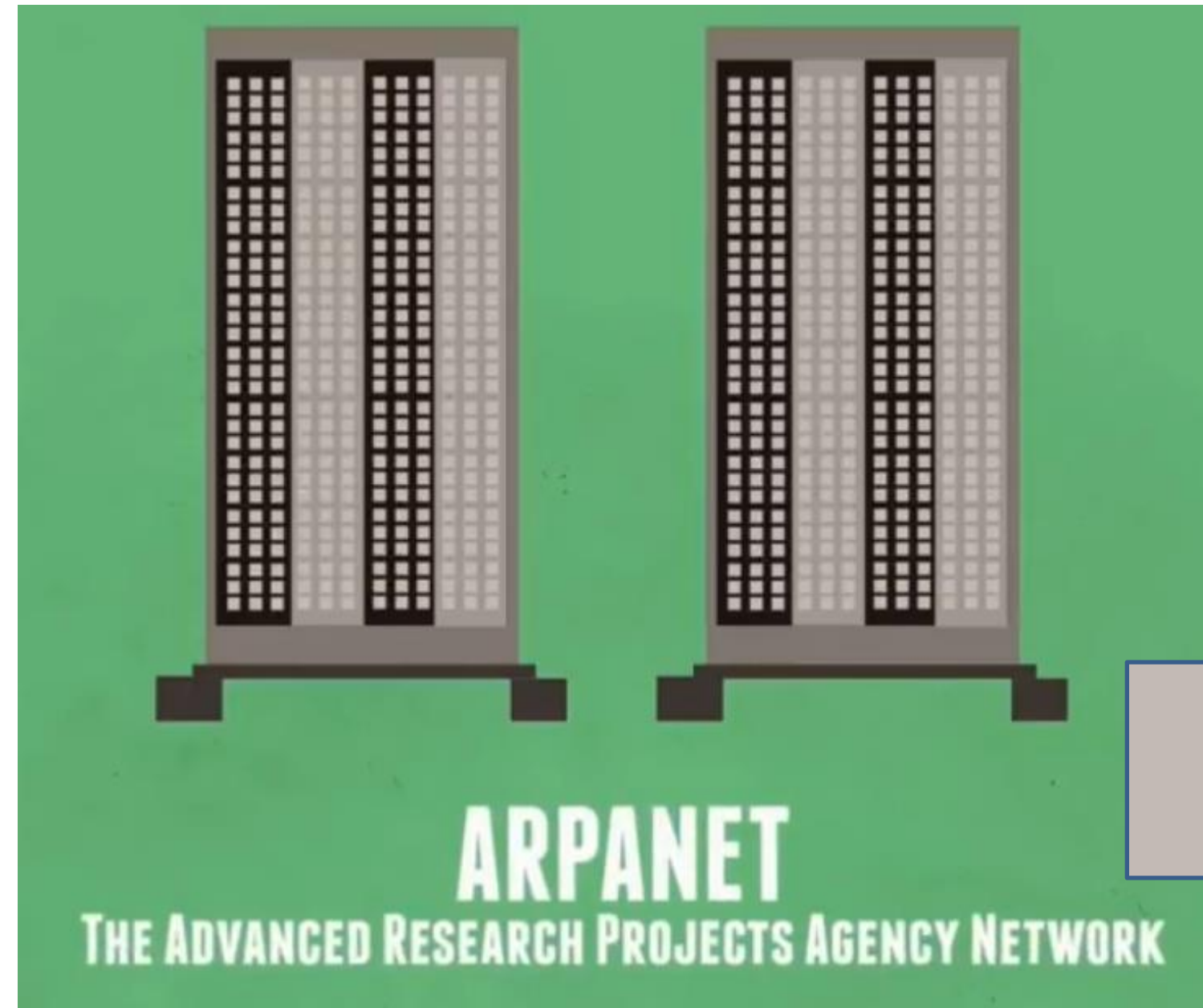
# Internet History - Early Days

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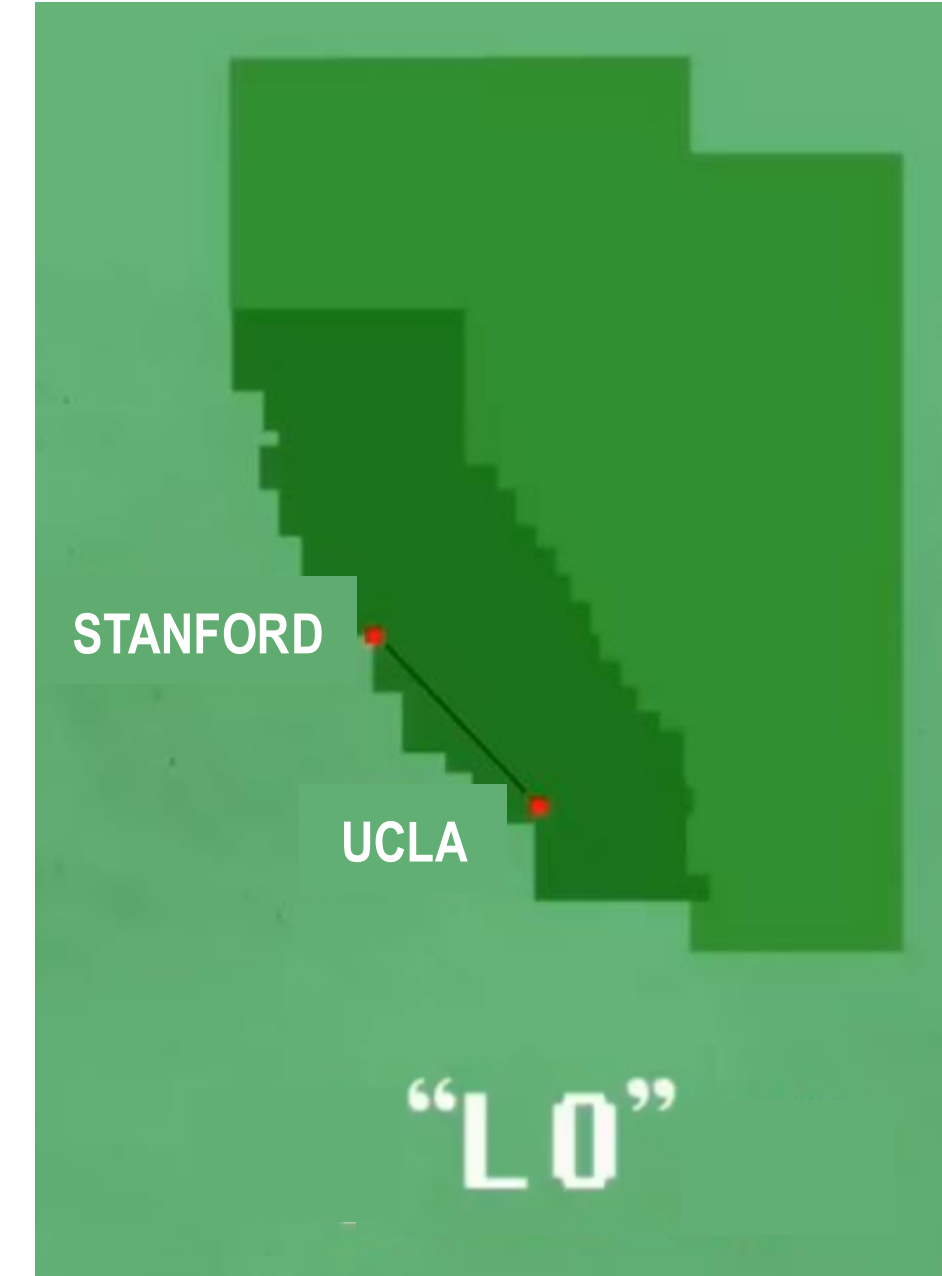
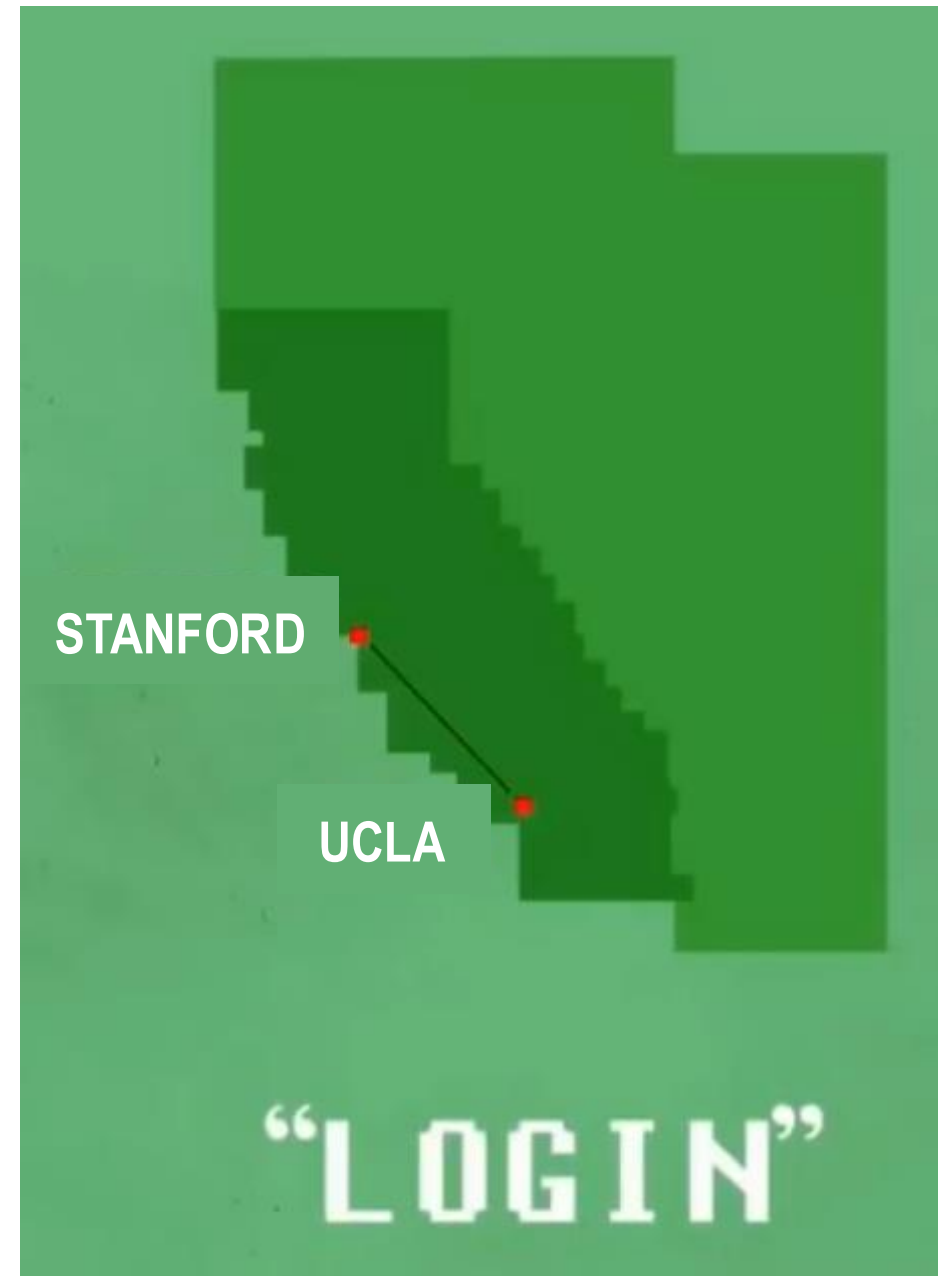
# Internet History - 1969

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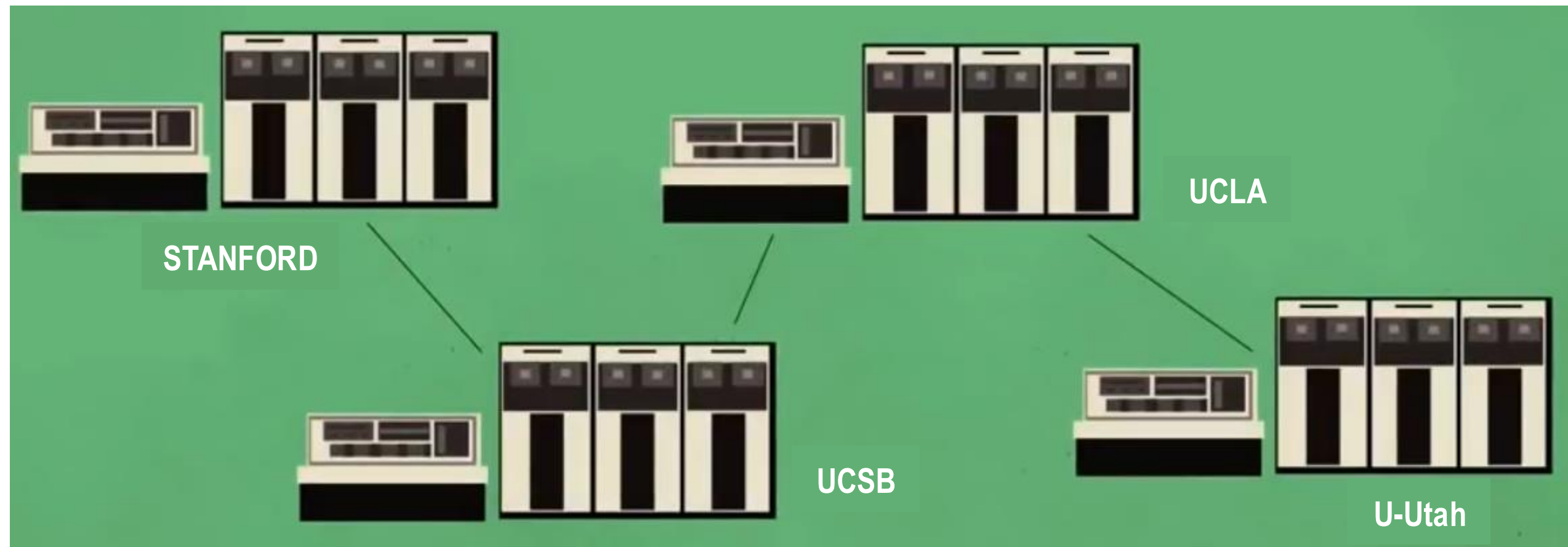


**Interface Message  
Processor (IMP)**

# Internet History - 1969



# Internet History - 1969



*Network Control Protocol (NCP)* software provided communication between the hosts.

# Internet History - 1971

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# Internet History - 1971

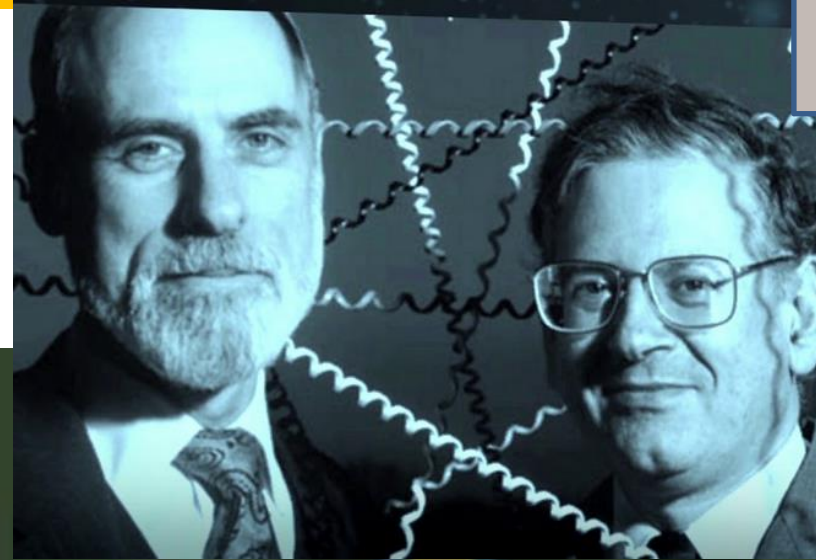
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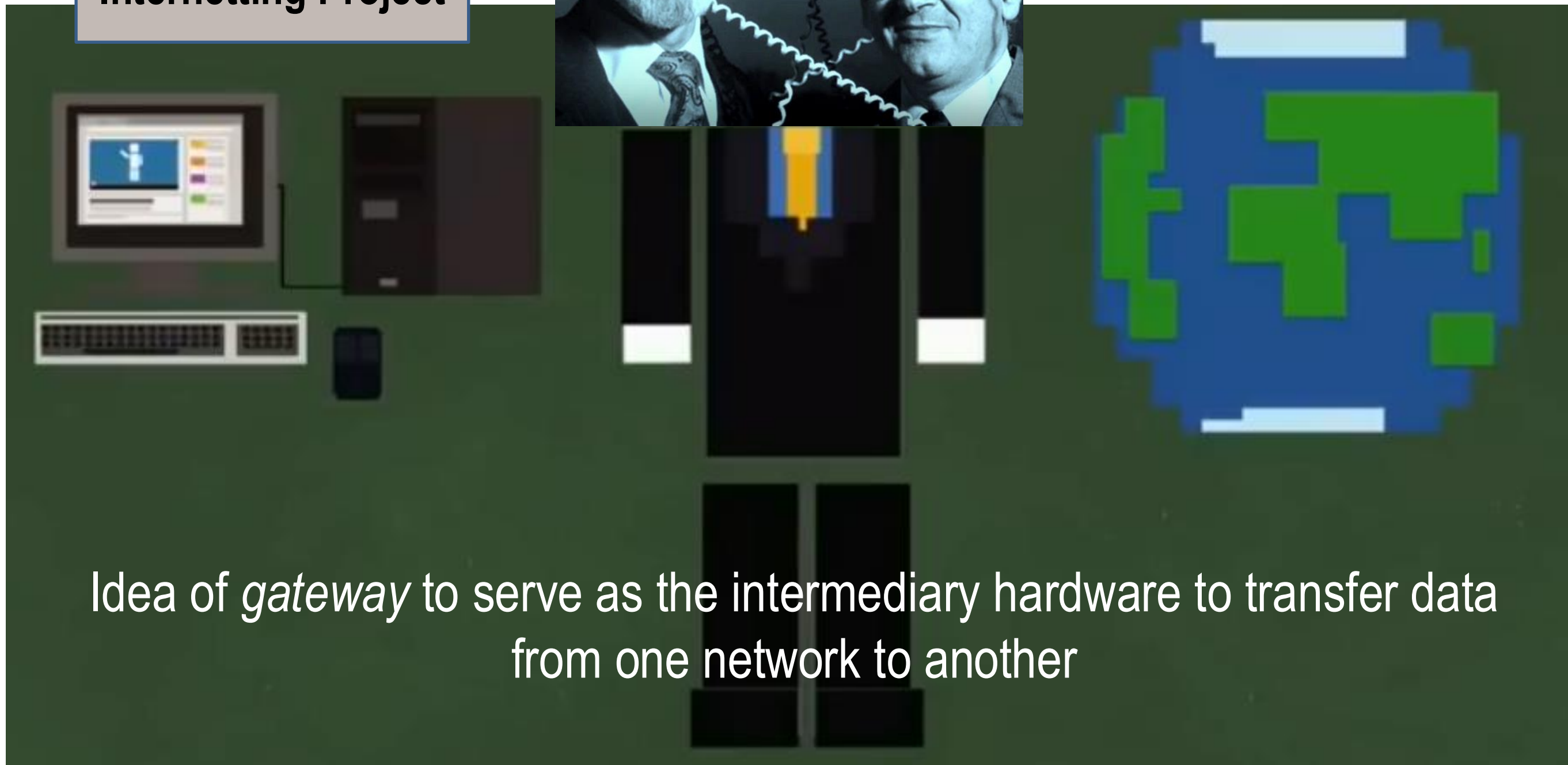
# Internet History - 1972



Vint Cerf and Bob Kahn

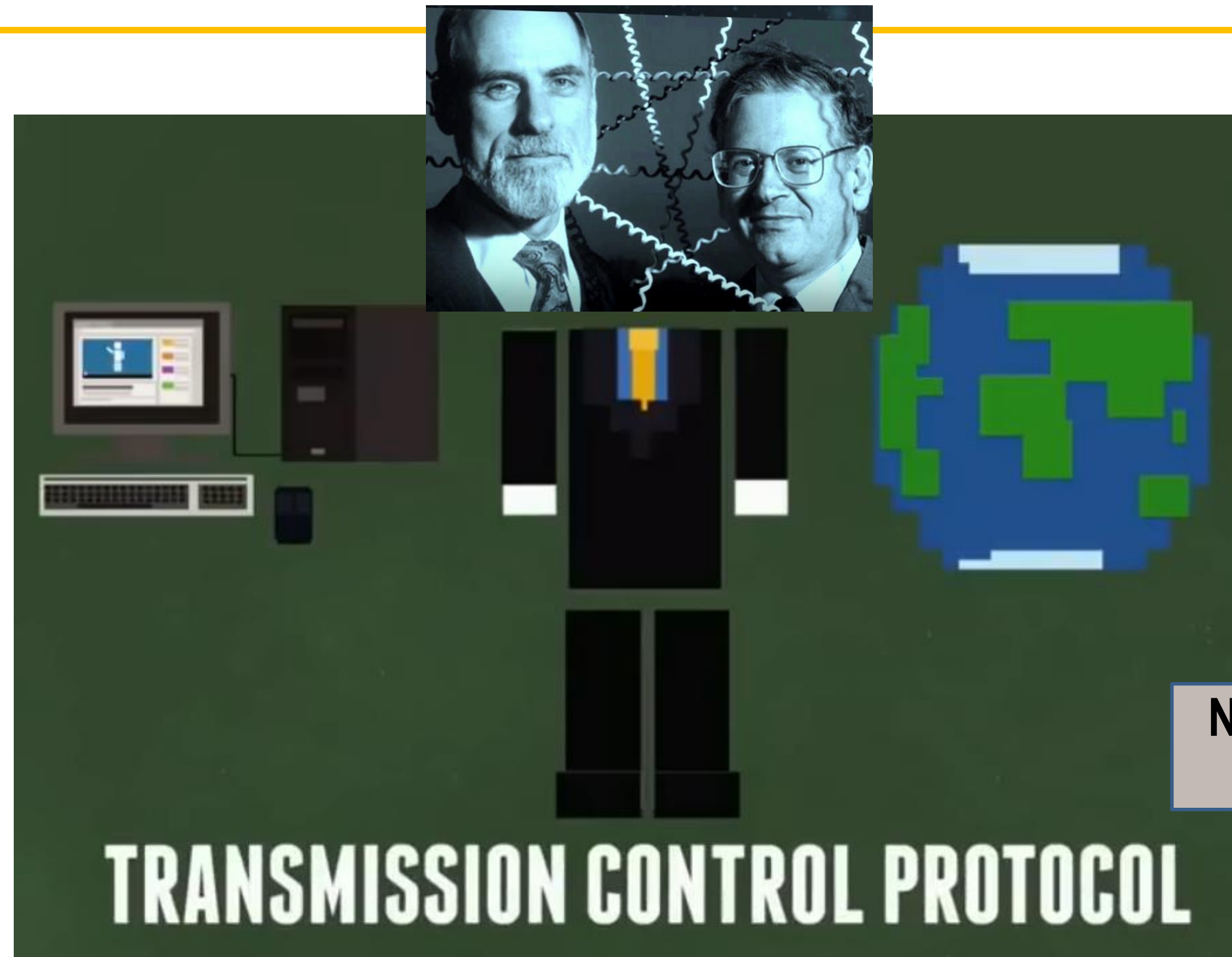


Internetting Project



Idea of *gateway* to serve as the intermediary hardware to transfer data from one network to another

# Internet History - 1973



Newer Version  
of NCP

# Internet History - October 1977

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- ▶ First demonstration of an internet consisting of three different networks (ARPANET, packet radio, and packet satellite).
- ▶ TCP into two protocols:
- ▶ Transmission Control Protocol (TCP)
- ▶ Internet Protocol (IP).
  - IP would handle datagram routing while TCP would be responsible for higher level functions such as segmentation, reassembly, and error detection.
- ▶ The new combination became known as TCP/IP.

# CSNET (Computer Science Network) - 1981

---



- ▶ Sponsored by the National Science Foundation (NSF).
- ▶ Conceived by universities that were ineligible to join ARPANET
- ▶ By the mid-1980s, most U.S. universities with computer science departments were part of CSNET.



# Internet History - 1981

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- ▶ The term Internet, originally associated with government-funded connected networks, now referred to the connected networks using TCP/IP protocols



Ad about Internet

# Internet History - 1983

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- ▶ TCP/IP became the official protocol for the ARPANET



# MILNET - 1983

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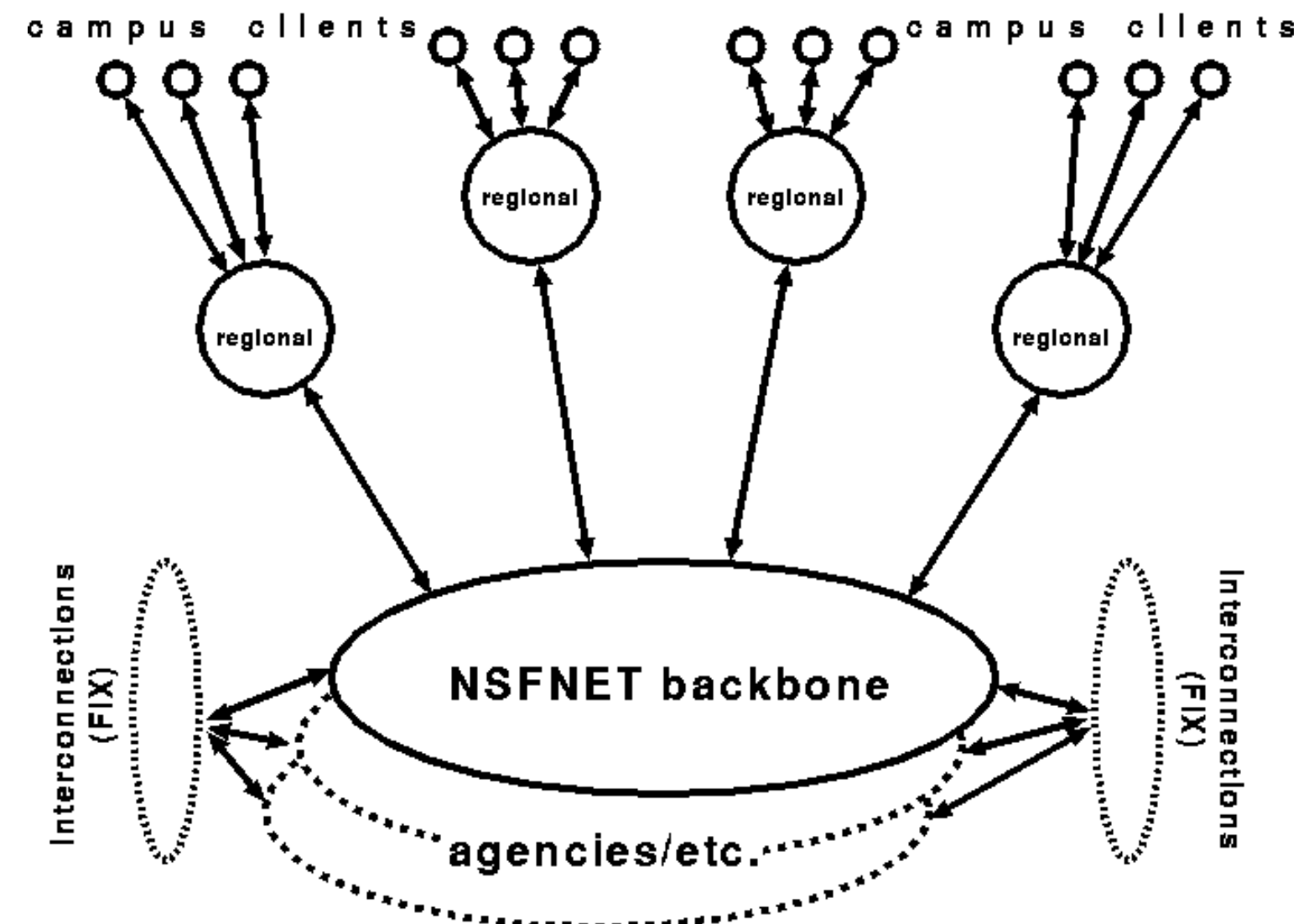
- ▶ ARPANET split into two networks: Military Network (MILNET) for military users and ARPANET for non-military users.



# NSFNET - 1986



- ▶ The NSF sponsored the National Science Foundation Network (NSFNET)
- ▶ A backbone that connected five supercomputer centers located throughout the US.



# Internet History - 1990

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- ▶ ARPANET was officially retired and replaced by NSFNET (National Science Foundation Network)



## 1990'S

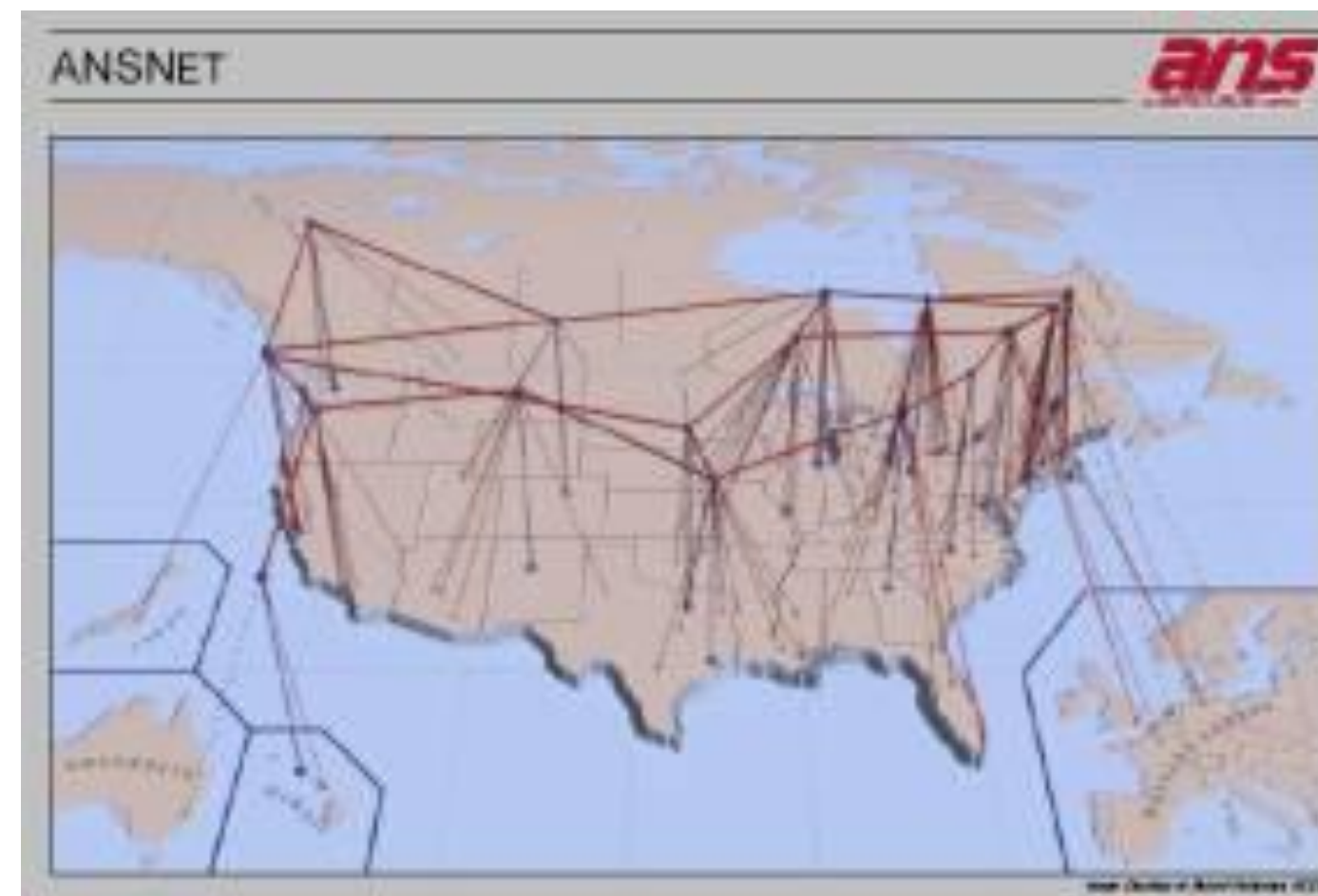
**Wave One:**

Web & Internet

# ANSNET - 1991



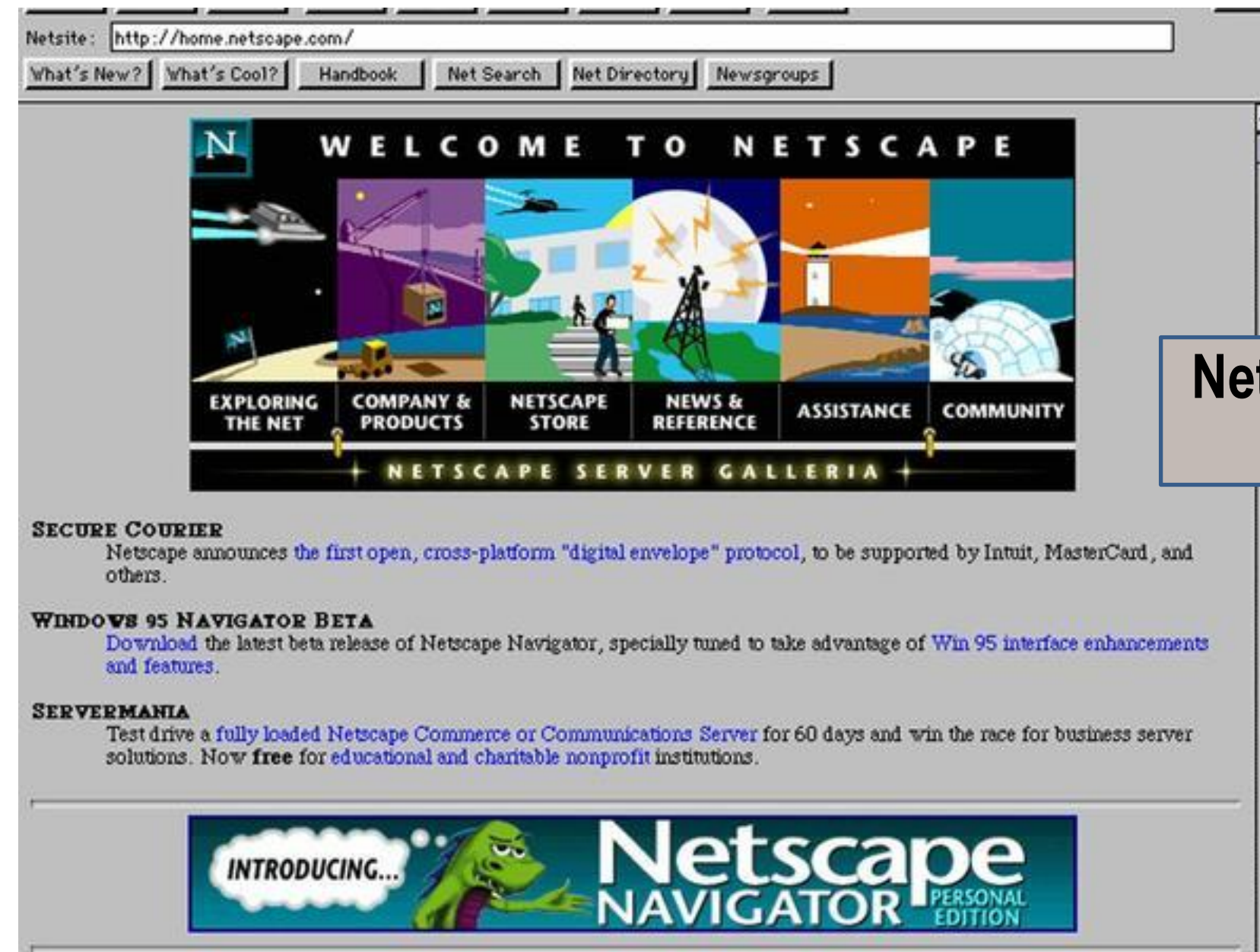
- ▶ IBM, Merit, and Verizon formed Advanced Network & Services (ANS) to build a new, high-speed Internet backbone called ANSNET.



# Internet History - 1995



- NSFNET reverted back to its original concept of a research network



Netscape Navigator  
Browser 1995

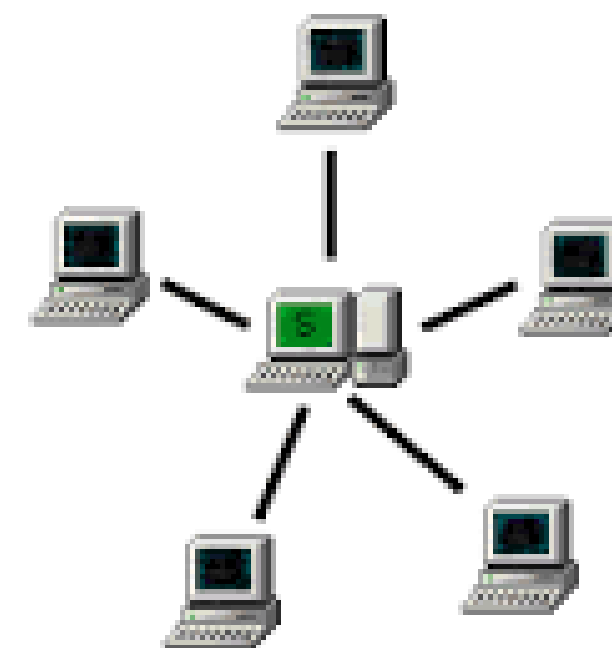
# Internet Today



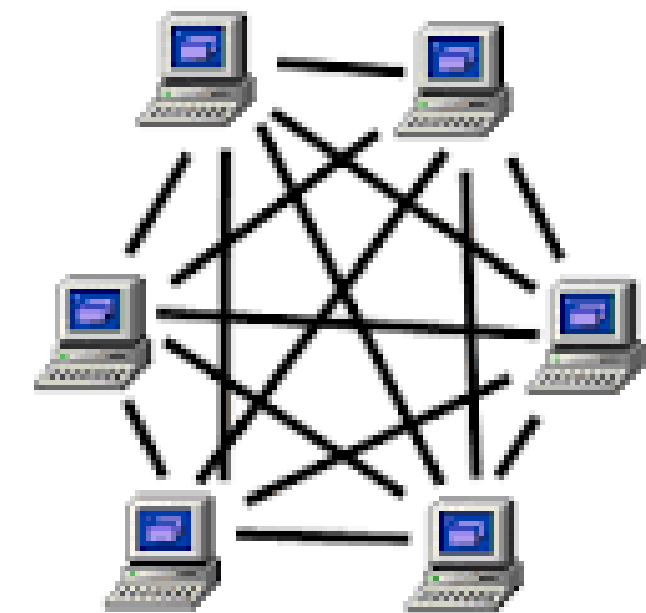
The first website ever, <http://info.cern.ch/>, went live on August 6, 1991, and was set up by British physicist Tim Berners-Lee, working for the CERN institute.



Server Based Network



Peer to Peer Network



# Question

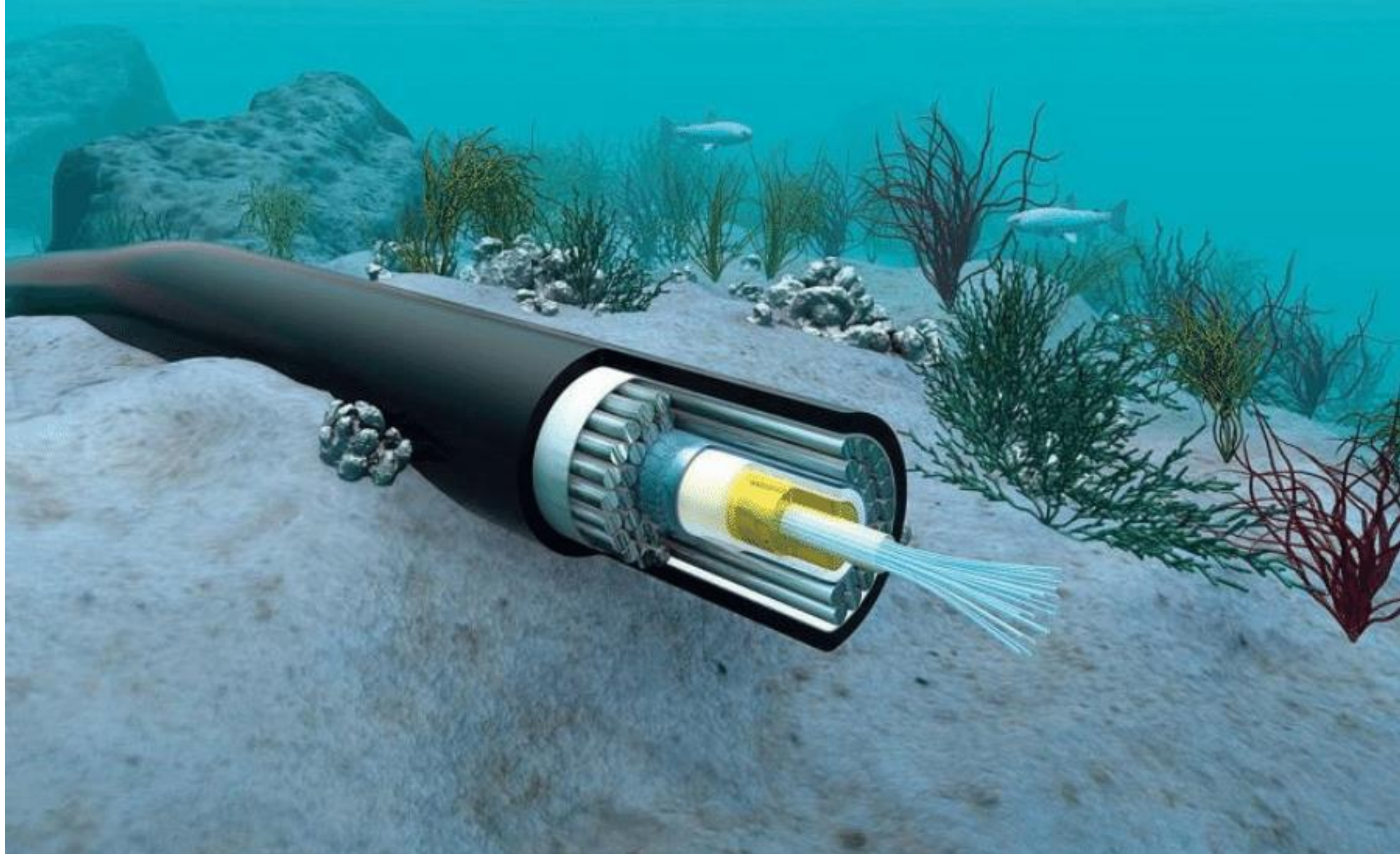
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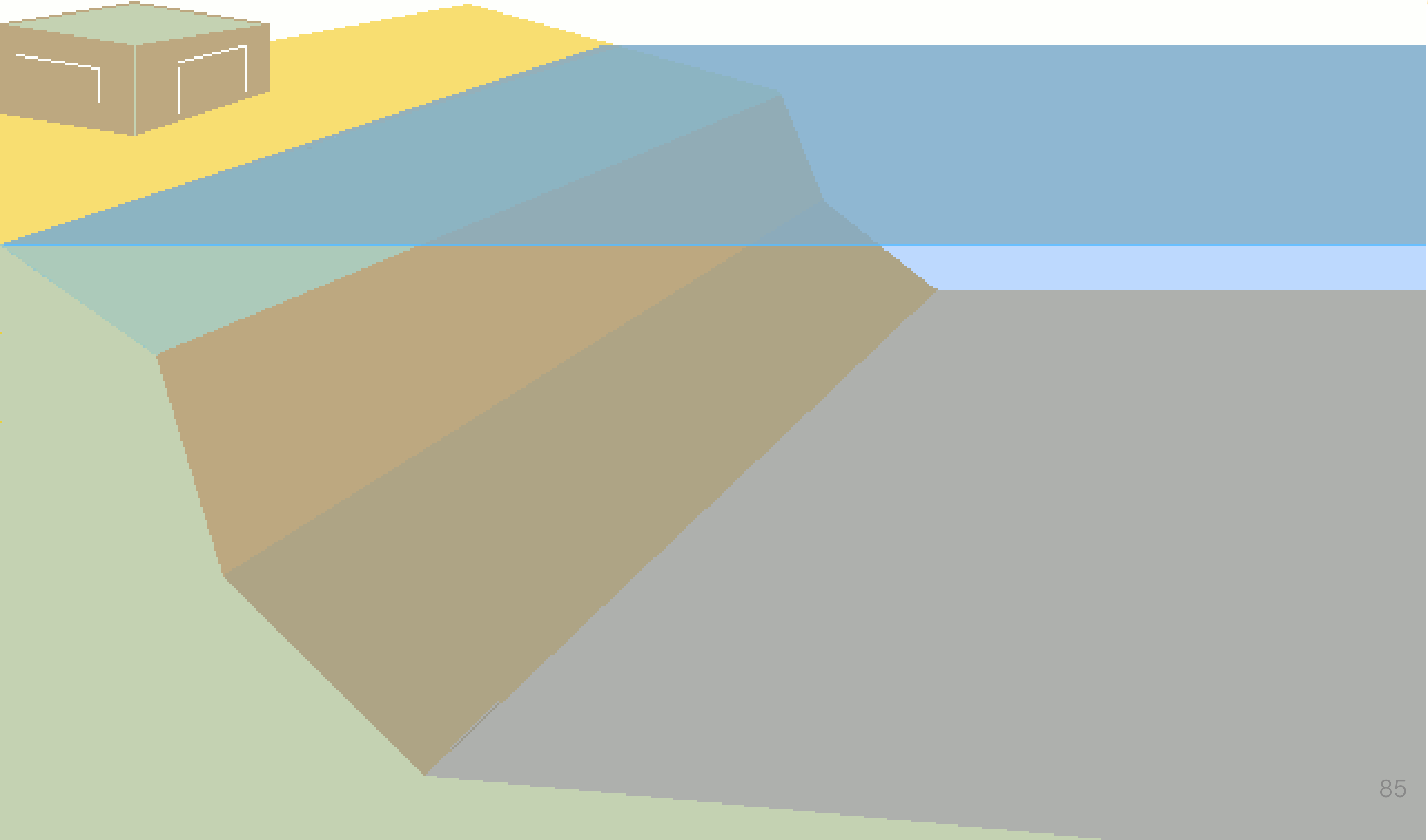


- ▶ How is the whole world connected literally?
- ▶ Major Role Player: Undersea Cables

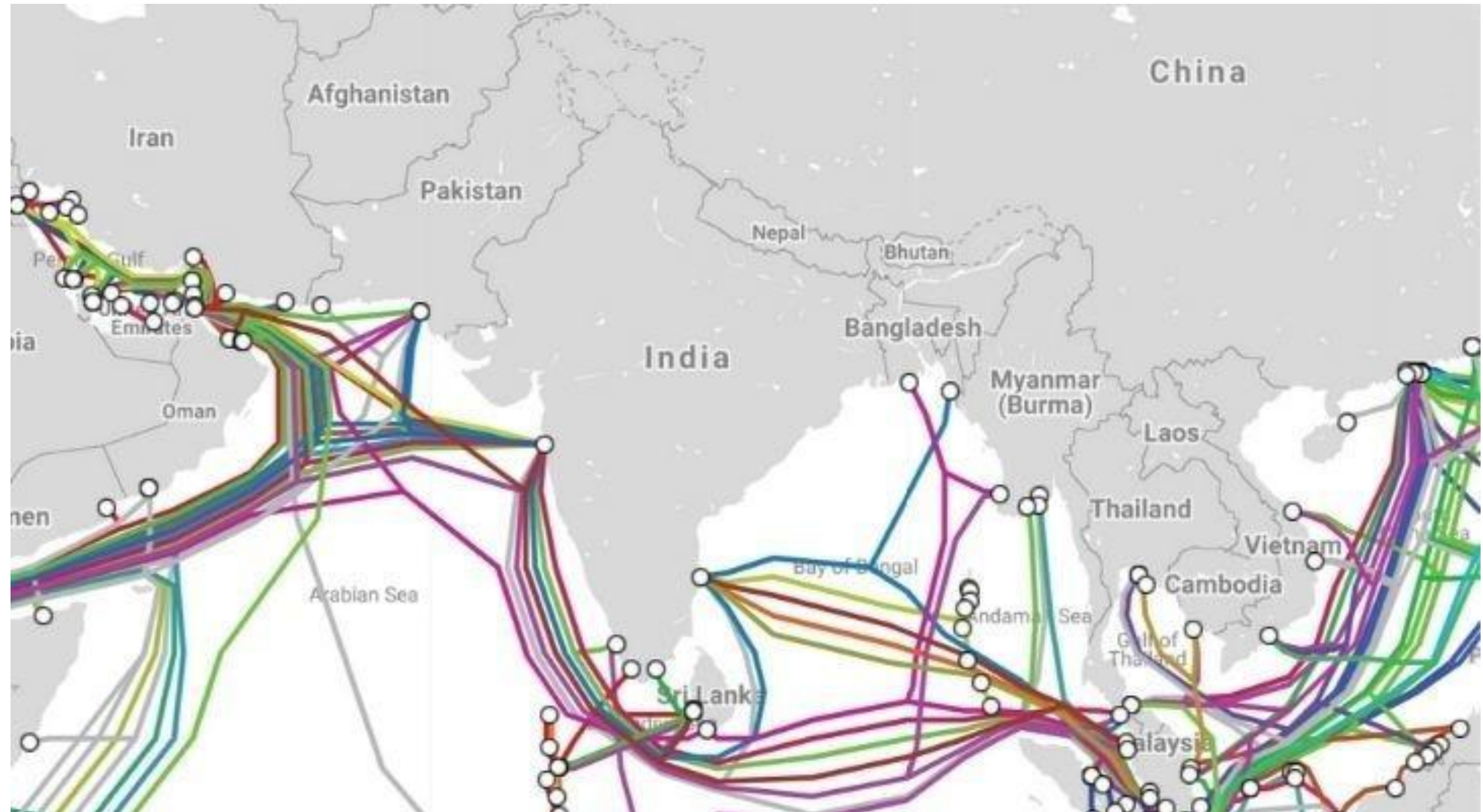
# Submarine Cable

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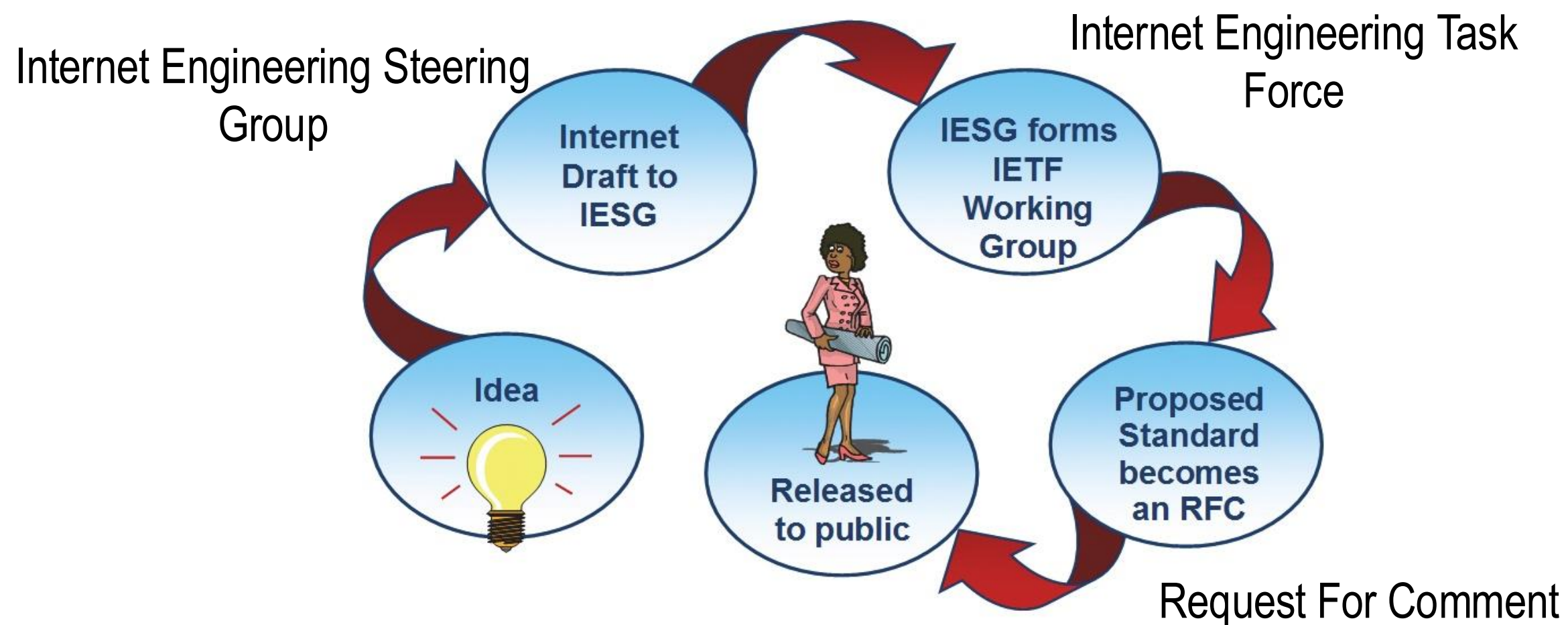
# Cable Stations in India



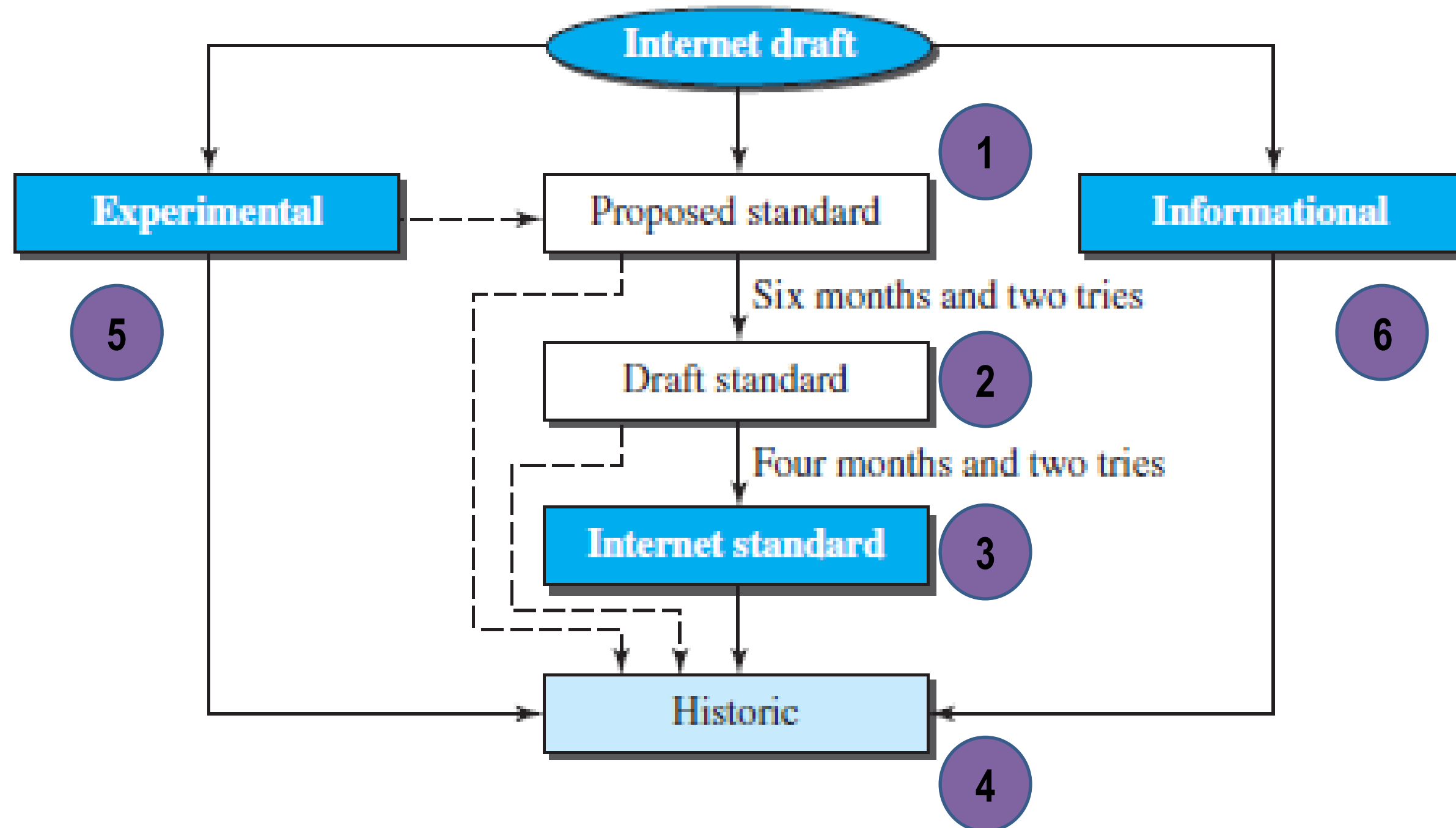
# Internet Standards



- ▶ A thoroughly tested specification that is useful to and adhered to by those who work with the Internet
- ▶ An Internet draft is a working document with no official status and a six-month lifetime



# Maturity Levels



# Requirement Levels

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- ▶ Required
  - An RFC is labeled required if it must be implemented by all internet systems to achieve minimum conformance eg. IP, ICMP
- ▶ Recommended
  - An RFC labeled recommended is not required for minimum conformance. Eg. telnet, FTP
- ▶ Elective
  - An RFC labeled elective is not required and not recommended

# Requirement Levels

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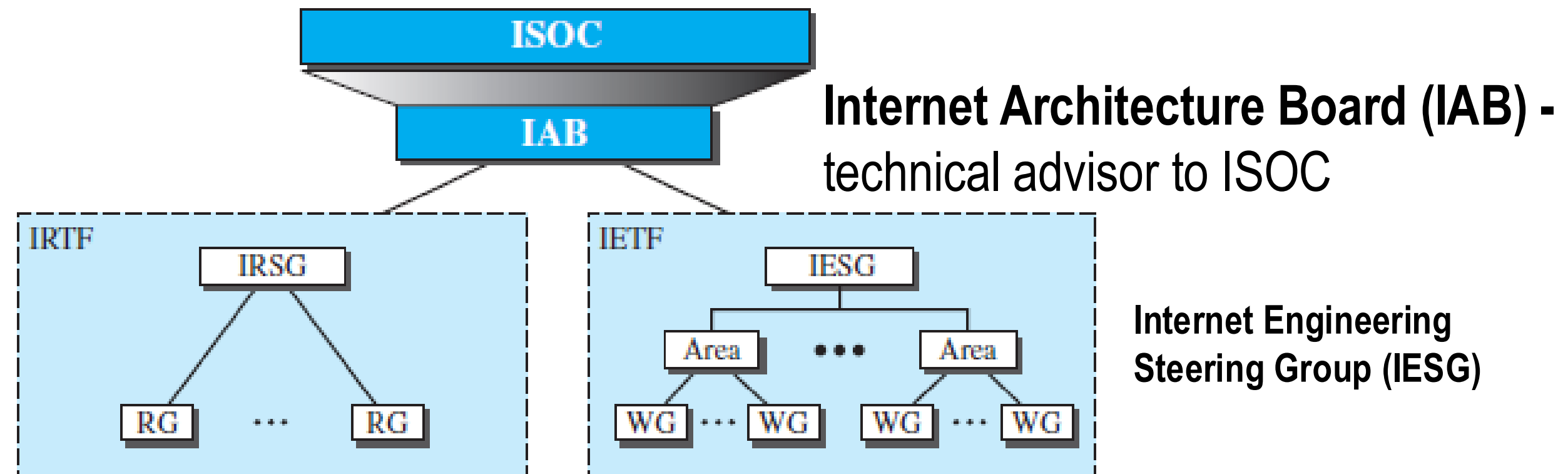


- ▶ Limited Use
  - An RFC labeled limited use should be used only in limited situations. Most of the experimental RFCs fall under this
- ▶ Not Recommended
  - An RFC labeled not recommended is inappropriate for general use.

# Internet Administration



The **Internet Society (ISOC)** is an international, nonprofit organization formed in 1992 to provide support for the Internet standards process



The **Internet Research Task Force (IRTF)** focuses on long-term research. Managed by **Internet Research Steering Group**

**Internet Engineering Task Force (IETF)** is responsible for identifying operational problems and proposing solutions to them.

**WG** - Working Group for 9 areas

# PROTOCOL LAYERING

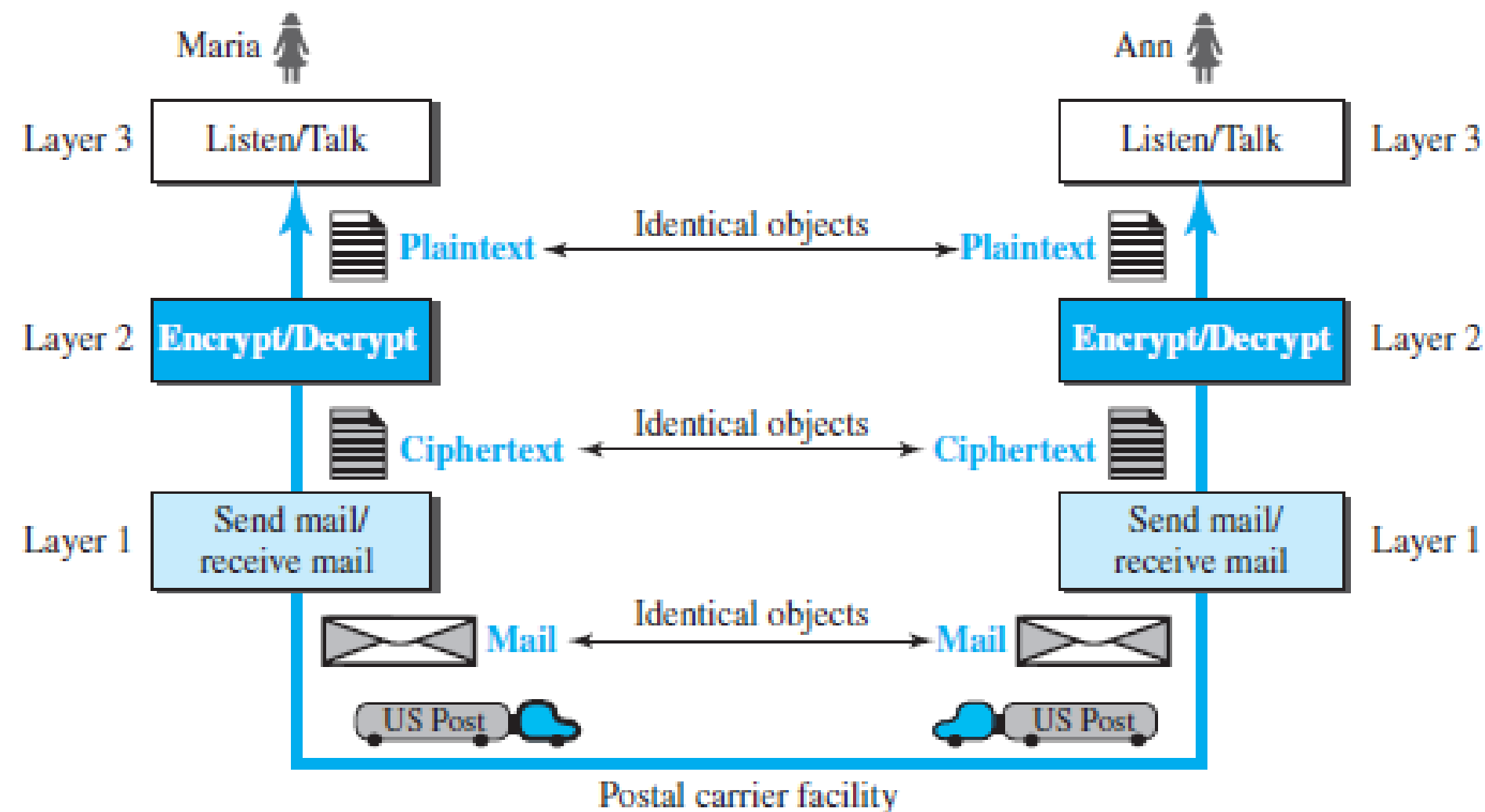
92



# Protocol Layering



- ▶ When the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or protocol layering.



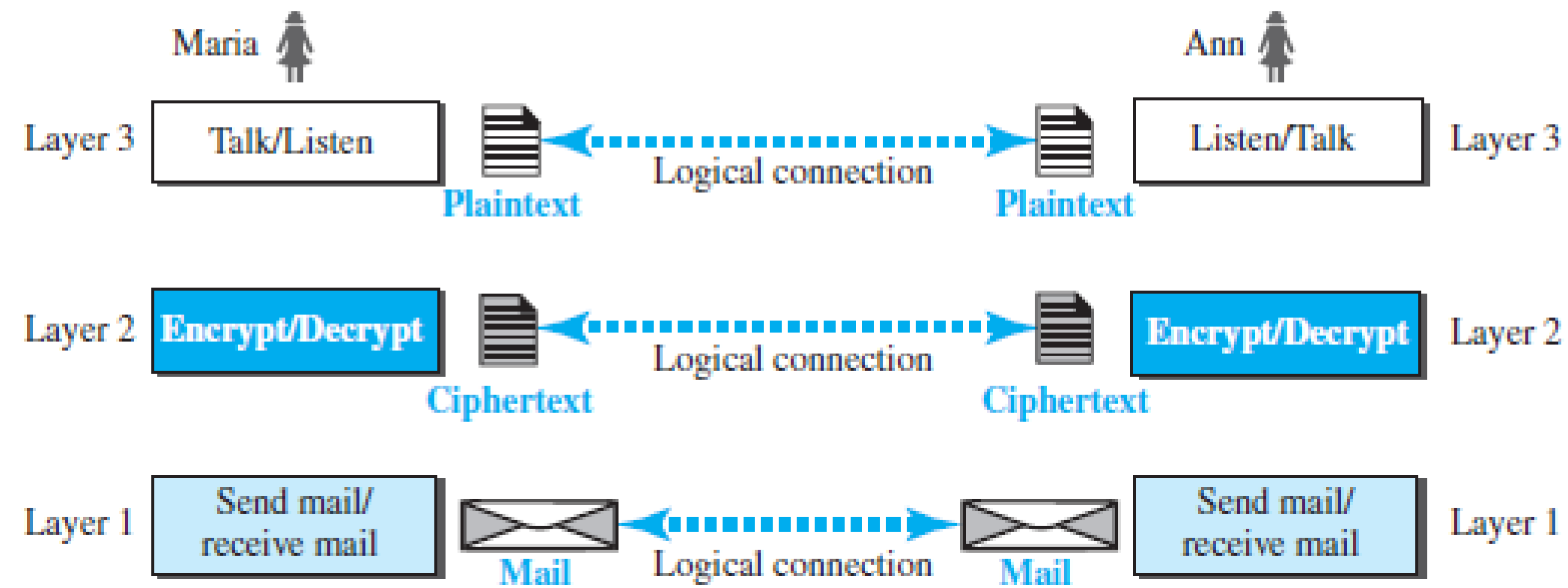
# Principles of Protocol Layering

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- ▶ First Principle
- ▶ If we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction.
- ▶ Second Principle
- ▶ The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical.

# Logical Connection



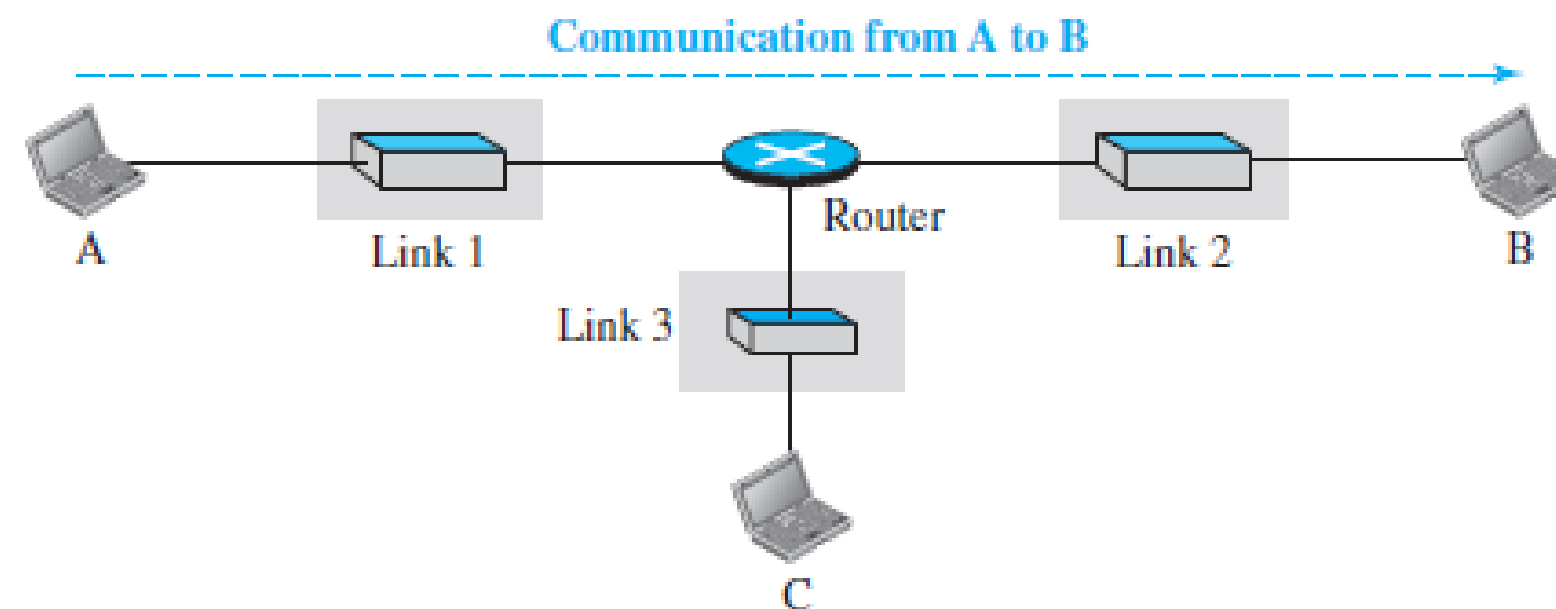
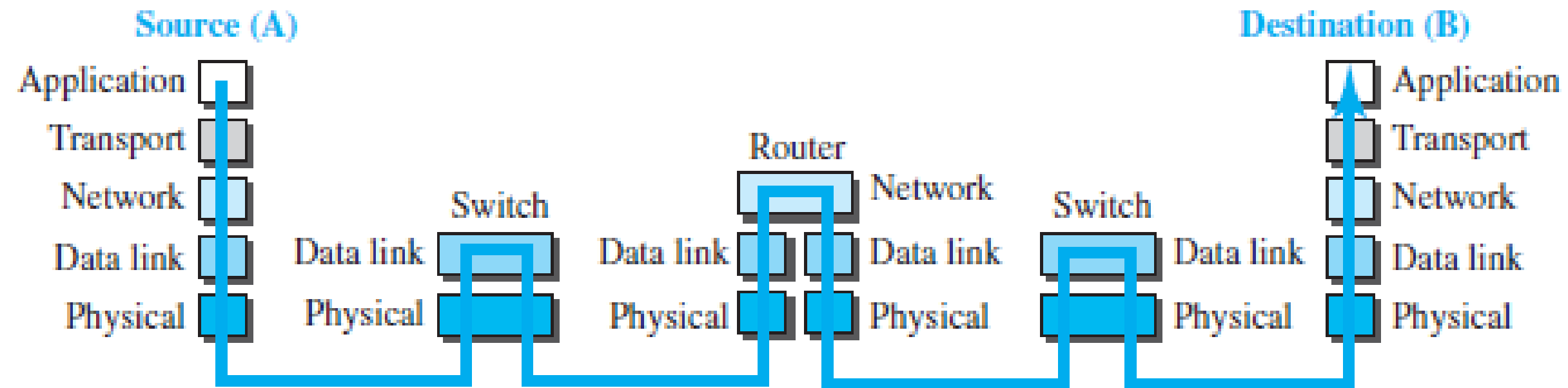
# TCP/IP Protocol Suite

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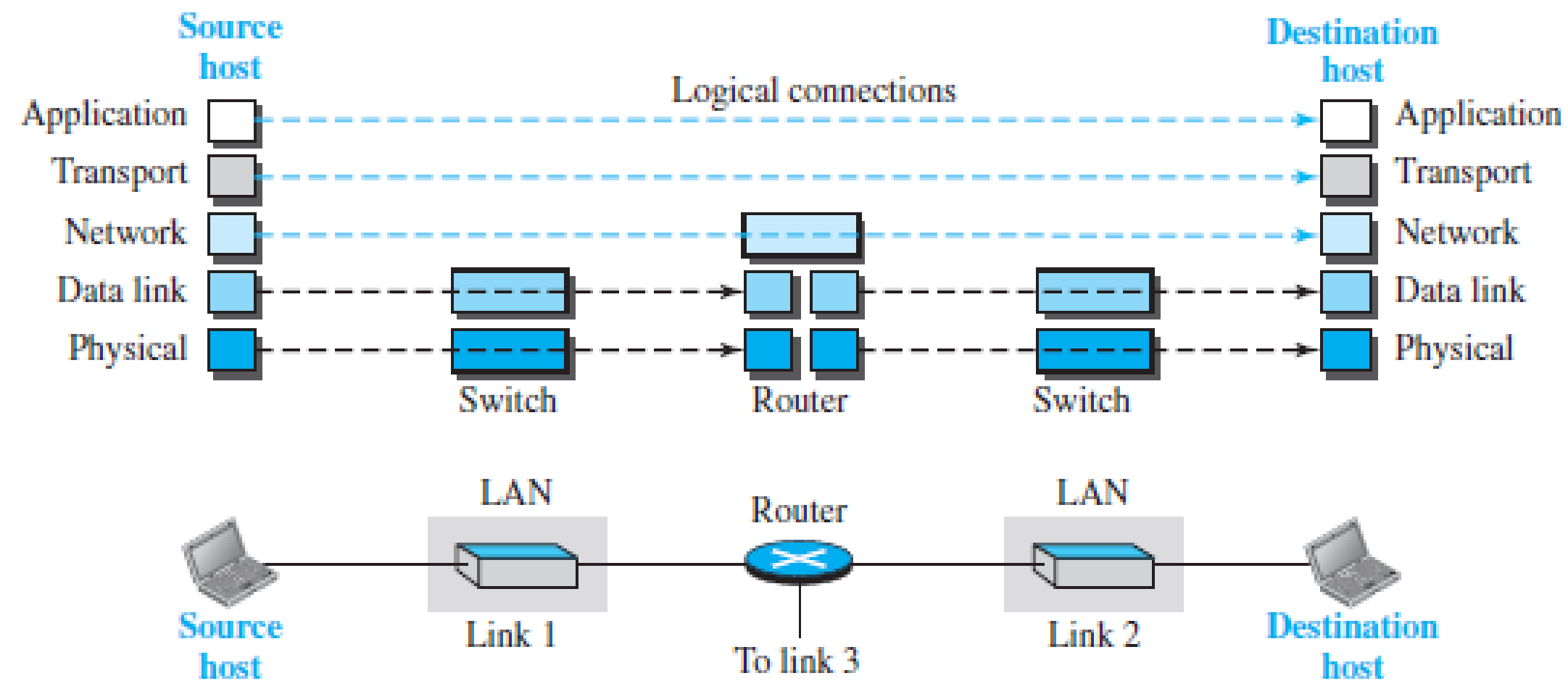


- ▶ TCP/IP is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- ▶ It is a hierarchical protocol made up of interactive modules
  - Each module provides a specific functionality
  - Each upper level protocol is supported by the services provided by one or more lower level protocols

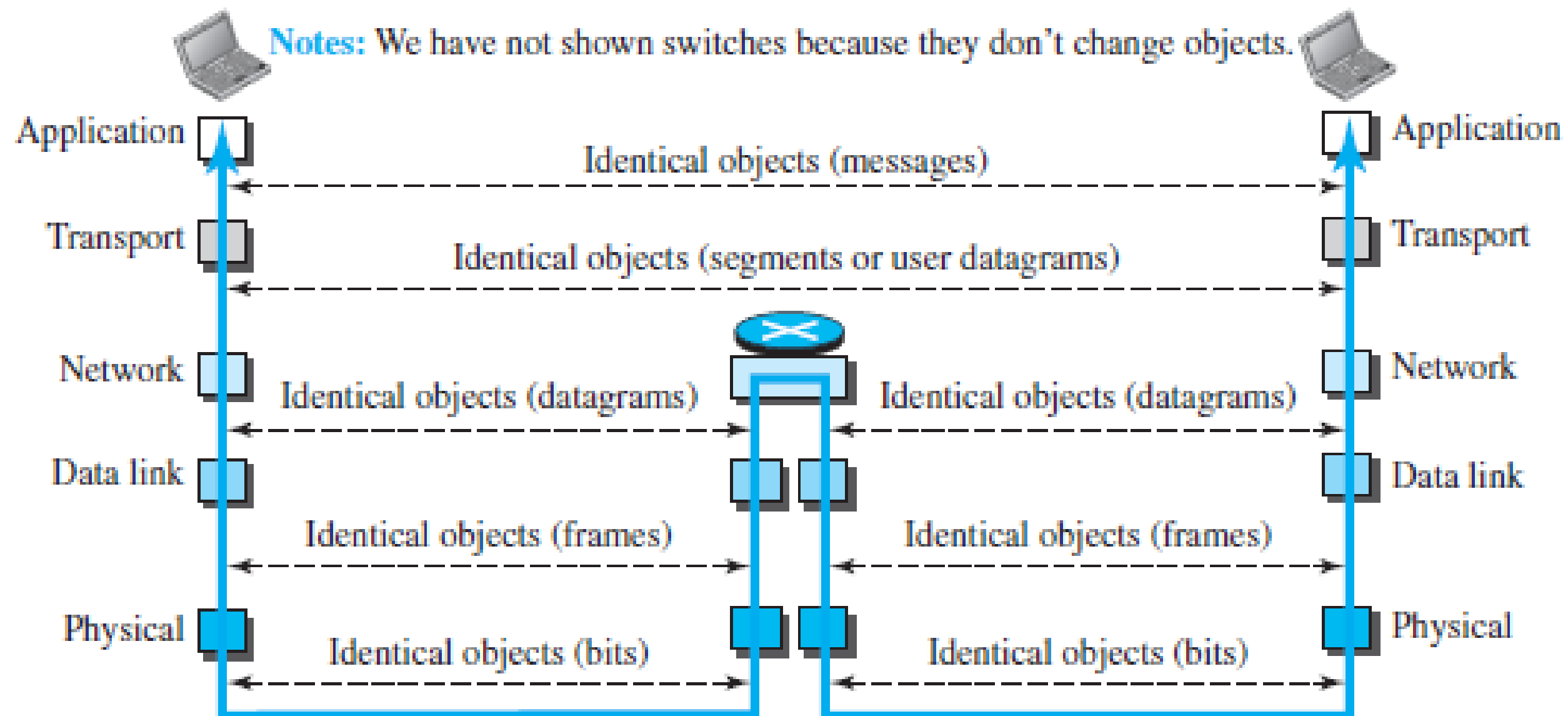
# Layered Architecture



# Layers in TCP/IP



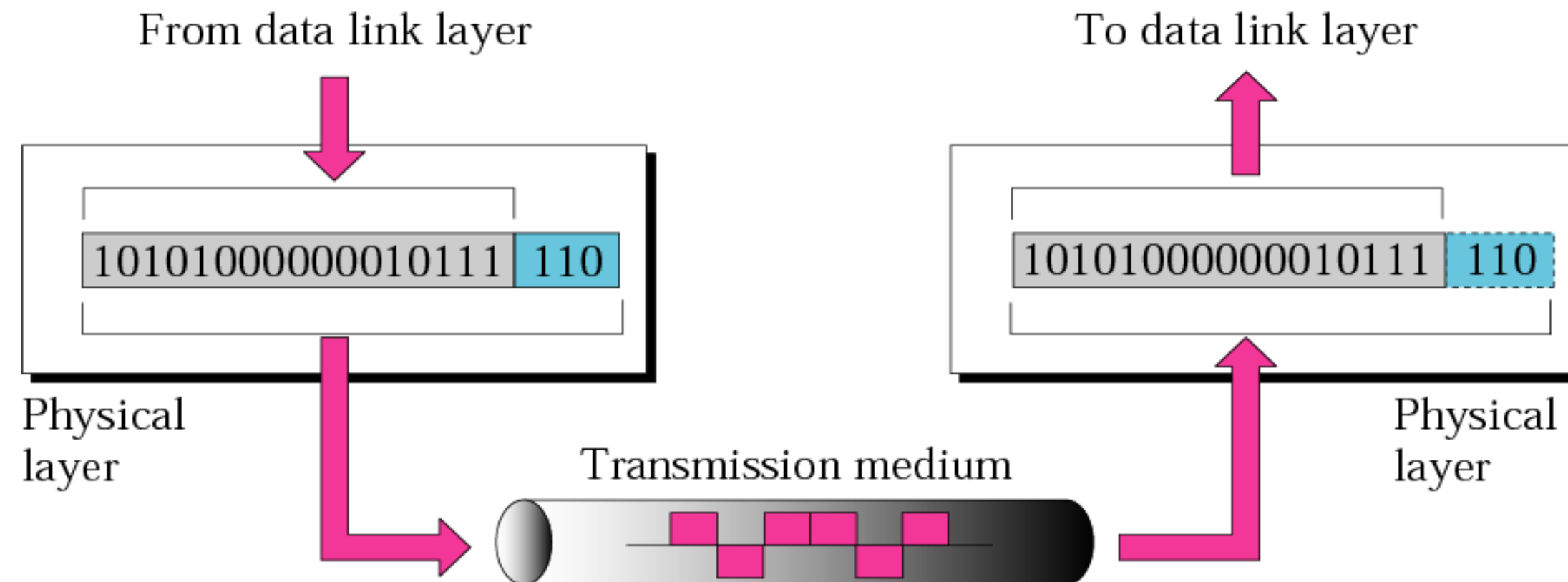
# Layers in TCP/IP



# Physical Layer



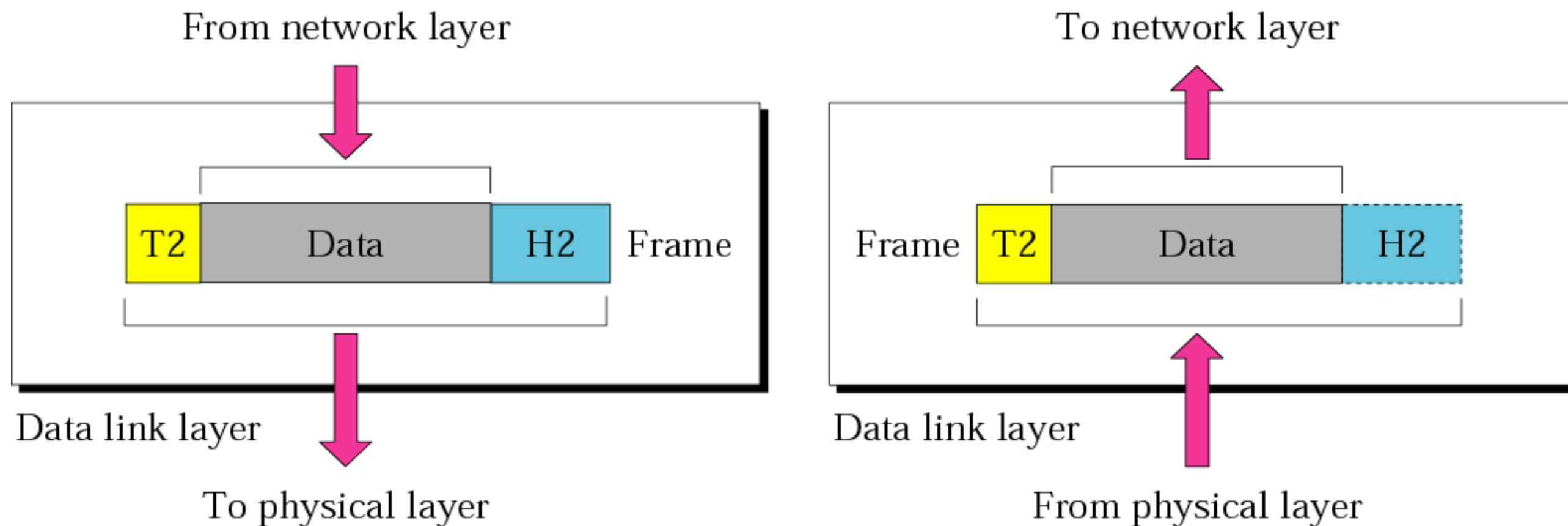
- ▶ Responsible for carrying individual bits in a frame across the link.
- ▶ The communication between two devices is still a logical communication.
  - Transmission medium does not carry bits; it carries electrical or optical signals.



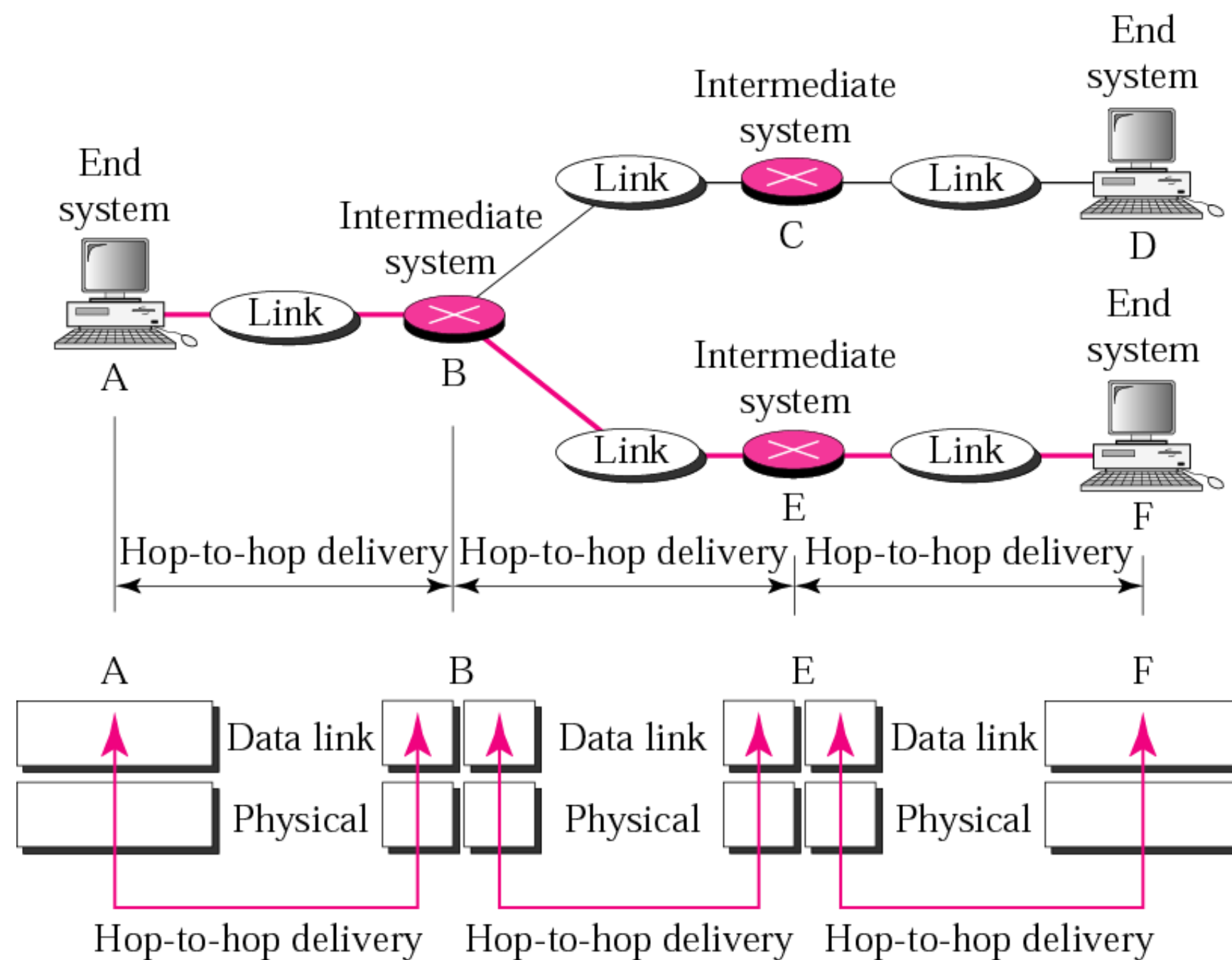
# Data-Link Layer



- ▶ When the next link to travel is determined by the router, the data-link layer takes the datagram and moves it across the link.
- ▶ It takes a datagram & encapsulates it in a packet called frame.



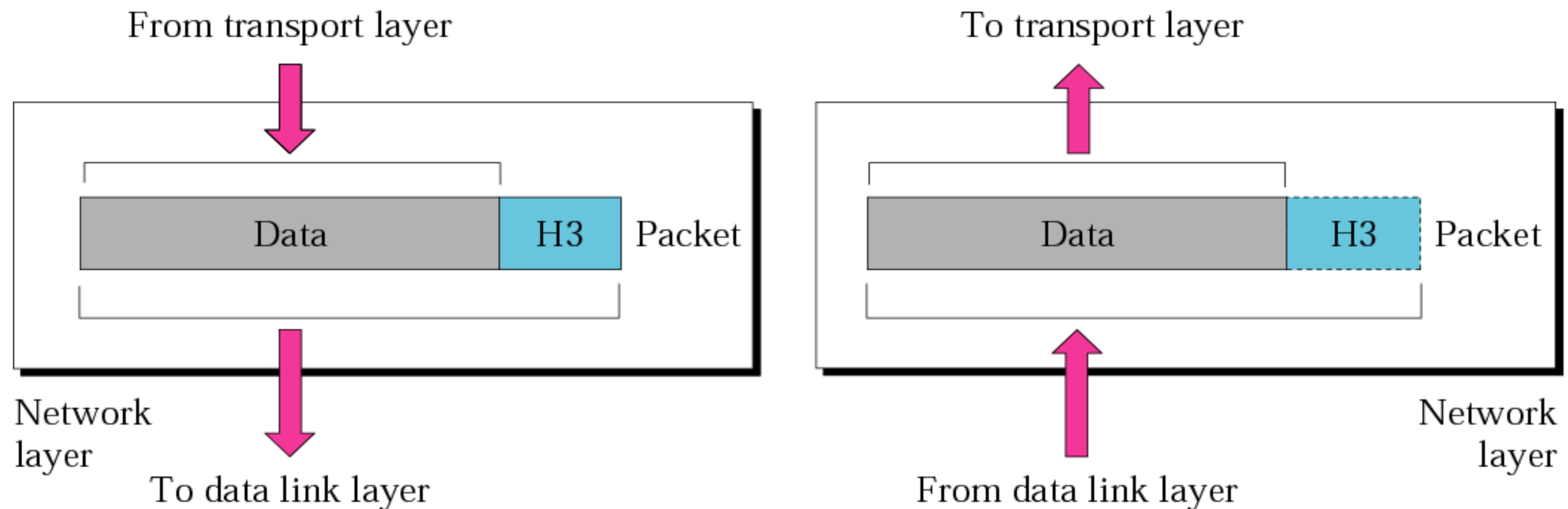
# Data-Link Layer



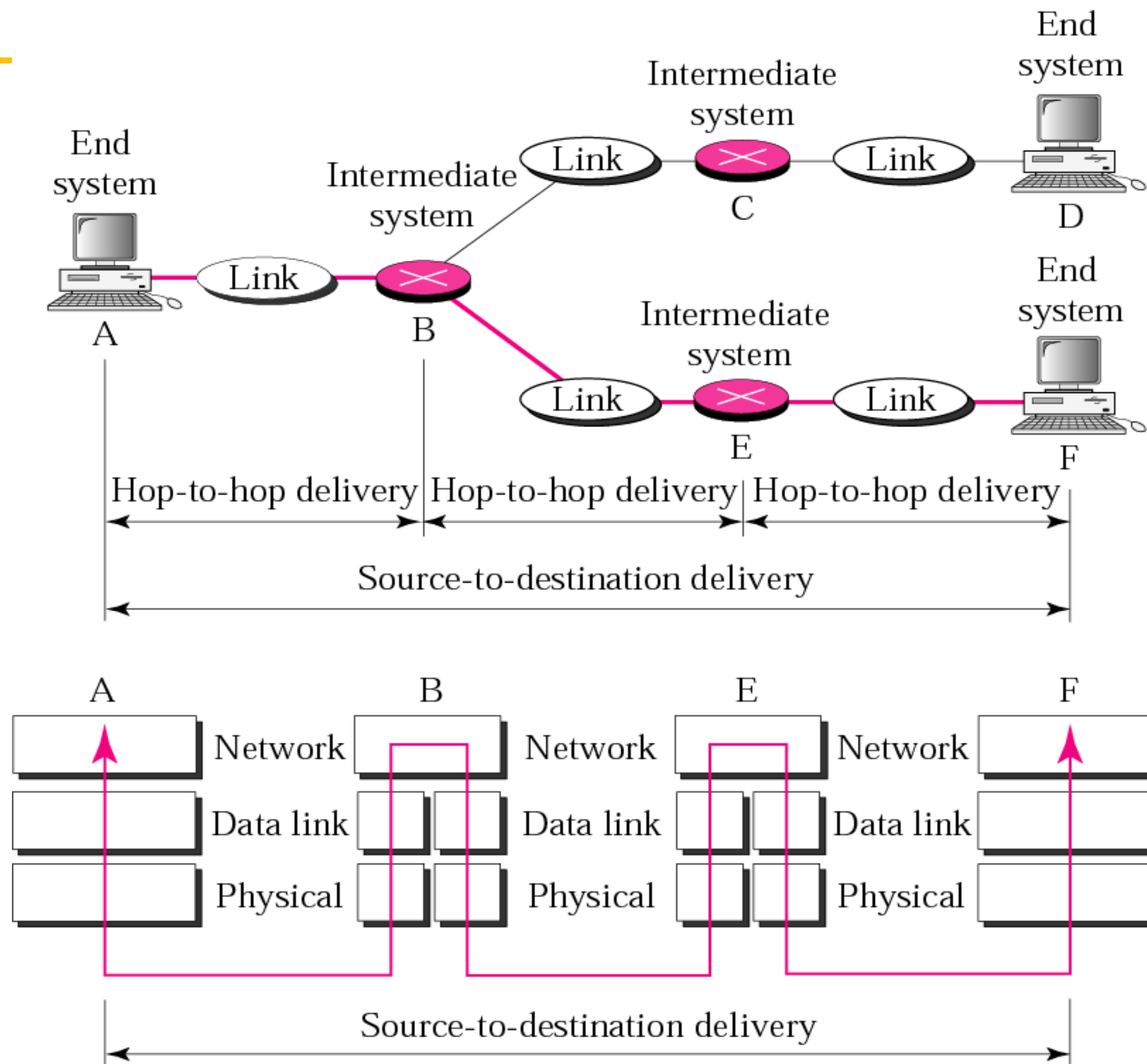
# Network Layer



- ▶ Responsible for creating host-to-host communication and routing packet through possible routes
- ▶ It includes Internet Protocol (IP), that defines the format of the packet, called a datagram and routes the packets



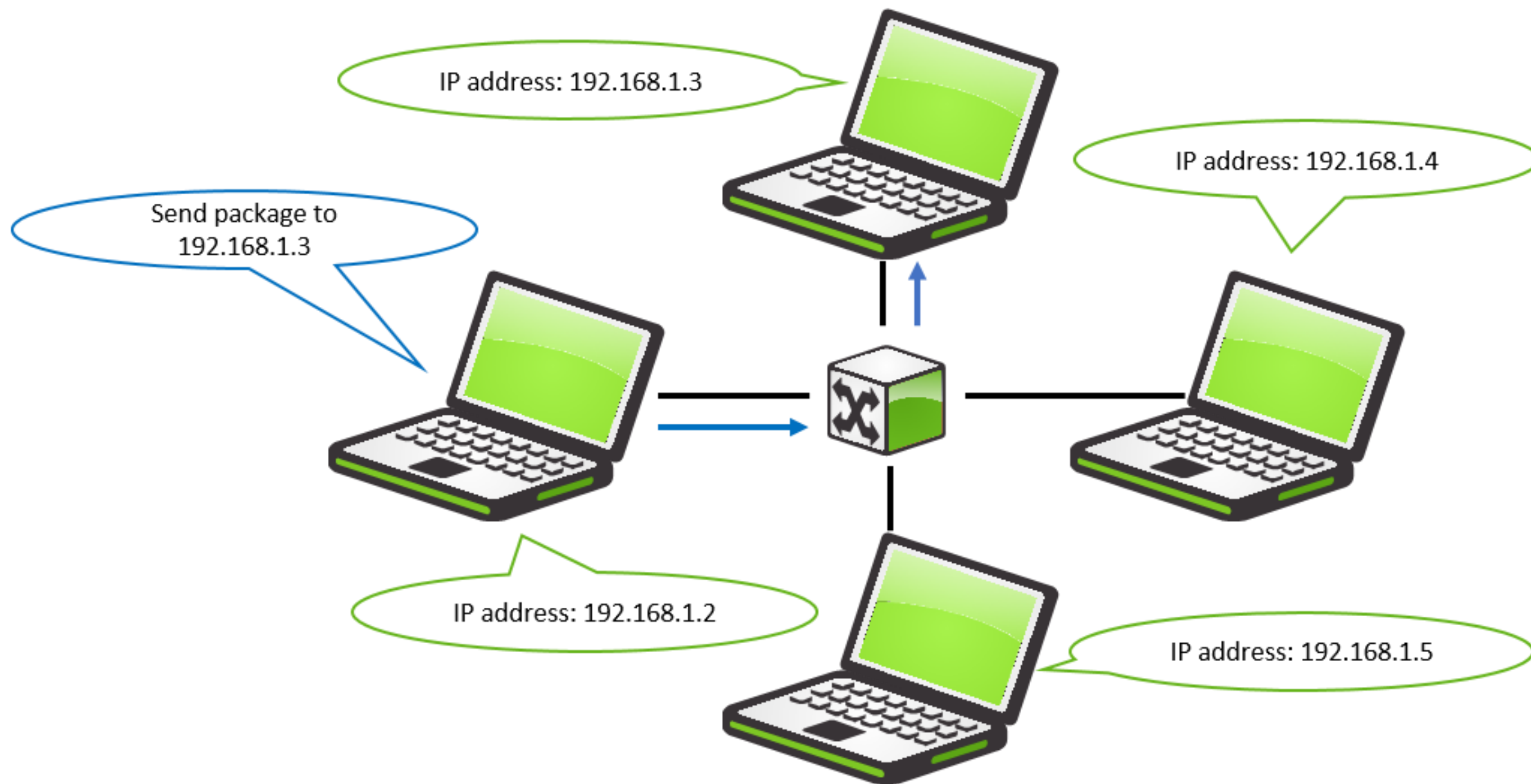
# Network Layer



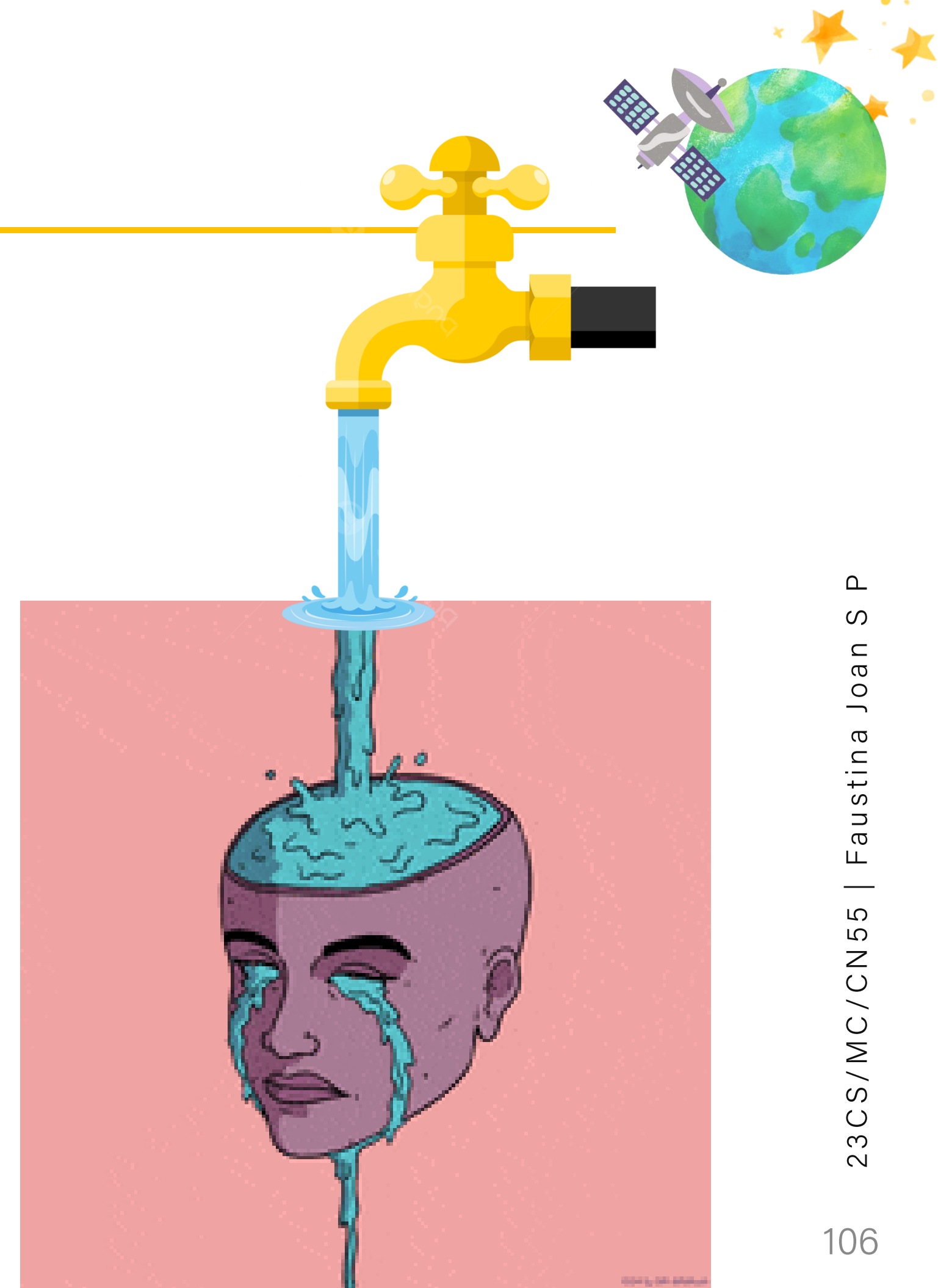
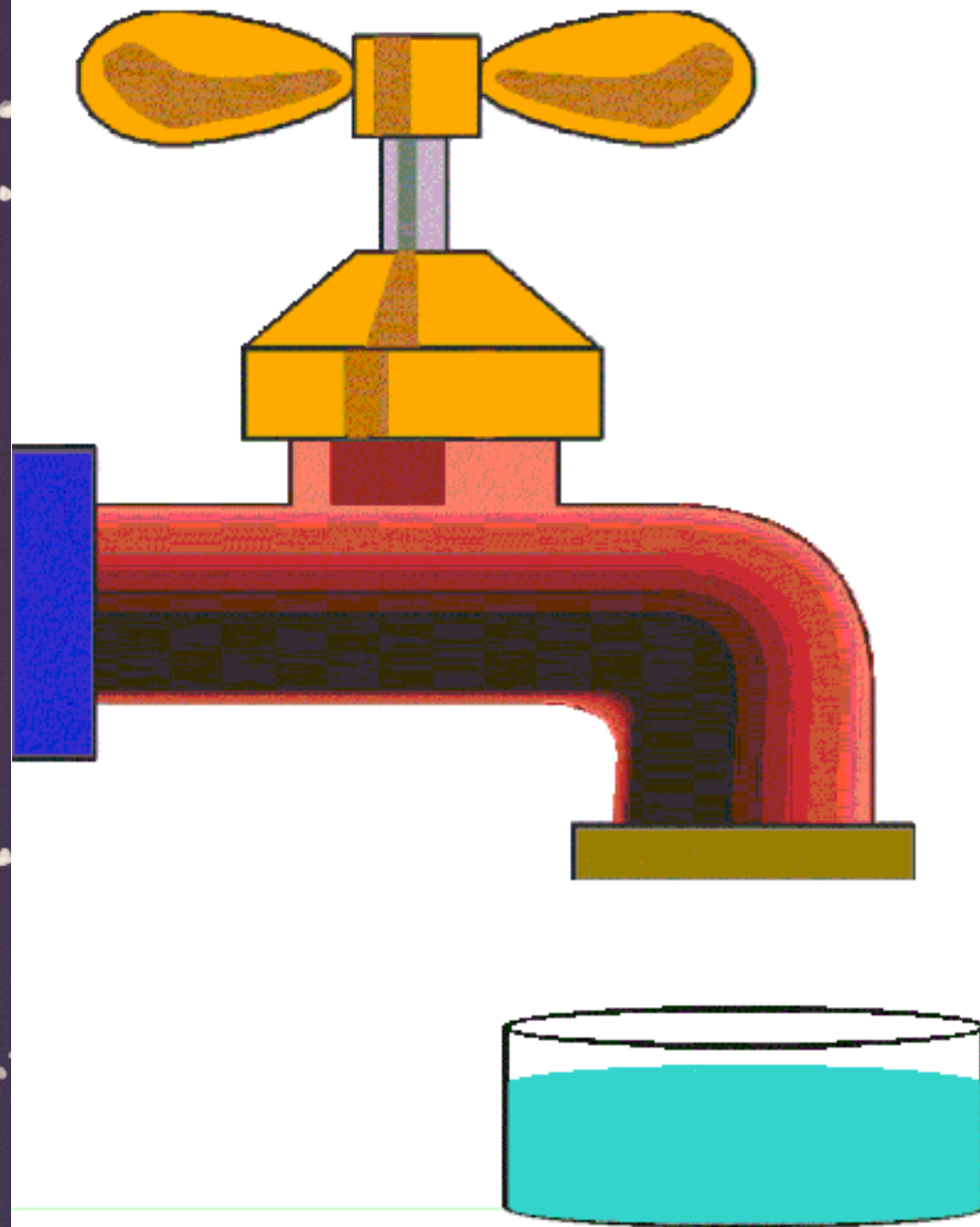
# Network Layer and IP



- IP defines the format and the structure of addresses used in this layer



# Flow Control

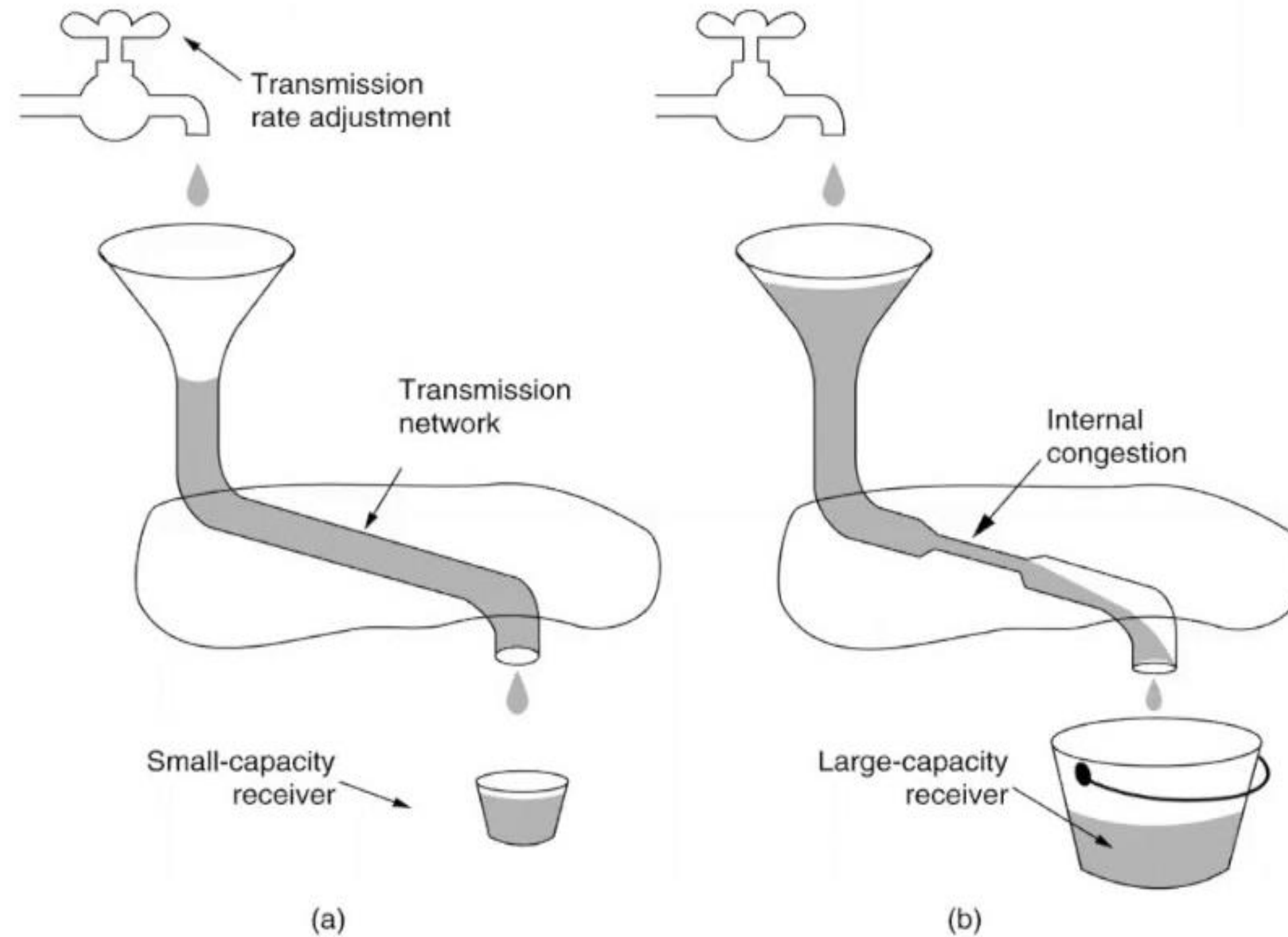


# Error Control

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# Congestion Control



(a) A fast network feeding a low capacity receiver.

(b) A slow network feeding a high-capacity receiver.

# Network Layer and IP

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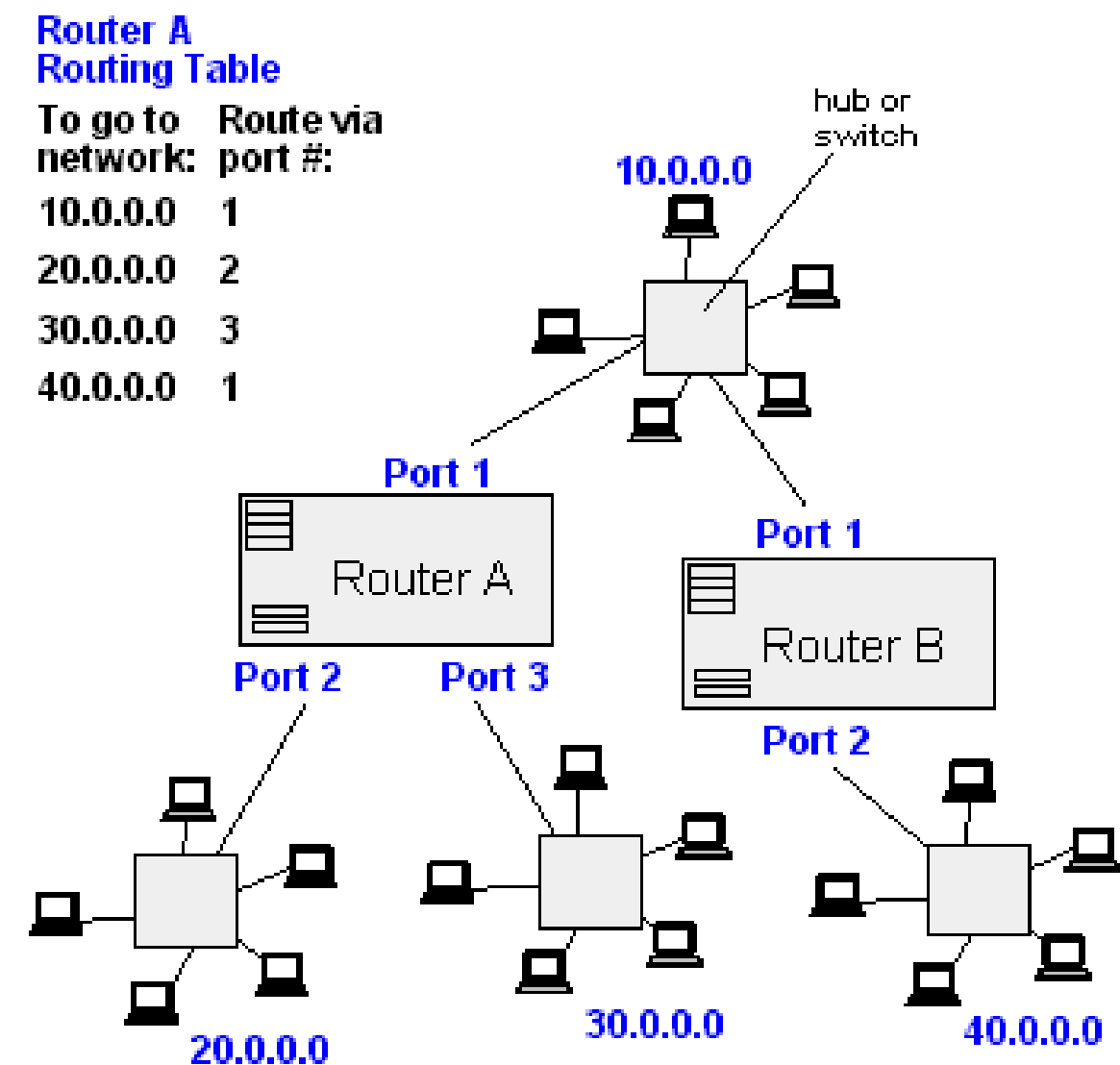


- ▶ IP is a connectionless protocol that provides no flow control, no error control, and no congestion control services.
  - The application should rely only on the transport-layer protocol if it needs any of these.

# Network Layer and Routing Protocols



- Includes unicast (one-to-one) & multicast (one-to-many) routing protocols.
  - Doesn't involve in routing but it creates forwarding tables for routers to help them in the routing process.



# Network Layer and Other Protocols

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- ▶ Internet Control Message Protocol (ICMP) helps IP to report some problems when routing a packet.
- ▶ Internet Group Management Protocol (IGMP) helps IP in multitasking.
- ▶ Dynamic Host Configuration Protocol (DHCP) helps IP to get the network-layer address for a host.
- ▶ Address Resolution Protocol (ARP) helps IP to find the link-layer address of a host or a router when its network-layer address is given.

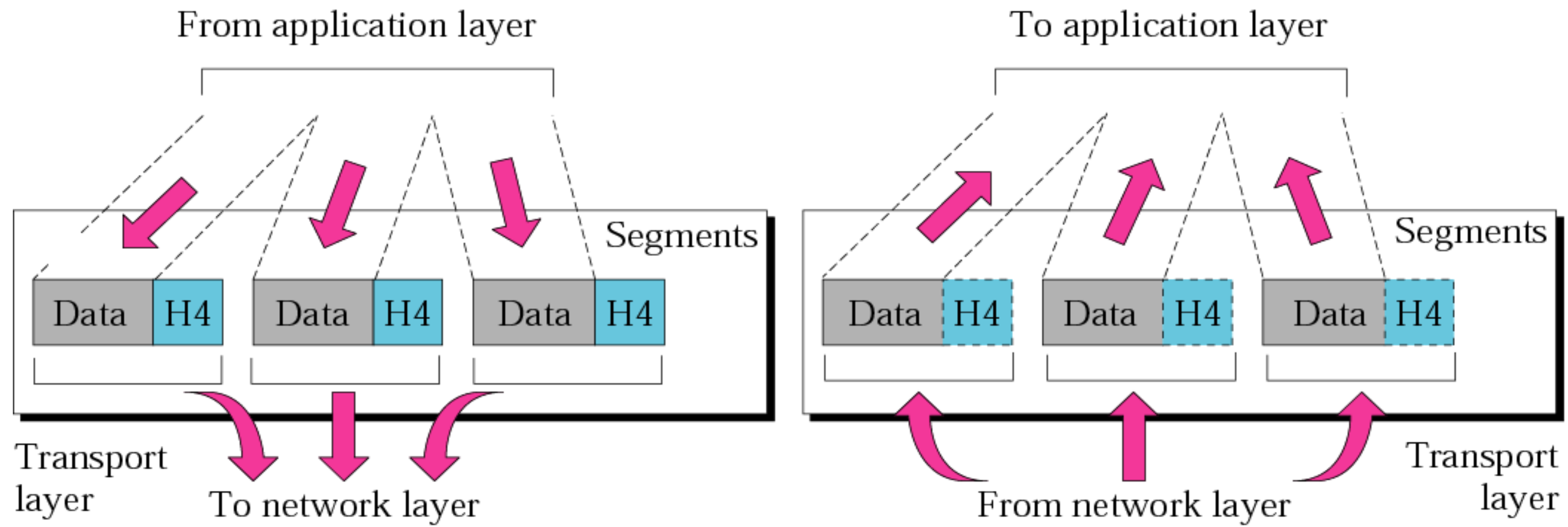
# Transport Layer

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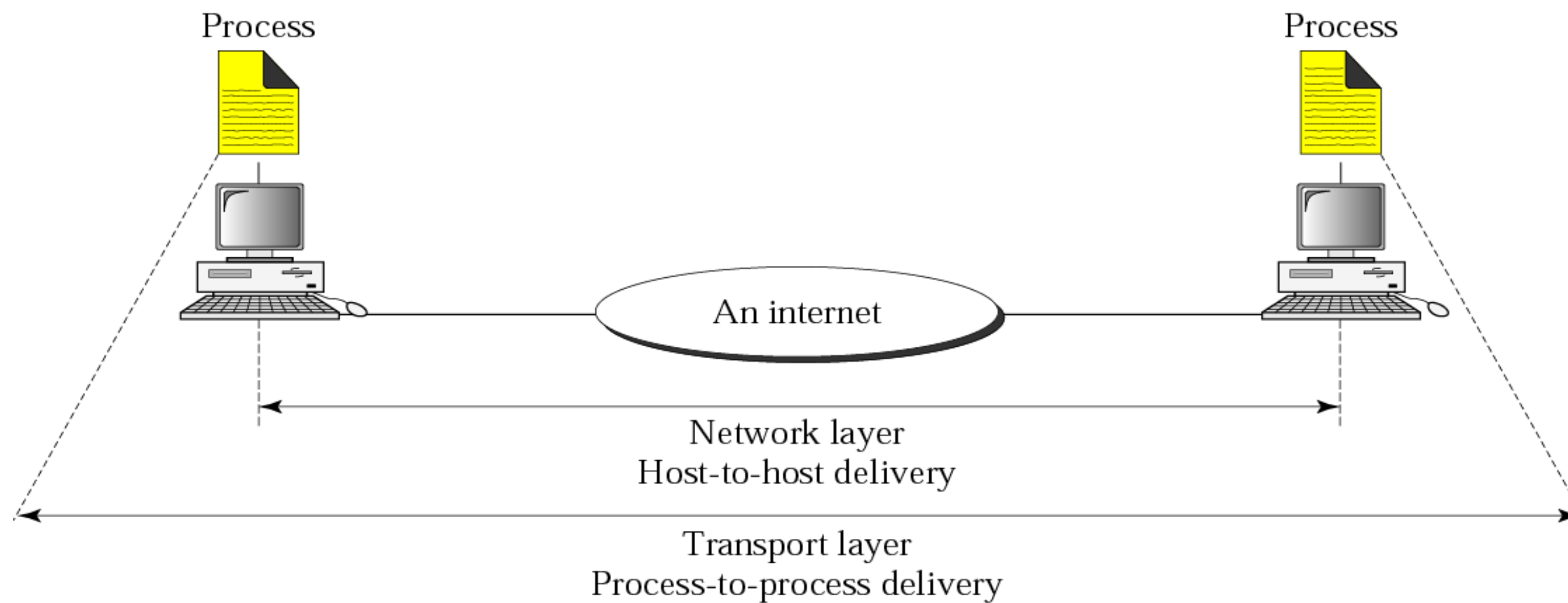


- ▶ Responsible to get a message from an application program process on the source and deliver it to the corresponding application program process on the destination.
- ▶ It encapsulates it in a transport layer packet (called a segment or a user datagram in different protocols) and sends it.

# Transport Layer



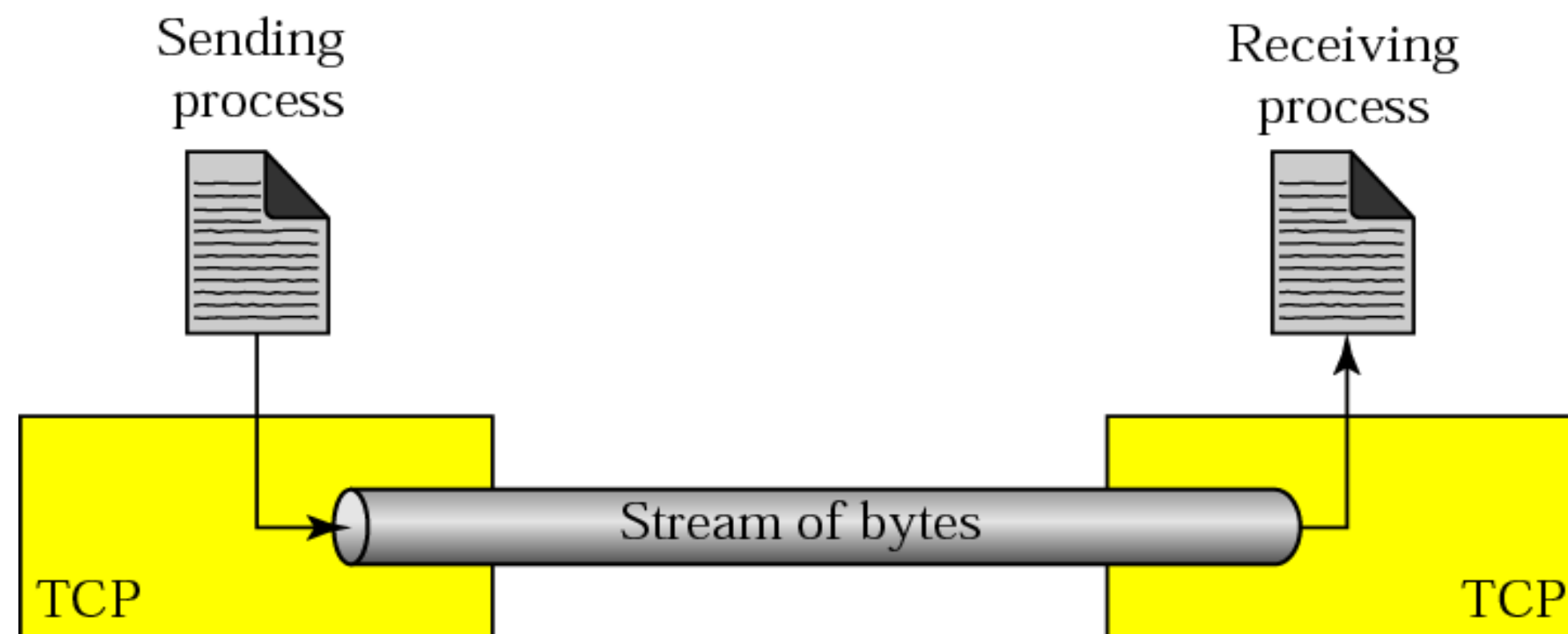
# Transport Layer



# Transport Layer and TCP



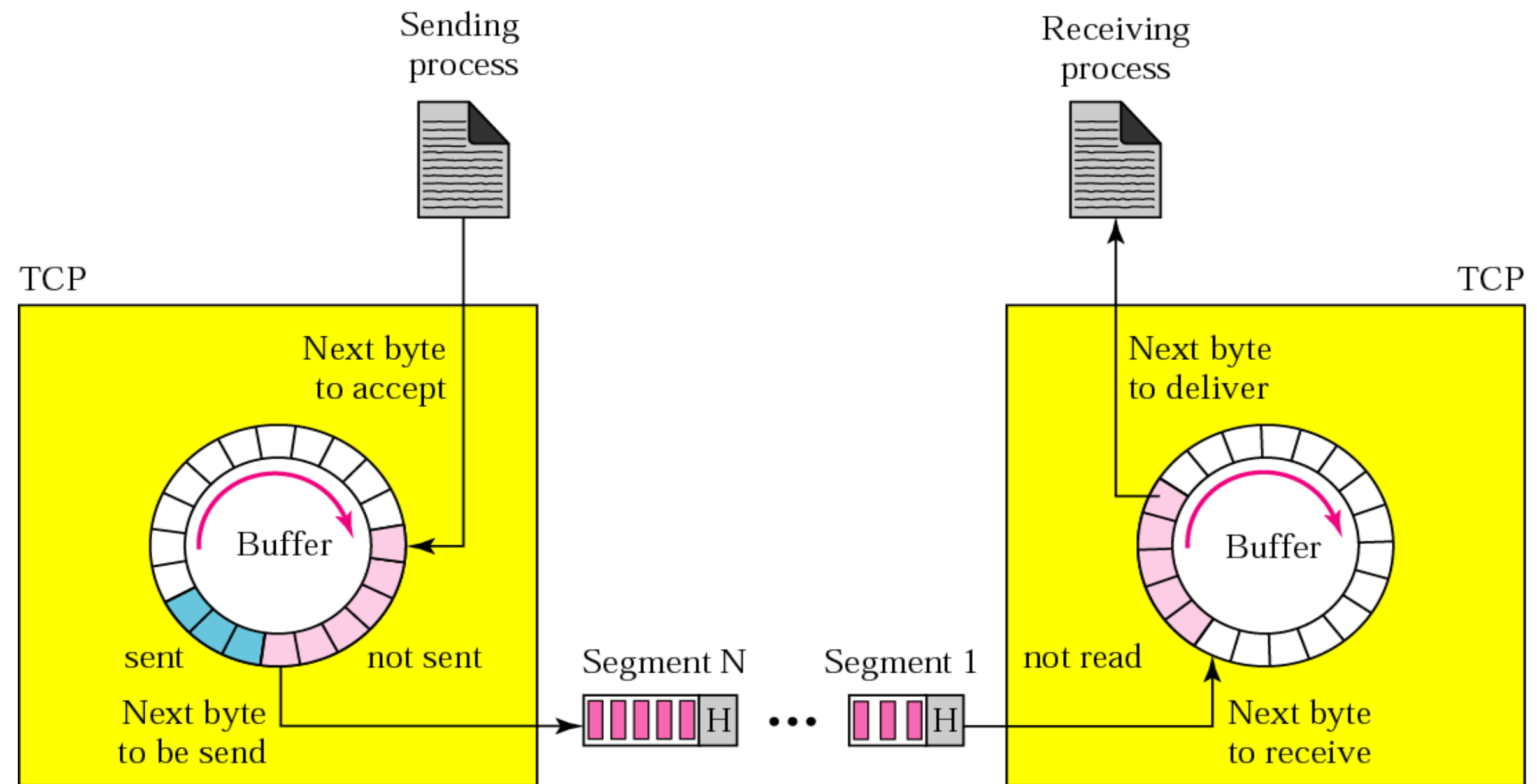
- ▶ Transmission Control Protocol (TCP), is a connection-oriented protocol
  - It first establishes a logical connection between transport layers at two hosts before transferring data.
  - It creates a logical pipe between two TCPs for transferring a stream of bytes.



# Transport Layer and TCP



- ▶ TCP provides
  - flow control
  - error control
  - congestion control



# Transport Layer and UDP

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- ▶ User Datagram Protocol (UDP) is a connectionless protocol
  - It transmits user datagrams without first creating a logical connection.
- ▶ In UDP, each user datagram is an independent entity without being related to the previous or the next one.
  - UDP is a simple protocol that does not provide flow, error, or congestion control.
- ▶ It is attractive when there's need to send short messages and cannot afford the retransmission of the packets.
- ▶ Stream Control Transmission Protocol (SCTP) is designed to respond to new applications that are emerging in the multimedia

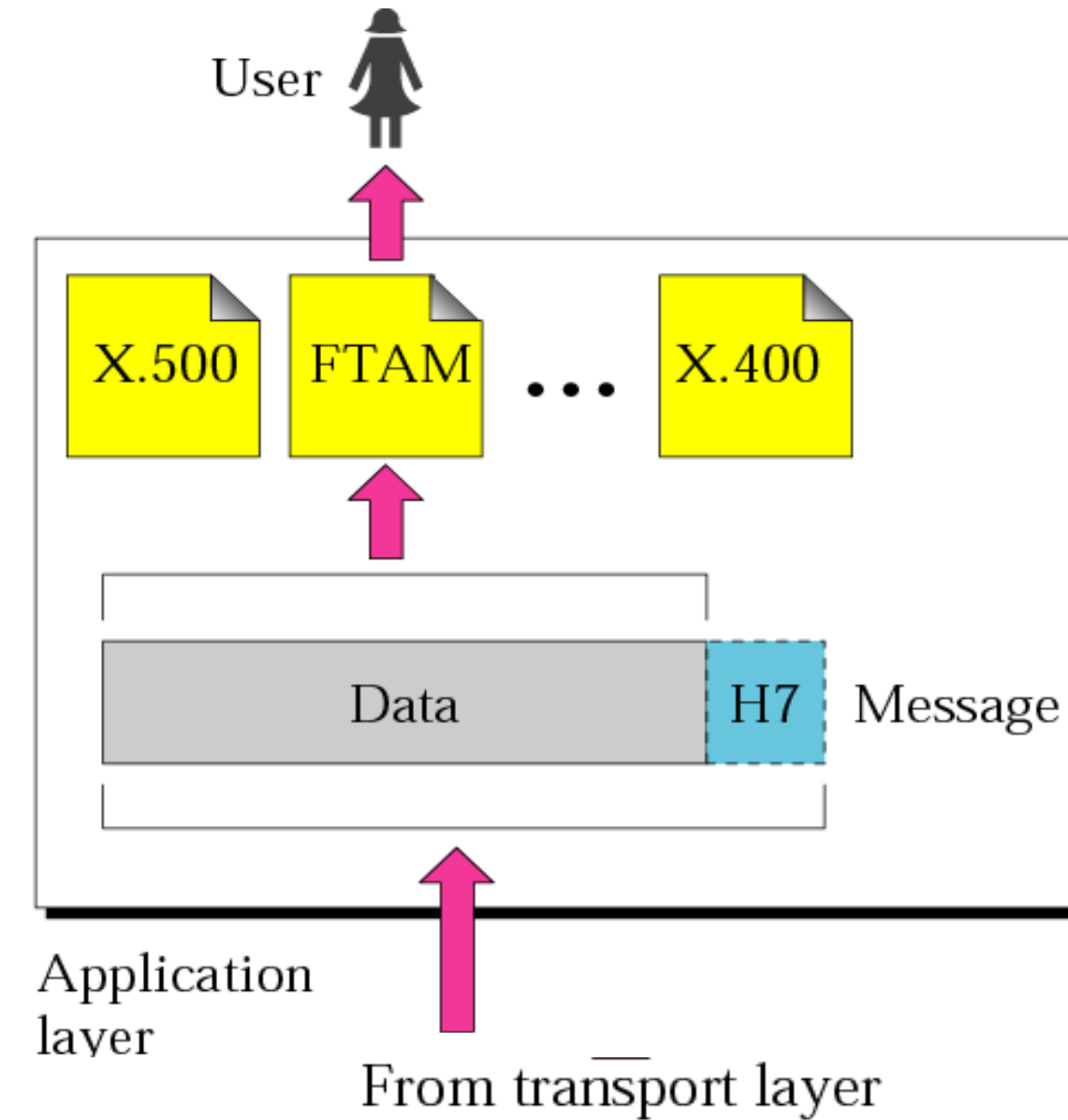
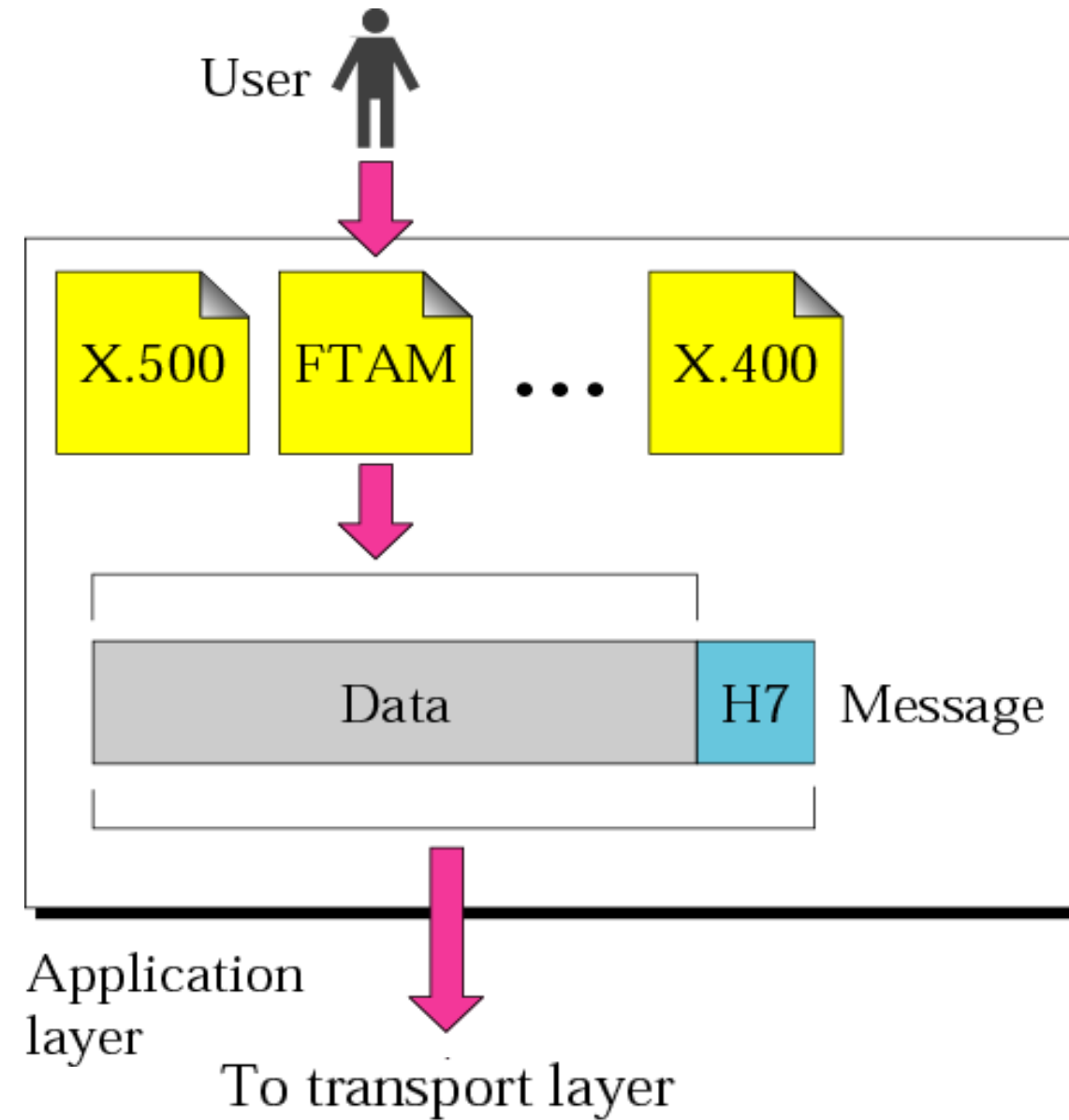
# Application Layer

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- ▶ Process-to-process communication (between two programs running at this layer) is the duty of the application layer.
  - To communicate, a process sends a request to the other process and receives a response.

# Application Layer



# Application Layer and Other Protocols

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- ▶ Hypertext Transfer Protocol (HTTP) is a vehicle for accessing the World Wide Web (WWW).
- ▶ Simple Mail Transfer Protocol (SMTP) is the main protocol used in electronic mail (e-mail) service.
- ▶ File Transfer Protocol (FTP) is used for transferring files from one host to another.
- ▶ Terminal Network (TELNET) and Secure Shell (SSH) are used for accessing a site remotely.

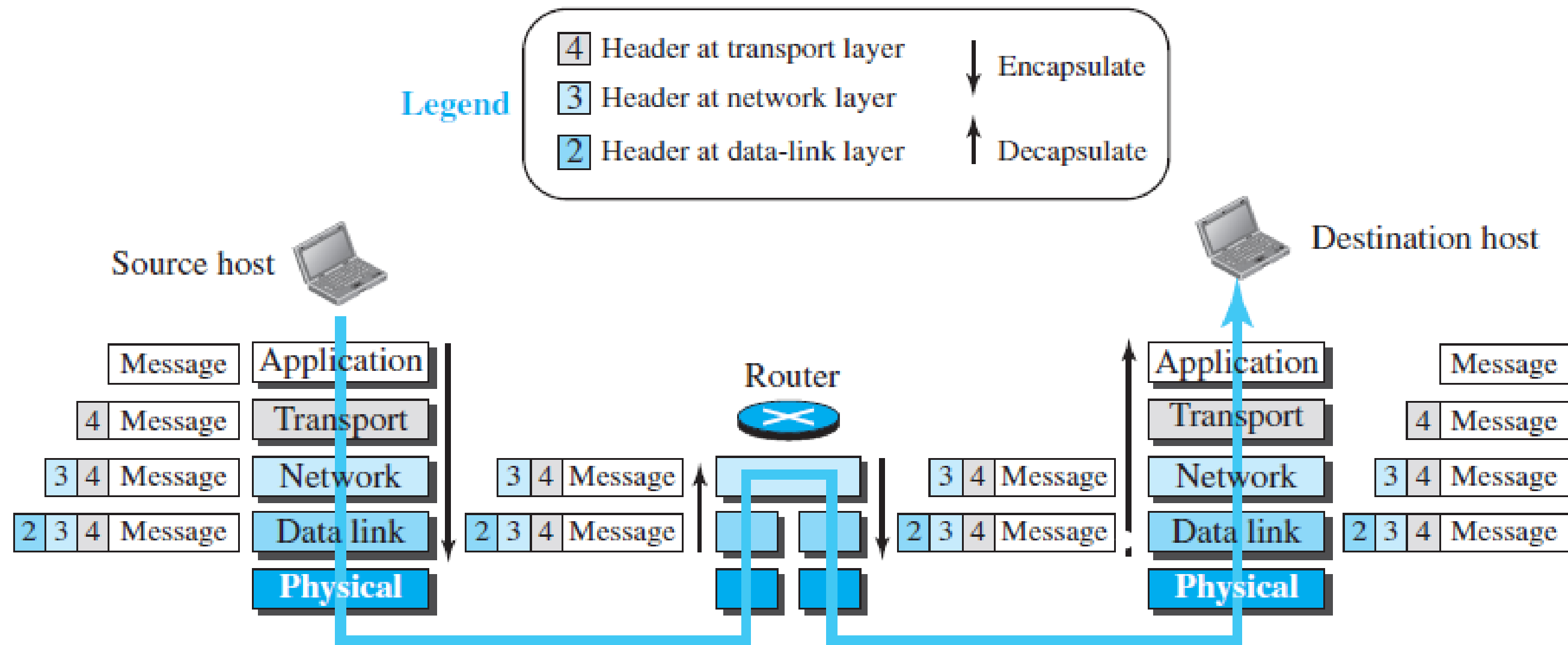
# Application Layer and Other Protocols

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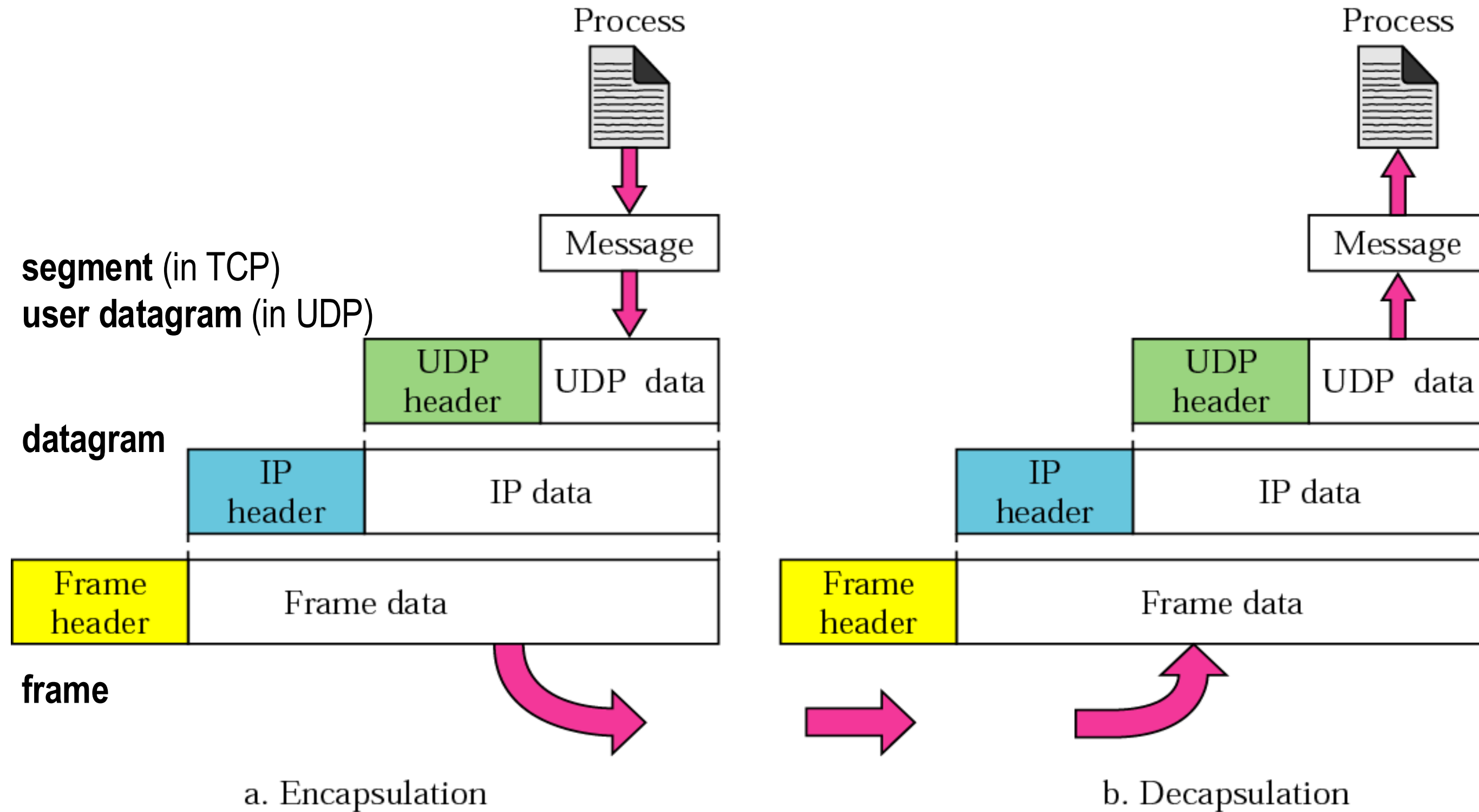


- ▶ Simple Network Management Protocol (SNMP) is used by an administrator to manage the Internet at global and local levels.
- ▶ Domain Name System (DNS) is used by other protocols to find the network-layer address of a computer.
- ▶ Internet Group Management Protocol (IGMP) is used to collect membership in a group.

# Encapsulation/Decapsulation



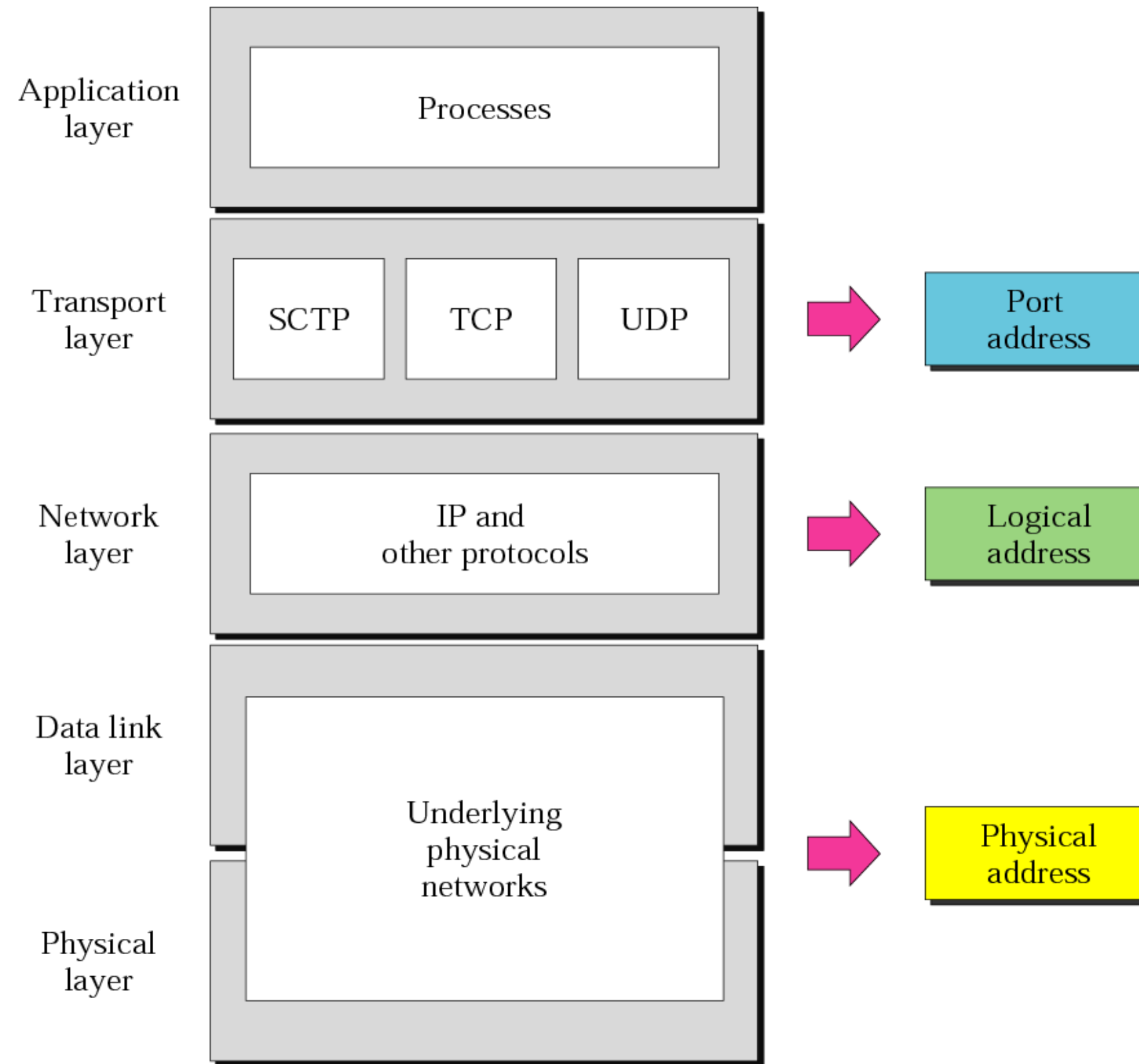
# Encapsulation/Decapsulation



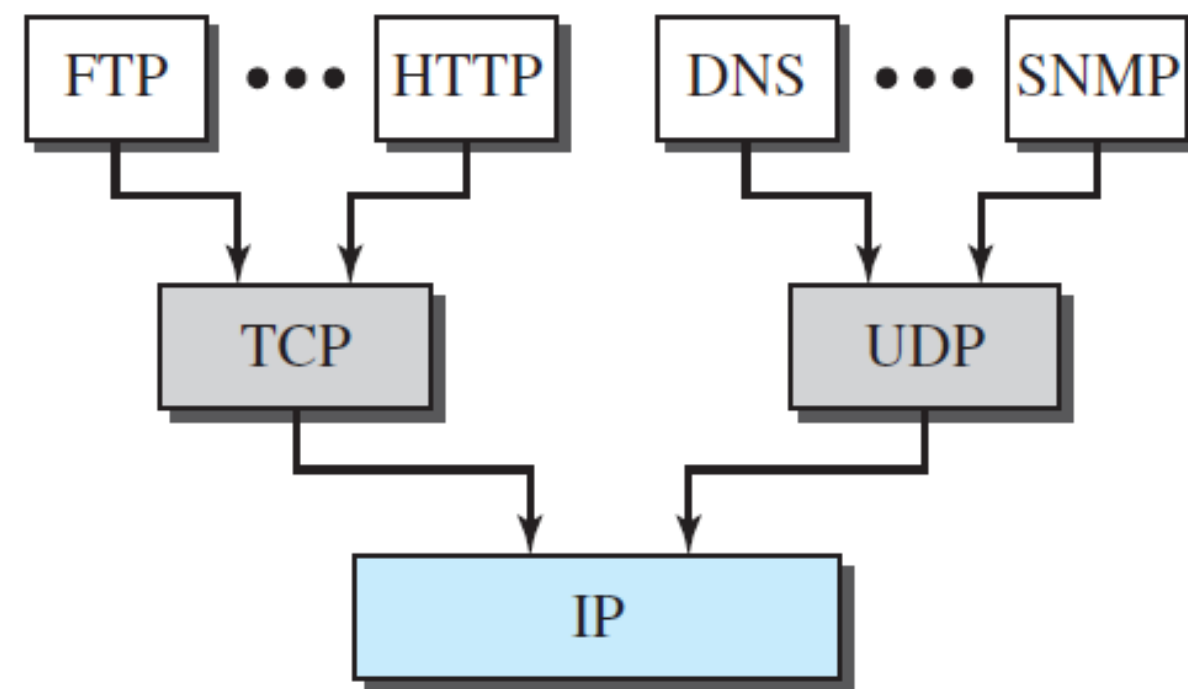
# Addressing



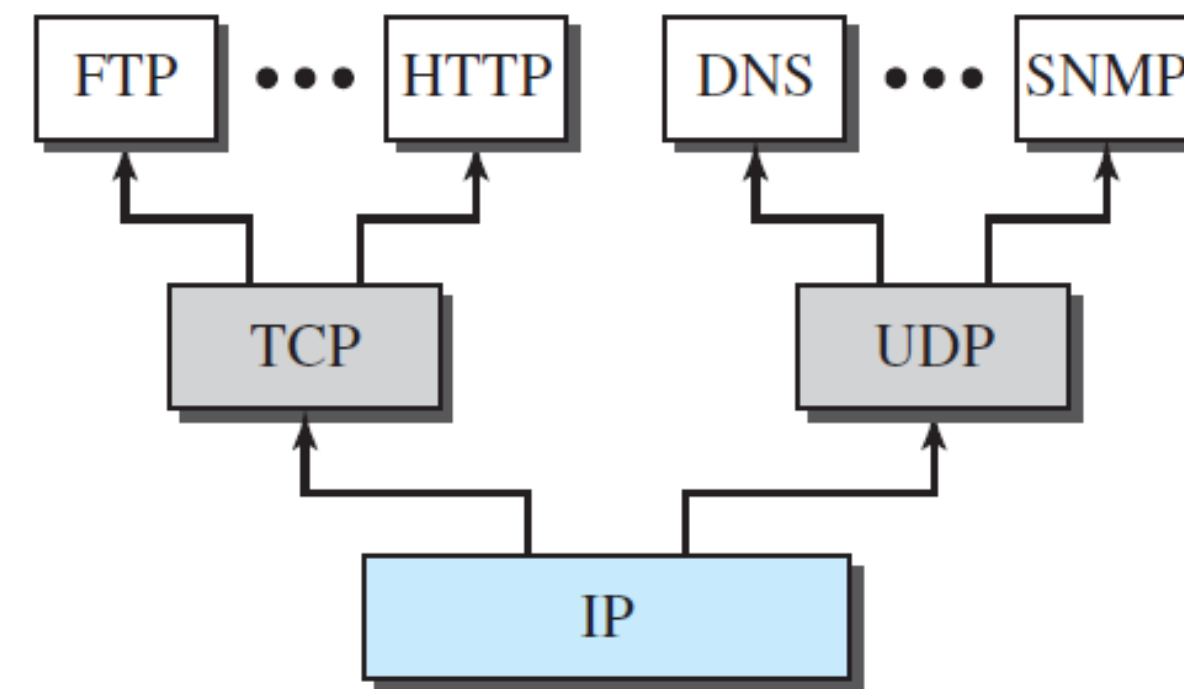
| Packet names            | Layers            | Addresses            |
|-------------------------|-------------------|----------------------|
| Message                 | Application layer | Names                |
| Segment / User datagram | Transport layer   | Port numbers         |
| Datagram                | Network layer     | Logical addresses    |
| Frame                   | Data-link layer   | Link-layer addresses |
| Bits                    | Physical layer    |                      |



# Multiplexing/Demultiplexing



a. Multiplexing at source



b. Demultiplexing at destination

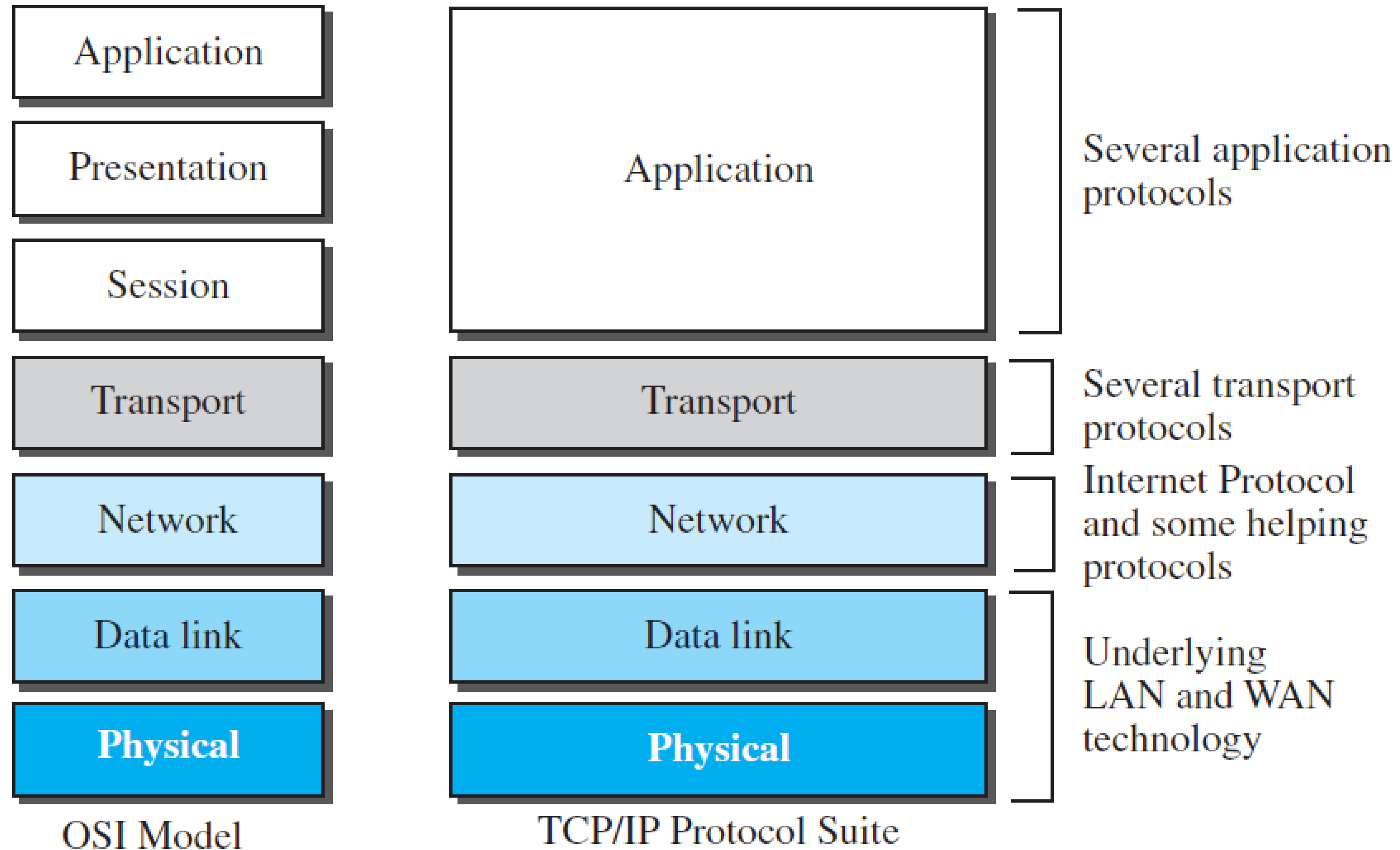
# OSI Model of ISO

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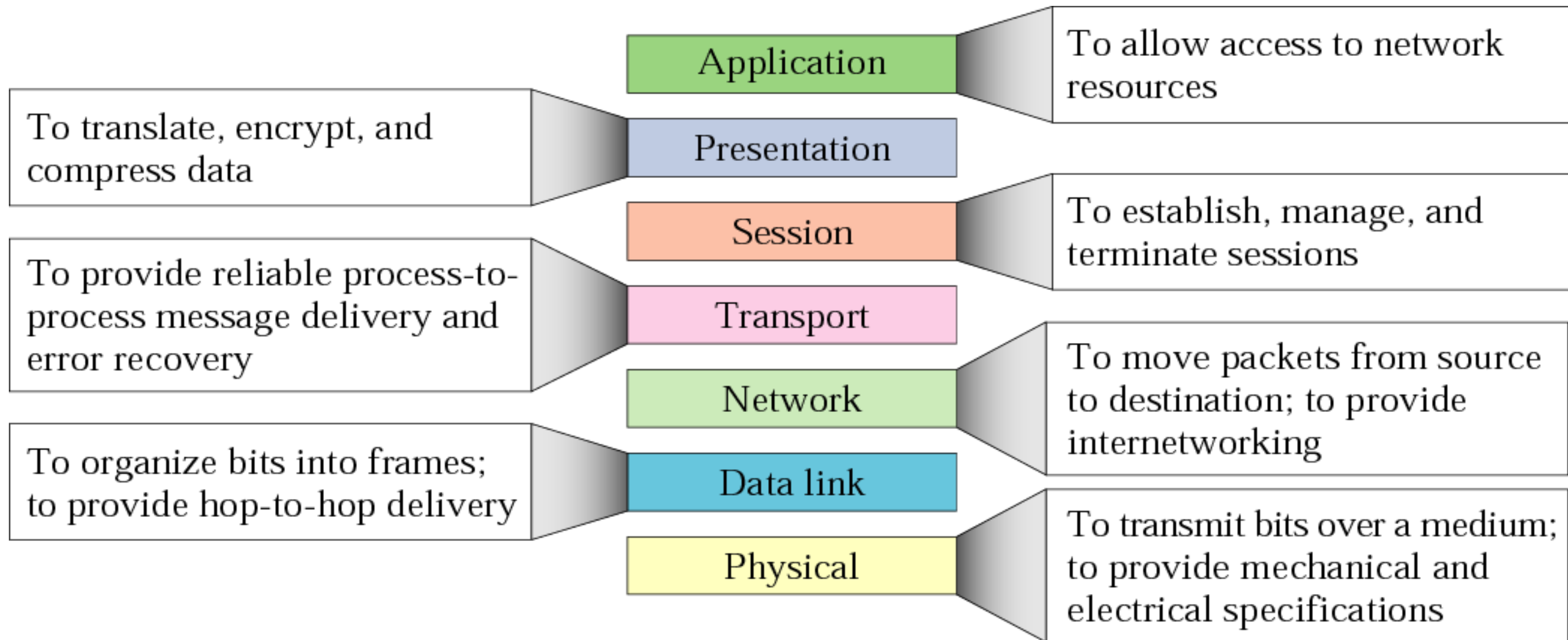


- ▶ Open Systems Interconnection (OSI) model, an International Organization for Standardization (ISO) standard.
- ▶ An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.

# OSI and TCP/IP



# OSI Model



# RECAP

